

Algebra

CHAPTER 1

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1

Set Theory and Real Number System

SETS

We often deal with groups or collection of objects in real life such as a set of books, a group of students, a list of states in a country, a collection of songs on a mobile phone, a collection of shirts etc.

In mathematics, we come across collection of numbers like even integers, odd integers, prime numbers, integers divisible by 3 etc.

Also, in geometry we deal with collections of geometrical figures like types of triangle, types of quadrilaterals etc.

A well defined collection of distinct objects, numbers, figures etc. is a set.

Thus, formally we define set as a collection of objects of any kind of our intuition or thought which are distinct and distinguishable.

By the word 'distinct' we mean that no object is repeated and by the word distinguishable, we mean that the object of the collection must be known i.e., given any object we must be able to decide whether that object belongs to the collection or not. Objects of a set have a common property and objects outside this collection do not have this property.

The objects of the collection are called the elements or members of the set.

A set is represented by listing all its elements between braces $\{ \}$ and by separating them from each other by commas (if there is more than one element). Sets are denoted by capital letters of English alphabet while the elements are denoted, in general, by small letters. If x is an element of a set A , we write $x \in A$ (read as 'x belongs to A'). If x is not an element of A , we write $x \notin A$ (read as 'x does not belong to A').

We have following sets of different types of numbers.

- N : the set of all natural numbers
- Z : the set of all integers
- Q : the set of all rational numbers
- R : the set of real numbers
- Z^+ : the set of positive integers
- Q^+ : the set of positive rational numbers
- R^+ : the set of positive real numbers

REPRESENTATION OF SET

Tabular Form or Roster Form

In this method of describing a set, the elements of the set are separated by commas within brackets.

For example,

- (i) The set of all odd positive integers less than 10 can be described as $\{1, 3, 5, 7, 9\}$.
- (ii) The set of all vowels can be described as $\{a, e, i, o, u\}$.
- (iii) The set of all natural numbers can be described as $\{1, 2, 3, \dots\}$. Here three dots $\dots$ stand for "and so on".
- (iv) The set of letters of the word EXAMINATION is $\{E, X, A, M, I, N, T, O\}$.

Property Method or Set-Builder Form

In this method, each element is generalised as a variable x (say) and then after colon ($:$) or a vertical line ($|$), the property which defines each element of the set is written in the form x .

For example,

- (i) The set $A = \{1, 3, 5\}$ can be written as
 $A = \{x: x \in N, x \text{ is an odd number } x < 5\}$
- (ii) The set A of all even natural numbers can be described as
 $A = \{x | x = 2n, n \in N\}$
- (iii) The set $A = \{\text{January, June, July}\}$ can be written as
 $A = \{x : x \text{ is a name of month which starts with letter 'J'}\}$

ILLUSTRATION 1.1

State which of the following statements are true and which ones are false. Justify your answer.

- (i) $21 \in \{x | x \text{ has exactly four positive factors}\}$
- (ii) $64 \in \{y | \text{the sum of all the positive factors of } y \text{ is } 2y\}$
- (iii) $2 \notin \{x | x^4 - 3x^3 + 4x^2 - 5x + 6 = 0\}$
- (iv) $23562 \notin \{y | y \text{ is divisible by } 9\}$

Sol.

- (i) The factors of 21 are 1, 3, 7 and 21. So, 21 is an element of the set.

Hence, the given statement is true.

- (ii) The factors of 64 are 1, 2, 4, 8, 16, 32 and 64.

$$\begin{aligned} \therefore \text{Sum of factors} &= 1 + 2 + 4 + 8 + 16 + 32 + 64 \\ &= 127 \neq 2 \times 64 \end{aligned}$$

So, 64 is not an element of the set.

Hence, the given statement is false.

- (iii) We have, $x^4 - 3x^3 + 4x^2 - 5x + 6 = 0$

$$\text{For } x = 2, (2)^4 - 3(2)^3 + 4(2)^2 - 5(2) + 6 = 0$$

$$\Rightarrow 16 - 24 + 16 - 10 + 6 = 0$$

$$\Rightarrow 4 = 0; \text{ which is not true.}$$

So, 2 is not an element of the set.

Hence, the given statement is true.

(iv) We have number, 23562

Sum of the digits of the number is $2 + 3 + 5 + 6 + 2 = 18$; which is divisible by 9.

So, 23562 is an element of the set.

Hence, the given statement is false.

DIFFERENT TYPES OF SETS

Finite Set

A set having finite number of elements is called a finite set.

For example,

(i) $A = \{1, 2, 3\}$

Here, A is a finite set as it has 3 (finite number) elements.

(ii) $B = \text{Set of vowels of English Alphabet} = \{a, e, i, o, u\}$

Here, B is a finite set with 5 elements.

(iii) $C = \text{Set of all rivers in India.}$

However, the list is long, but the number is finite. So, C is a finite set.

Infinite Set

A set is said to be an infinite set if the number of elements in the set is not finite.

For example,

(i) $N = \text{Set of all positive integers} = \{1, 2, 3, 4, \dots\}$

There are infinite positive integers and hence, N is an infinite set.

Similarly, sets W , Z , Q and R are all infinite sets.

(ii) $A = \text{Set of all points on a straight line}$

Since there are infinite points on a straight line, set A is an infinite set.

Null set (or Empty set or Void set)

A set having no element is called a null set or an empty set or a void set. It is denoted by ϕ or $\{\}$.

For example,

(i) $A = \text{The set of odd numbers divisible by 2}$

(ii) $B = \text{The set of days between Sunday and Monday}$

Here, A and B are null sets.

(iii) $C = \{x : x \in N \text{ and } 1 < x < 2\}$.

Here, C is a null set because there is no natural number lying between 0 and 1.

Note:

The set $\{0\}$ is not an empty set as it contains one element i.e., 0. Also, the set $\{\phi\}$ is not an empty set as it contains one element i.e., ϕ .

Singleton Set

A set having only one element is called a singleton set.

For example,

(i) $A = \text{The set of present President of India.}$

(ii) $B = \{x : x \text{ is an even prime number}\}$

Equal Sets

Two sets A and B are said to be equal if each element of A is an element of B and each element of B is an element of A . Thus, two sets A and B are equal if they have exactly the same elements but the order in which the elements in the two sets are written is immaterial.

If sets A and B are equal, then we write $A = B$.

For example,

(i) $A = \{4, 5, 6, 7\}$, $B = \{5, 6, 4, 7\}$.

Here, A and B have exactly the same elements.

Therefore, $A = B$.

(ii) $A = \text{Set of letters of the word LOYAL}$

$B = \text{Set of letters of the word ALLOY}$

Here, $A = \{L, O, Y, A\}$ and $B = \{A, L, O, Y\}$

Therefore, $A = B$.

Note that two sets A and B are equal if

$$x \in A \Rightarrow x \in B$$

$$\text{and } x \in B \Rightarrow x \in A.$$

Set of Sets

A set S having all its elements as sets is called a set of sets or a family of sets or a class of sets.

For example,

(i) $\{\{1, 2\}, \{2, 5\}, \{3, 6, 8\}\}$ is a set of sets having three elements $\{1, 2\}$, $\{2, 5\}$ and $\{3, 6, 8\}$ which are themselves sets.

(ii) $\{\phi\}$ is a singleton set of set having null set ϕ as its element.

(iii) $S = \{\{1, 2\}, 3, \{4\}\}$ is not a set of sets as $3 \in S$ is not a set.

ILLUSTRATION 1.2

Are the following pairs of sets equal?

(i) $A = \{x \mid x \text{ is prime factor of } 6\}$;

$$B = \{x \mid x \text{ is a solution of } x^2 - 5x + 6 = 0\}$$

(ii) $A = \{x \mid x \text{ is a letter in the word REPLACED}\}$;

$$B = \{y \mid y \text{ is a letter in the word PARCELED}\}$$

(iii) $A = \{x \mid x \text{ is a natural number, } x > 1\}$;

$$B = \{x \mid x \text{ is natural number, } x \geq 1\}$$

Sol.

(i) $A = \{x \mid x \text{ is prime factor of } 6\} = \{2, 3\}$

$$B = \{x \mid x \text{ is solution of } x^2 - 5x + 6 = 0\}$$

$$x^2 - 5x + 6 = 0$$

$$\text{or } (x - 2)(x - 3) = 0$$

$$\text{or } x = 2, 3$$

$$\therefore B = \{2, 3\}$$

Therefore, $A = B$.

(ii) $A = \{x \mid x \text{ is a letter in the word REPLACED}\}$

$$= \{R, E, P, L, A, C, D\}$$

$$B = \{y \mid y \text{ is a letter in the word PARCELED}\}$$

$$= \{P, A, R, C, E, L, D\}$$

Therefore, $A = B$.

(iii) $A = \{x \mid x \text{ is natural number, } x > 1\} = \{2, 3, 4, \dots\}$

$$B = \{x \mid x \text{ is natural number, } x \geq 1\} = \{1, 2, 3, \dots\}$$

We observe that $1 \in B$ but $1 \notin A$.

Therefore, $A \neq B$.

SUBSET

Consider sets, $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6, 7\}$

We observe that each element of set A is an element of set B . In other words, all the elements of set A are also elements of set B . Actually, set A is contained in the set B .

Now, consider sets, $A = \{x \mid x \text{ is a girl student in your class}\}$, $B = \{x \mid x \text{ is student in your class}\}$

Obviously, each girl student in your class is student of your class. Thus, set A is contained in set B .

In both of the above cases, set A is called the subset of B . Mathematically, it is written as $A \subseteq B$.

Thus, $A \subseteq B \Leftrightarrow x \in A \Rightarrow x \in B$.

Here, $\subseteq$ symbolizes the subset or equal sets.

Now, consider sets $A = \{1, 2, 3\}$ and $B = \{2, 3, 1\}$.

Here, sets A and B are equal or coincide. We can write $A = B$ or $A \subseteq B$ or $B \subseteq A$.

If at least one element of the set A is not the element of the set B , then A not a subset of B i.e., $A \not\subseteq B$.

PROPER SUBSET OF A SET

Set A is said to be a proper subset of set B if A is a subset of B and $A \neq B$.

This fact is expressed by writing $A \subset B$ or $B \supset A$ (read as ' A is a proper subset of B ').

Thus, the number of elements in B is greater than that in A . For example,

- (i) Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 1, 5\}$. Then $A \subset B$ or $B \supset A$.
- (ii) $N \subset Z \subset Q \subset R$

Equality of Sets

Two sets A and B are equal if A is a subset of B and B is a subset of A .

Thus, $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.

Note:

$A \subset B \Leftrightarrow A \subseteq B$ and $A \neq B$.

Properties of Subsets

- Every set is its own subset.
- Empty set is a subset of each set.
- Let A, B, C be three sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Superset of a Set

A set A is said to be a superset of set B , if B is a subset of A i.e., each element of B is an element of A . If A is a superset of B , we write $A \supset B$.

The statement $B \subseteq A$ can also be expressed equivalently by writing $A \supset B$ (read as ' A is a superset of B ').

ILLUSTRATION 1.3

If $A \not\subseteq B$ and $B \not\subseteq C$ then $A \not\subseteq C$. Is this statement true?

Sol. Let $A = \{1, 2\}$, $B = \{0, 6, 8\}$, and $C = \{0, 1, 2, 6, 9\}$

Here, $A \not\subseteq B$ and $B \not\subseteq C$, but $A \subset C$.

So, the given statement is false.

ILLUSTRATION 1.4

If $x \in A$ and $A \not\subseteq B$, then $x \in B$. Is this statement true?

Sol. Let $A = \{3, 5, 7\}$ and $B = \{3, 4, 6\}$

Now, $5 \in A$ and $A \not\subseteq B$

However, $5 \notin B$

So, the given statement is false.

POWER SET

The set or family of all the subsets of a given set A is said to be the power set of A and is denoted by $P(A)$.

Symbolically, $P(A) = \{X \mid X \subseteq A\}$

Thus, $X \in P(A) \Leftrightarrow X \subseteq A$

Also, $\phi \in P(A)$ and $A \in P(A)$ for all sets A .

The elements of the power set $P(A)$ are the subsets of A .

For example,

- (i) If $A = \{1\}$, then $P(A) = \{\phi, \{1\}\}$
- (ii) If $B = \{1, 2\}$, then $P(B) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$
- (iii) If $C = \{1, 2, 3\}$, then $P(C) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$

We observe that number of elements in power set of a given set is 2^n , where n is the number of elements in the given set.

Note:

If $A \subseteq B \Rightarrow P(A) \subseteq P(B)$.

UNIVERSAL SET

In any discussion regarding set theory, we need a set such that all the sets under consideration in that discussion are its subsets. Such a set is called the universal set for that discussion.

Any set which is superset of all the sets under consideration is called the universal set and is denoted by S or U .

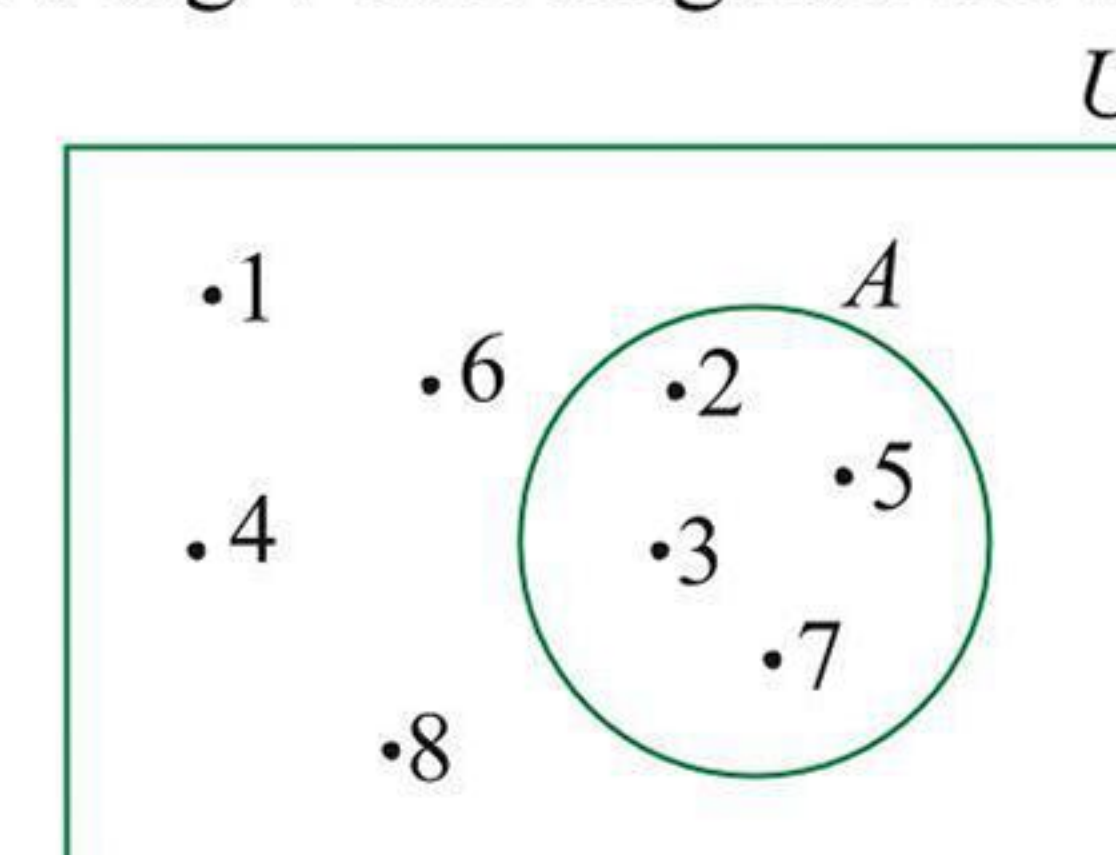
A universal set can be chosen arbitrarily for any discussion of given sets; but once chosen, it is fixed for that discussion of the sets.

e.g., for sets $A = \{1, 2, 3\}$, $B = \{3, 4, 6, 7, 8, 9\}$ and $C = \{0, 1\}$, we can consider $U = \{0, 1, 2, 3, 4, 6, 7, 8, 9\}$ as universal set. This is the smallest universal set in this discussion. In this case, any set for which U is its subset can also be taken as universal set.

VENN DIAGRAM

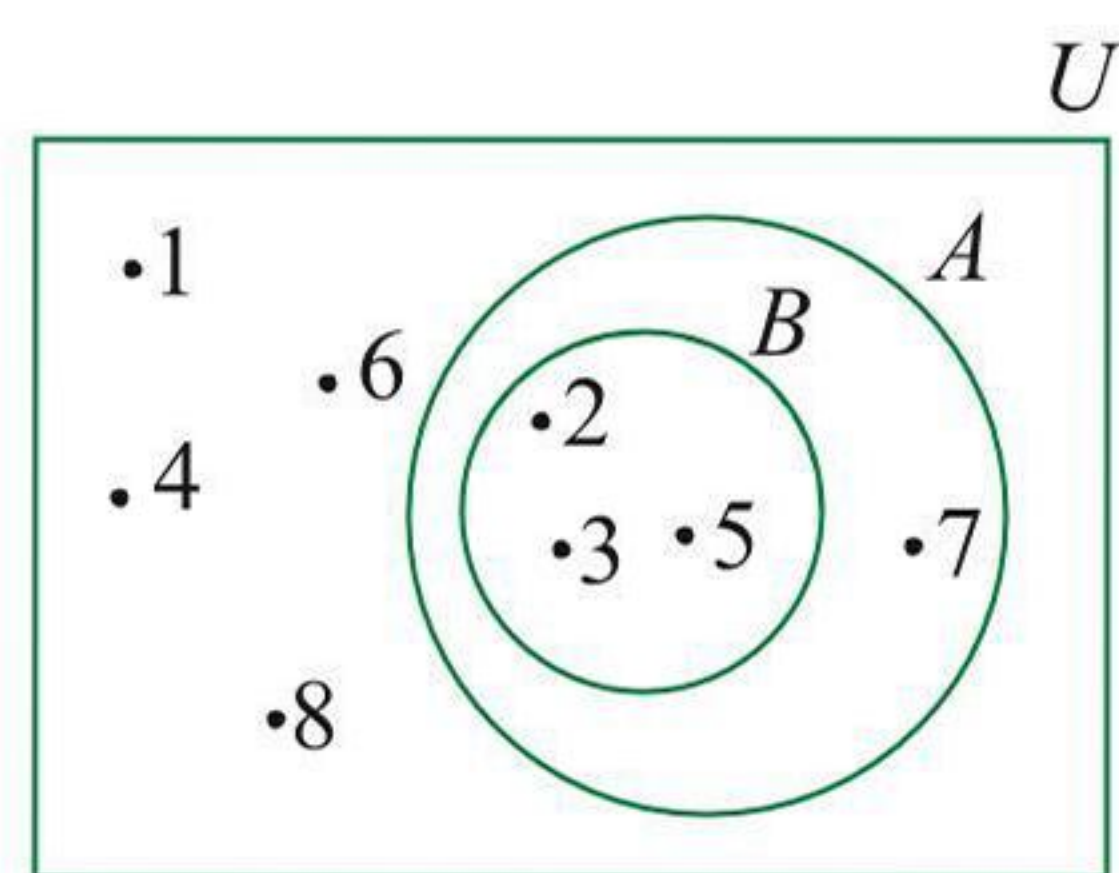
Most of the relationships between sets can be represented by means of diagrams which are known as Venn diagrams. Venn diagrams are named after the English logician, John Venn. These diagrams consist of rectangles and closed curves usually circles. The universal set is represented usually by a rectangle and its subsets by circles. In Venn diagrams, the elements of the sets are written in their respective circles.

Consider $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and one subset $A = \{2, 3, 5, 7\}$. We have following Venn diagram for these sets.



Now, consider one another subset $B = \{2, 3, 5\}$ and clearly, $B \subset A$.

We have the following Venn diagram to represent this information.



Here, circle for set B is contained in the circle for set A .

OPERATION ON SETS

Like operations on real numbers such as addition, subtraction, multiplication and division, we have some operations on the sets.

Union of Two Sets

The union of two sets A and B is the set of all those elements which are either in A or in B or in both.

This set is denoted by $A \cup B$ (read as ‘ A union B ’).

Symbolically, $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Clearly, $x \in A \cup B \Leftrightarrow x \in A \text{ or } x \in B$

For example,

- (i) $A = \{1, 2, 3\}$ and $B = \{2, 1, 5, 6\}$
Then $A \cup B = \{1, 2, 3, 5, 6\}$
- (ii) $A = \{x : x \text{ is a prime number less than } 10\}$ and
 $B = \{x : x \in \mathbb{Z}, -5 < x < 5\}$
Then $A = \{2, 3, 5, 7\}$ and $B = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$
 $\therefore A \cup B = \{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 7\}$
- (iii) $A = \{1, 2, 3\}$, $B = \{3, 5\}$, $C = \{4, 7, 8\}$
Then $A \cup B \cup C = \{1, 2, 3, 4, 5, 7, 8\}$

Properties of union of sets

- If $x \notin A \cup B$, then certainly $x \notin A$ and $x \notin B$.
- If $A \subseteq B$, then $A \cup B = B$.
- $A \cup B = B \cup A$ (commutative)
- $A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C)$ (Associative)
Also, $A \cup B \cup C = B \cup C \cup A = C \cup B \cup A$ etc.
- $A \subseteq A \cup B$ and $B \subseteq A \cup B$
- $\phi \cup A = A \cup \phi = A$

Union and the word ‘or’

The word ‘or’ tells us that there is a union of two sets.

For example,

$\{\text{singers}\} \cup \{\text{instrumentalists}\} \equiv \{\text{people who sing or play an instrument}\}$

$\{\text{vowels}\} \cup \{\text{letters in word ARMED}\} \equiv \{\text{letters that are vowels or are in word ARMED}\}$

Union and the word ‘at least’

The word ‘at least’ tells us that there is a union of two sets.

Let sets A , B and C be the sets of students who play cricket, football and hockey, respectively.

Then set $A \cup B$ is the set of students who play at least one of the sports cricket and football.

Similarly, $B \cup C$ is the set of students who play at least one of the sports football and hockey.

Also, $A \cup B \cup C$ is the set of students who play at least one of the sports, cricket, football and hockey.

Intersection of Two Sets

The intersection of two sets A and B is the set of all the elements which are common in A and B . This set is denoted by $A \cap B$ (read as ‘ A intersection B ’).

Symbolically, $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Clearly, $x \in A \cap B \Leftrightarrow x \in A \text{ and } x \in B$.

For example,

- (i) Consider sets, $A = \{1, 2, 3\}$ and $B = \{1, 2, 5, 6\}$
Here, elements 1 and 2 are common in sets A and B .
So, $A \cap B = \{1, 2\}$
- (ii) Consider sets $A = \{1, 2, 3\}$ and $B = \{5, 6, 7, 8\}$.
We observe that there is no element common in sets A and B .
So, $A \cap B = \phi$
- (iii) Consider sets $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5, 6\}$.
We observe that all the elements of set A are in set B . In fact, set A is subset of set B .
So, $A \cap B = \{1, 2, 3\} = A$
- (iv) Consider sets $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$ and $C = \{1, 2\}$.
Then $A \cap B \cap C = \{2\}$

Properties of intersection of sets

- $x \notin A \text{ and } x \notin B \Leftrightarrow x \notin A \cap B$
If $x \notin (A_1 \cap A_2 \cap A_3 \cap \dots)$ then x is not the element of at least one of the sets.
- If $A \subseteq B$ then $A \cap B = A$.
- $A \cap B = B \cap A$ (commutative)
- $A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$ (associative)
Also, $A \cap B \cap C = B \cap C \cap A = C \cap B \cap A$ etc.
- $A \cap B \subseteq A$ and $A \cap B \subseteq B$
- $\phi \cap A = A \cap \phi = \phi$
- $(A \cup B) \cap A = A$ and $(A \cup B) \cap B = B$
- $(A \cap B) \cup A = A$ and $(A \cap B) \cup B = B$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (Distributive law)
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive law)

Intersection and the word ‘and’

The word ‘and’ tells us that there is an intersection of two sets.

For example,

$\{\text{singers}\} \cap \{\text{instrumentalists}\} \equiv \{\text{people who sing and play an instrument}\}$

$\{\text{vowels}\} \cap \{\text{letters of word EXAMINATION}\} \equiv \{\text{letters that are vowels and are in word EXAMINATION}\}$

Intersection and the word ‘both’

The word ‘both’ tells us that there is an intersection of two sets.

Let sets A , B and C be the sets of students who play cricket, football and hockey, respectively.

Then set $A \cap B$ is the set of students who play both the sports cricket and football.

Similarly, $B \cap C$ is the set of students who play both the sports football and hockey.

Also, $A \cap B \cap C$ is the set of students who play all the sports cricket, football and hockey.

Disjoint sets

If $A \cap B = \phi$, then sets A and B are called disjoint sets.

Difference of Two Sets

The difference of two sets A and B (also called 'relative complement' of B in A) is the set of all those elements of A which are not elements of B . It is denoted by $A - B$. Thus $A - B$ can be obtained by discarding the elements of B , present in A .

Symbolically, $A - B = \{x \mid x \in A \text{ and } x \notin B\}$

For example,

- (i) Consider sets, $A = \{1, 3, 5, 6, 7\}$ and $B = \{2, 3, 4, 5\}$.

Here, elements 3 and 5 are common in sets A and B .

So, $A - B = \{1, 6, 7\}$ and $B - A = \{2, 4\}$

- (ii) Consider sets $A = \{1, 2, 3\}$ and $B = \{5, 6, 7, 8\}$.

We observe that there is no element common between sets A and B .

So, $A - B = \{1, 2, 3\} = A$ and $B - A = \{5, 6, 7, 8\} = B$

- (iii) Consider sets $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5, 6\}$.

We observe that all the elements of set A are in set B . i.e., set A is subset of set B .

So, $A - B = \phi$ and $B - A = \{4, 5, 6\}$

Properties of difference of sets

- If $A \cap B = \phi$ (i.e., sets A and B have no common elements), then $A - B = A$ and $B - A = B$.
- If $A \subseteq B$, then $A - B = \phi$.
- x is not the element of set $A - B$ if $x \in A$ and $x \in B$.
- $A - B \neq B - A$

Difference and the word 'only'

Consider A = set of students who play cricket
and B = set of students who play football.

Now, $A - B$ = set of students who play cricket but not football
= set of students who play cricket only

Similarly, $B - A$ = set of students who play football only

COMPLEMENT OF A SET

The complement of a set A (also called 'absolute complement' of A) is the set of all those elements of the universal set U which are not the elements of A . It is denoted by A^c or A' .

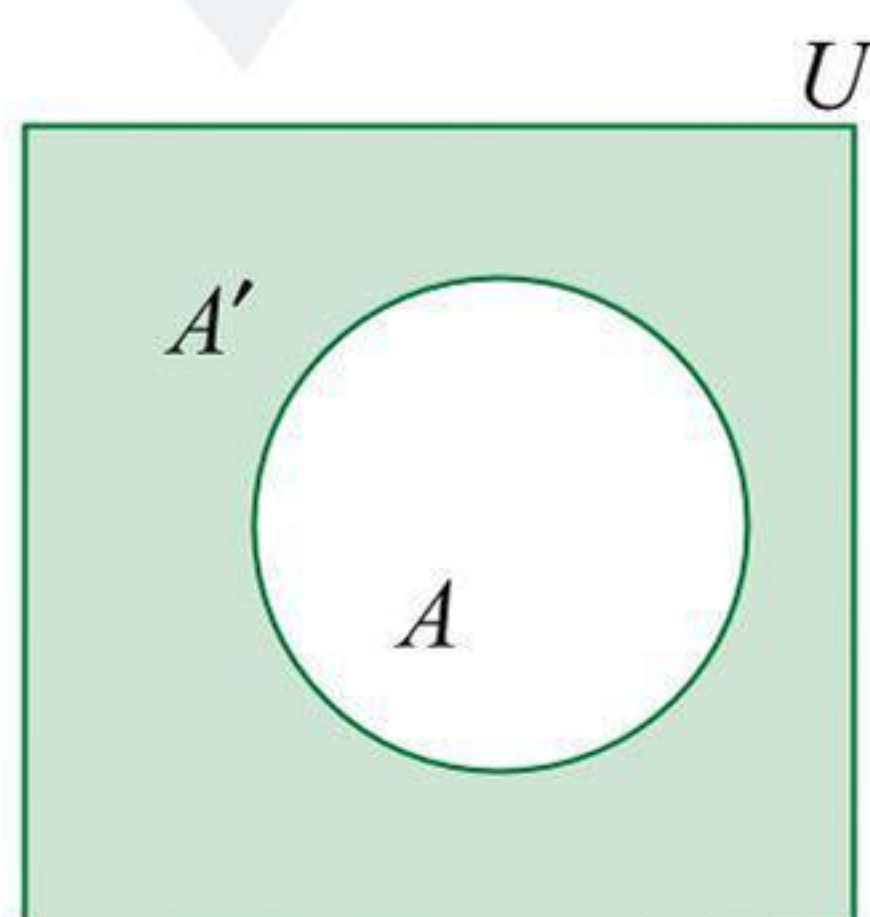
For example, if set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and set $A = \{1, 2, 5, 7, 9\}$ then set $A' = \{3, 4, 6, 8, 10\}$.

Clearly, $A' = U - A$.

Symbolically, set $A' = \{x \mid x \in U \text{ and } x \notin A\}$

Thus, if $x \in A \Leftrightarrow x \notin A'$.

Complement of a set can be represented by Venn diagram as shown in the given figure.



In above figure, shaded region represents set A' .

Properties of complement of sets

- $A \cap A' = \phi$ (as nothing is common between A and A')
- $A \cup A' = U$
- $U' = \phi$
- $(A')' = A$
- $A \subseteq B \Leftrightarrow B' \subseteq A'$
- $A - B = B' - A'$
- $A - B = A \cap B'$
- $B - A = A' \cap B$

Complement and the word 'not'

The word 'not' corresponds to the complement of a set.

For example, let set $V = \{\text{letters that are vowel}\}$

Then complement of set V is $V' = \{\text{letters that are not vowels}\}$
= $\{\text{letters that are consonants}\}$

De Morgan's Laws

- $(A \cup B)' = A' \cap B'$
- $(A \cap B)' = A' \cup B'$
- $A - (B \cup C) = (A - B) \cap (A - C)$
- $A - (B \cap C) = (A - B) \cup (A - C)$

SYMMETRIC DIFFERENCE OF SETS

The symmetric difference of two sets A and B , denoted as $A \Delta B$, is the set of elements that lie either in set A or in set B only, but not in both A and B .

i.e., $A \Delta B = (A - B) \cup (B - A) = A \cup B - A \cap B$

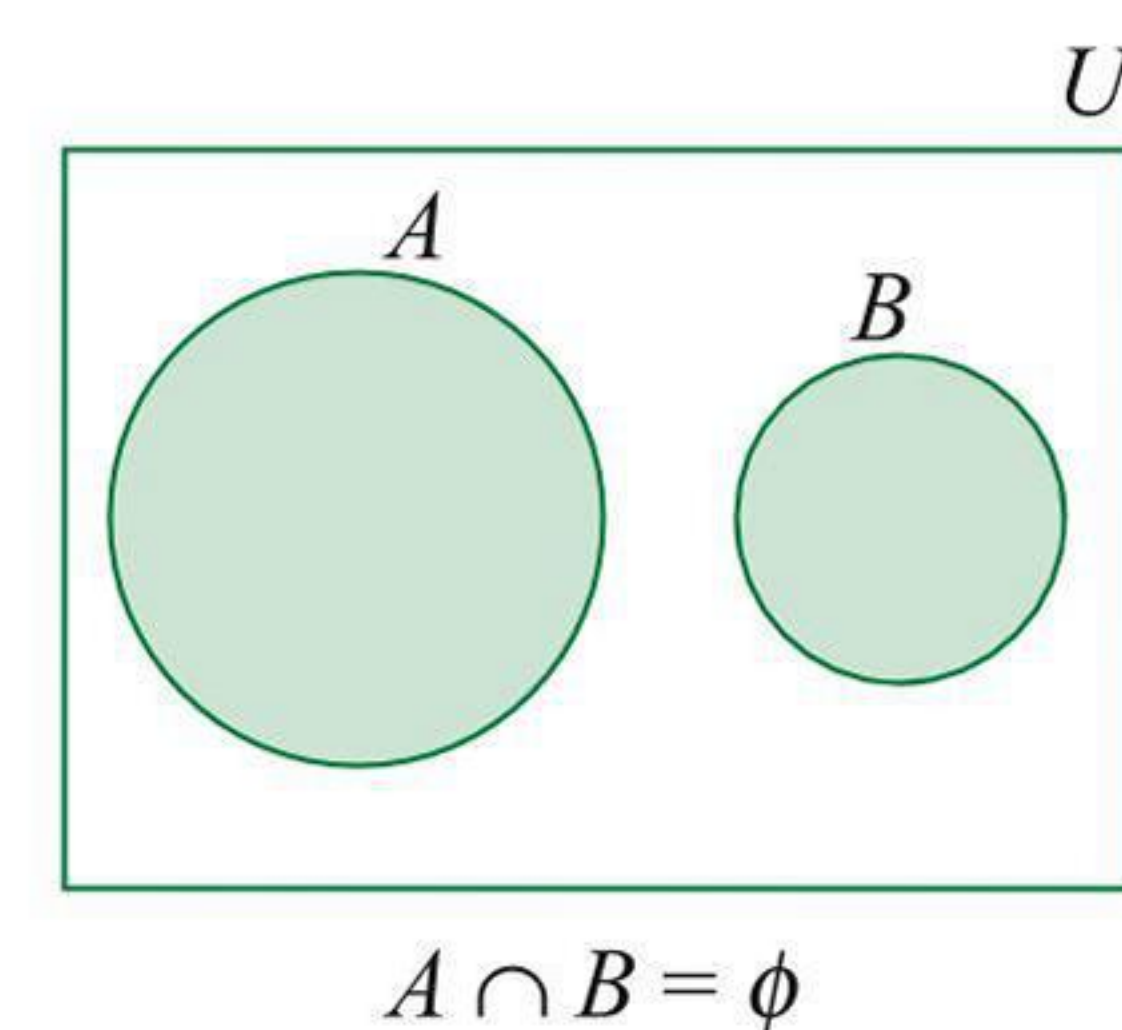
It follows that $A \Delta \phi = A$ and $A \Delta A = \phi$.

Also, we have the following properties:

- $A \Delta B = B \Delta A$ [Commutative property]
- $A \Delta (B \Delta C) = (A \Delta B) \Delta C$ [Associative property]
- If $A \Delta B = A \Delta C$, then $B = C$
- $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$

VENN DIAGRAMS OF OPERATIONS OF SETS

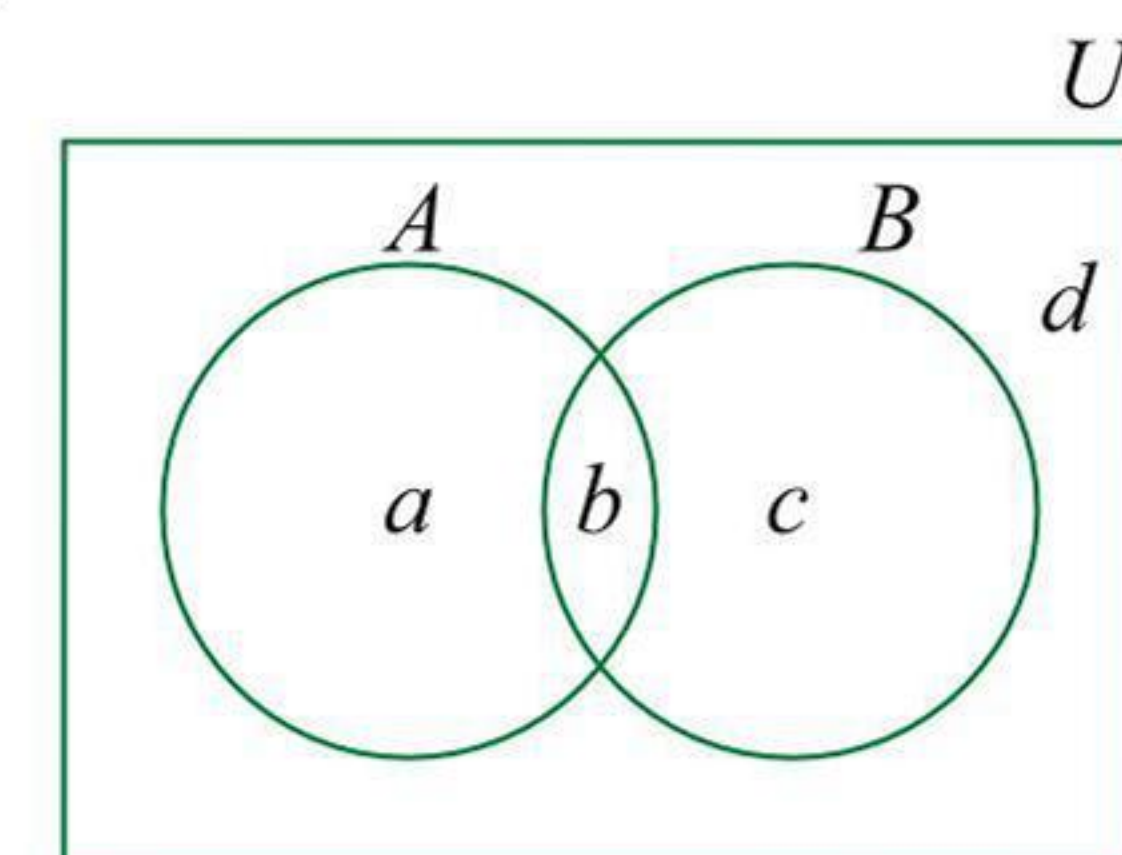
Case I: A and B are disjoint.



Here, shaded part represents $A \cup B$ and $A \Delta B$.

The unshaded part represents $(A \cup B)'$ or $A' \cap B'$.

Case II: $A \cap B \neq \phi$



Here region 'b' represents $A \cap B$.

Region 'a' represents $A \cap B'$ or $A - B$.

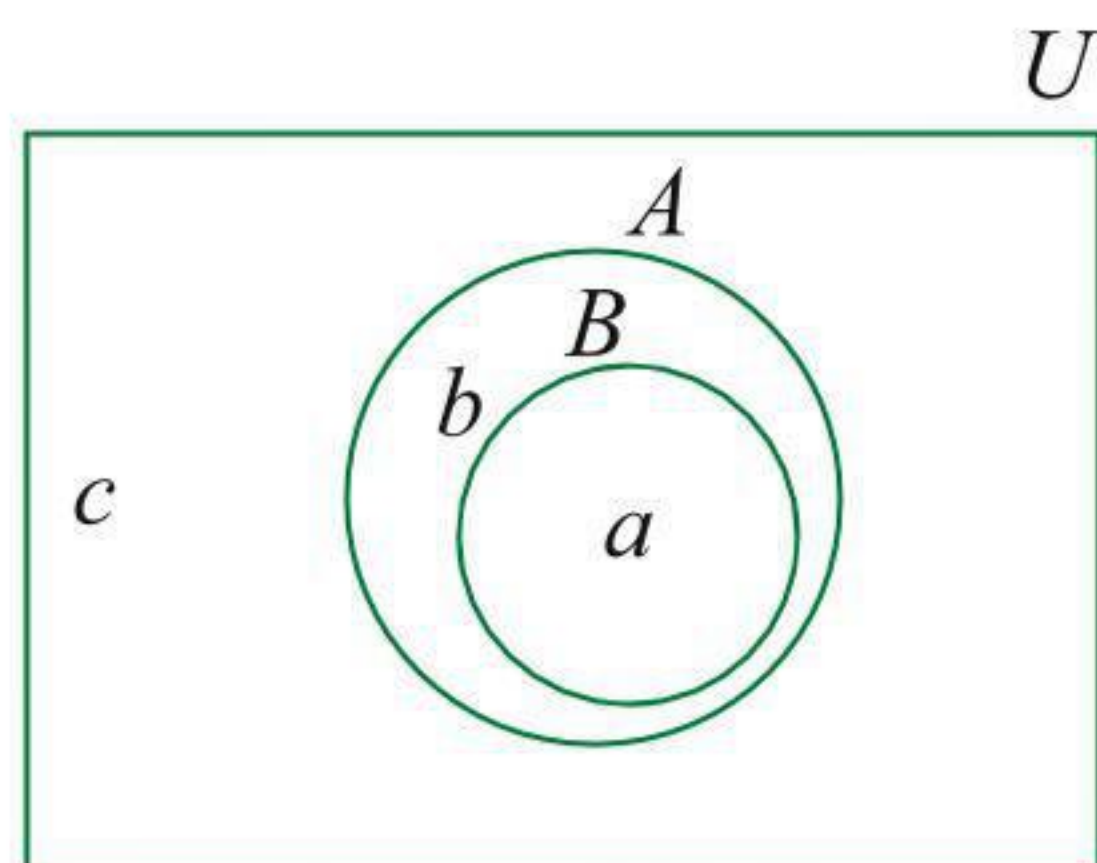
Region 'c' represents $A' \cap B$ or $B - A$.

Regions 'a', 'b' and 'c' collectively represent $A \cup B$.

Region 'd' represents $(A \cup B)'$ or $A' \cap B'$.

Regions 'a' and 'c' together represent $A \Delta B$.

Case III: $B \subset A$



Here, set A represents $A \cup B$.

Set B represents $A \cap B$.

Region 'b' (outside set B but inside set A) represents $A \cap B'$ or $A - B$.

Region 'c' represents $(A \cup B)'$ or $A' \cap B'$.

ILLUSTRATION 1.5

Consider the following sets:

A = set of natural numbers which are multiples of 2

B = set of natural numbers which are multiples of 3

C = set of natural numbers which are multiples of 5

Then find the following sets:

- (i) $A \cup B$ (ii) $B \cup C$ (iii) $A - B$ (iv) $B - C$
 (v) $A \cap C$ (vi) $A \cap B \cap C$ (vii) $(A \cup B) - C$ (viii) $A \Delta B$

Sol. We have

A = set of natural numbers which are multiples of 2

B = set of natural numbers which are multiples of 3

C = set of natural numbers which are multiples of 5

$A \cup B$ = set of natural numbers which are multiples of 2 or 3

= {2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, ...}

$B \cup C$ = set of natural numbers which are multiples of 3 or 5

= {3, 5, 6, 9, 10, 12, 15, 18, 20, ...}

$A - B$ = {2, 4, 6, 8, 10, 12, 14, 16, 18, ...}

- {3, 6, 9, 12, 15, 18, ...}

= set of natural numbers which are multiples of 2 but not multiples of 3

= {2, 4, 8, 10, 14, 16, 20, ...}

In this set, elements 6, 12, 18, ... are excluded which are multiples of 2 and 3 both.

Similarly,

$B - C$ = {3, 6, 9, 12, 15, 18, 21, 24, 27, 30, ...}

- {5, 10, 15, 20, 25, 30, ...}

= {3, 6, 9, 12, 18, 21, 24, 27, 33, ...}

In this set, elements 15, 30, ... are excluded which are multiples of 3 and 5 both.

$A \cap C$ = set of natural numbers which are multiples of 2 and 5 both

= set of natural numbers which are multiples of 10

= {10, 20, 30, 40, 50, ...}

$A \cap B \cap C$ = set of natural numbers which are multiples of 2, 3 and 5

= set of natural numbers which are multiples of 30

= {30, 60, 90, 120, 150, ...}

$(A \cup B) - C$ = set of natural numbers which are multiples of 2 or 3 but not 5

= {2, 3, 4, 6, 8, 9, 12, 14, 16, 18, 21, ...}

In this set, elements 5, 10, 15, 20, 25, 30, ... are excluded.

$A \Delta B$ = set of natural numbers which are multiples of 2 or 3 but not of 6

= {2, 3, 4, 8, 9, 10, 14, 15, 16, 20, 21, 22, 26, ...}

ILLUSTRATION 1.6

If $A - B = A$ and $B - A = B$, then what can we conclude?

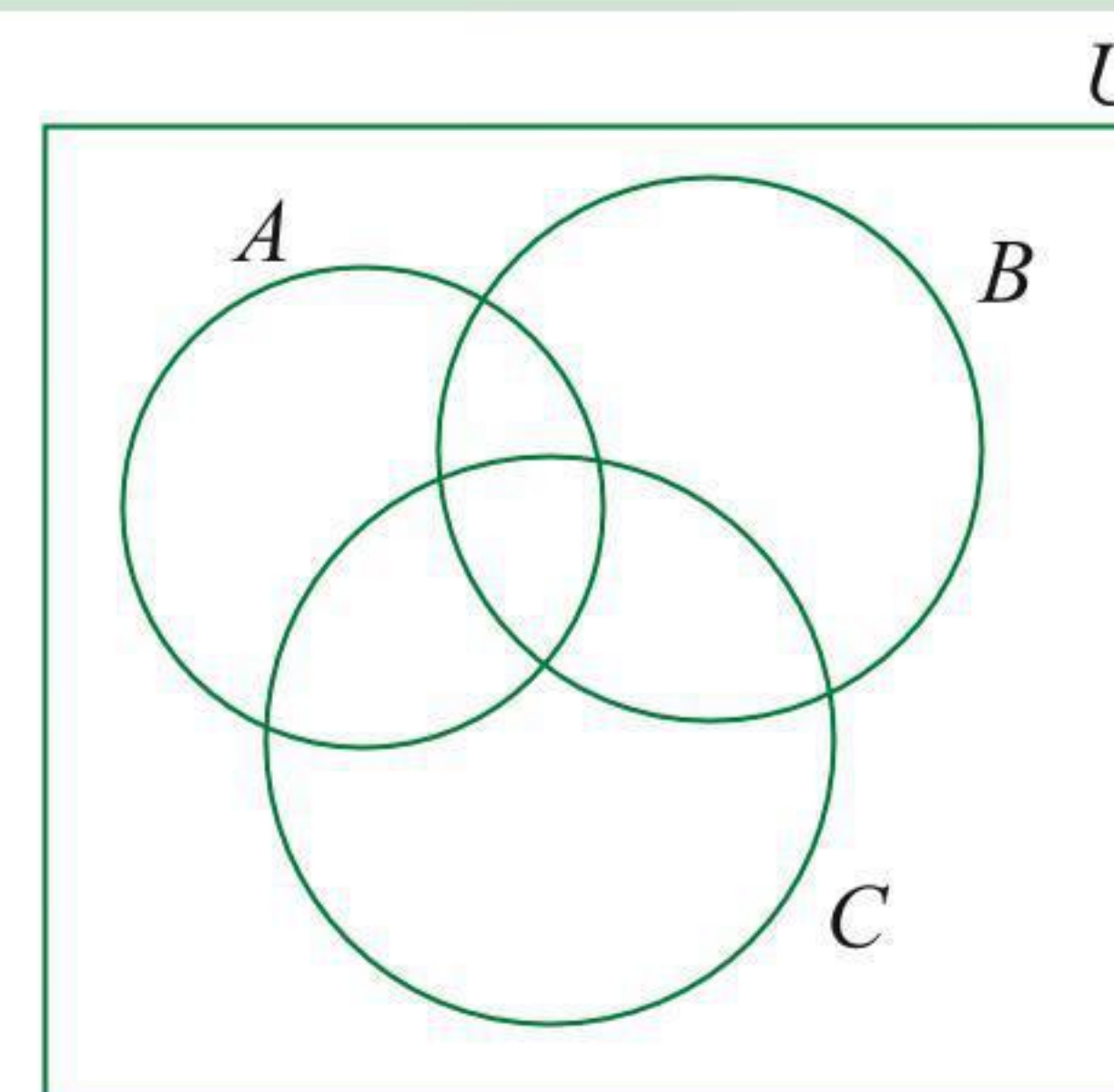
Sol. We have $A - B = A$, which means that no element of set A is common with elements of set B .

Also, $B - A = B$, which means no element of set B is common with elements of set A .

So, $A \cap B = \phi$, i.e., sets A and B are disjoint.

ILLUSTRATION 1.7

In the following Venn diagram, mark the region of the set $(A \Delta B) \Delta C$.



Sol.

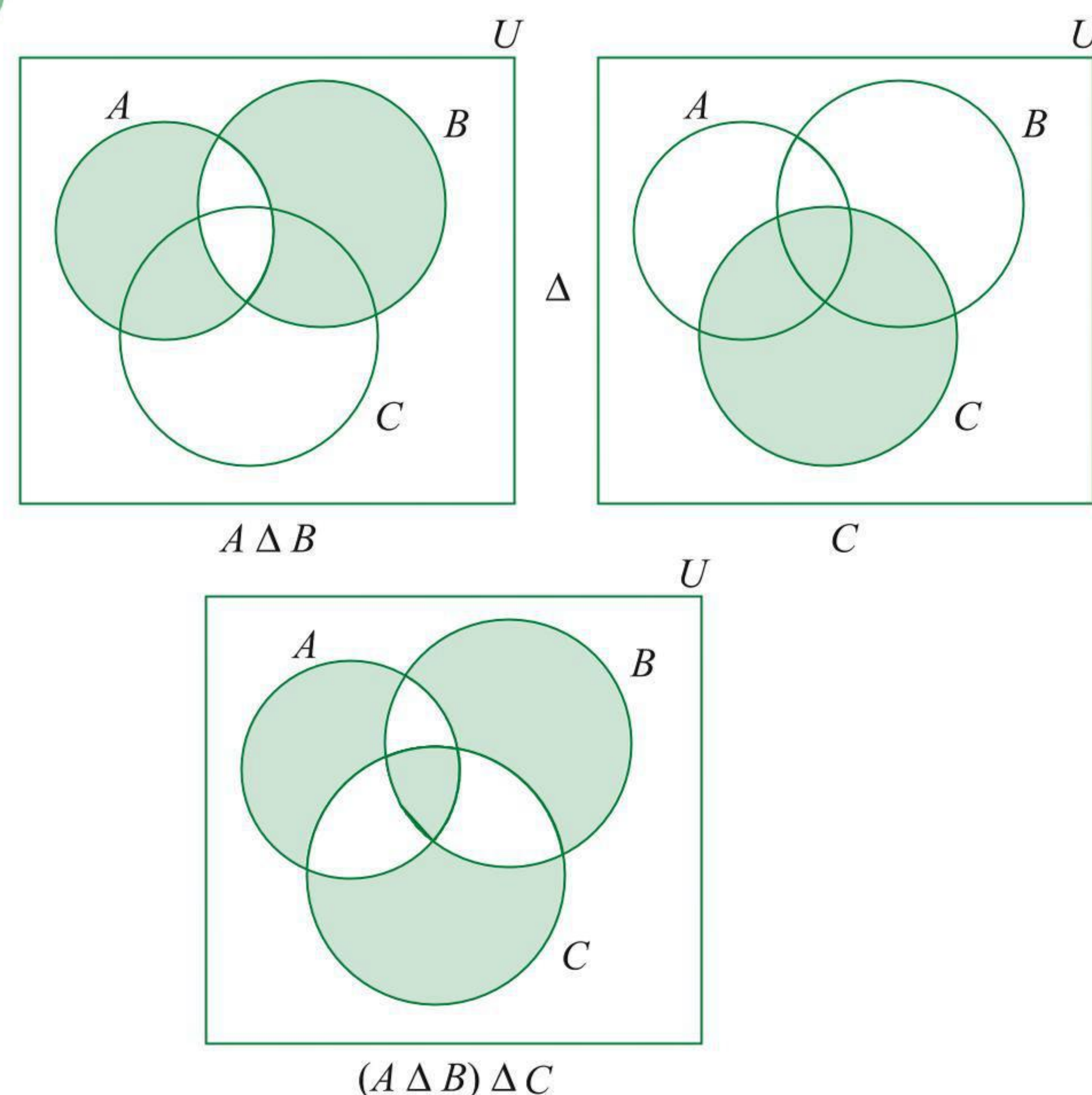


ILLUSTRATION 1.8

If $A \cap B = A \cup B$, then what can we conclude?

Sol. We have $A \cap B = A \cup B$

Consider the following cases:

(i) $A \subset B$

For this case, $A \cap B = A$ and $A \cup B = B$. So,

$$A \cap B \neq A \cup B$$

(ii) $B \subset A$

For this case, $A \cap B = B$ and $A \cup B = A$. So,

$$A \cap B \neq A \cup B$$

(iii) A and B are disjoint

For this case, $A \cap B = \phi$ and $A \cup B =$ set of all elements of set A and set B .

So, $A \cap B \neq A \cup B$

(iv) Sets A and B have some elements common

For this case, $A \cap B$ have some elements but $A \cup B$ is the set of elements of set A and set B .

So, $A \cap B \neq A \cup B$

From the above cases, we conclude that sets A and B are same. So, $A = B$.

ILLUSTRATION 1.9

Show that $A \cap B = A \cap C$ need not imply $B = C$.

Sol. Let $A = \{0, 1\}$, $B = \{0, 2, 3\}$, and $C = \{0, 4, 5\}$

Accordingly, $A \cap B = \{0\}$ and $A \cap C = \{0\}$

Here, $A \cap B = A \cap C = \{0\}$

However, $B \neq C$ ($2 \in B$ and $2 \notin C$)

ILLUSTRATION 1.10

Consider the following sets:

$A =$ set of all rectangles in the plane

$B =$ set of all squares in the same plane

$C =$ set of all parallelograms in the same plane

Find the following sets:

- (i) $A - B$ (ii) $C - A$ (iii) $A \cap C$ (iv) $B \cap C$
 (v) $B \cup C$ (vi) $A \cap B \cap C$

Sol. We have

$A =$ set of all rectangles in the same plane

$B =$ set of all squares in the same plane

$C =$ set of all parallelograms in the same plane

Since all squares are rectangles, $B \subseteq A$.

Also, all squares and rectangles are parallelograms.

So, $A, B \subseteq C$

$\therefore B \subseteq A \subseteq C$

(i) $A - B =$ set of all rectangles which are not squares

(ii) $C - A =$ set of all parallelograms which are not rectangles or squares
 $=$ set of all parallelograms which are not rectangles

(iii) $A \cap C =$ set of all rectangles $= A$

(iv) $B \cap C =$ set of all squares $= B$

(v) $B \cup C =$ set of all parallelograms $= C$

(vi) $A \cap B \cap C =$ set of all squares $= B$

ILLUSTRATION 1.11

Show that if $A \subset B$, then $C - B \subset C - A$.

Sol. Let $A \subset B$

We have to show that $C - B \subset C - A$

Let $x \in C - B$

$\Rightarrow x \in C$ and $x \notin B$

$\Rightarrow x \in C$ and $x \notin A$ [$\because A \subset B$]

$\Rightarrow x \in C - A$

$\therefore C - B \subset C - A$

ILLUSTRATION 1.12

Assume that $P(A) = P(B)$. Then show that $A = B$.

Sol. Suppose $A \neq B$.

Without loss of generality, there exists $x \in A$ such that $x \notin B$.

Then $\{x\} \in P(A)$ whereas, $\{x\} \notin P(B)$.

Thus, $P(A) \neq P(B)$, which contradicts the given condition.

So, $A = B$.

Conversely, if $P(A) = P(B)$ then all their singletons are the same.

Thus, $A = B$.

INTERVALS AS SUBSET OF SET R

We know that R is the set of all real numbers. Now, let us define different subsets of R .

Let $a, b \in R$ and $a < b$.

Interval	Type of Interval	Representation on real number line
$\{x \mid a < x < b\}$ or $x \in (a, b)$	Open interval	
$\{x \mid a \leq x \leq b\}$ or $x \in [a, b]$	Closed interval	
$\{x \mid a \leq x < b\}$ or $x \in [a, b)$	Closed-open interval	
$\{x \mid a < x \leq b\}$ or $x \in (a, b]$	Open-Closed interval	

The number $(b - a)$ is called the length of any of the intervals (a, b) , $[a, b]$, $[a, b)$ or $(a, b]$.

Clearly, $(a, b) \subset [a, b]$ but $[a, b] \not\subset (a, b)$.

The set $\{x \mid x \in R, -6 < x \leq 9\}$, written in set-builder form, can be written in the form of interval as $(-6, 9]$ and the interval $[-2, 5)$ can be written in set builder form as $\{x \mid -2 \leq x < 5\}$.

Note:

- The set $(-\infty, \infty)$ defines the set of all real numbers extending from $-\infty$ to ∞ .
- The set $(0, \infty)$ defines the set of all positive real numbers denoted by R^+ .
- The set $(-\infty, 0)$ defines the set of all negative real numbers denoted by R^- .
- The set $[0, \infty)$ defines the set of all non-negative real numbers $R^+ \cup \{0\}$.
- The set $(-\infty, 0]$ defines the set of all non-positive real numbers $R^- \cup \{0\}$.
- Complement of interval (a, b) is $(-\infty, a] \cup [b, \infty)$.
- Complement of interval $[a, b]$ is $(-\infty, a) \cup (b, \infty)$.

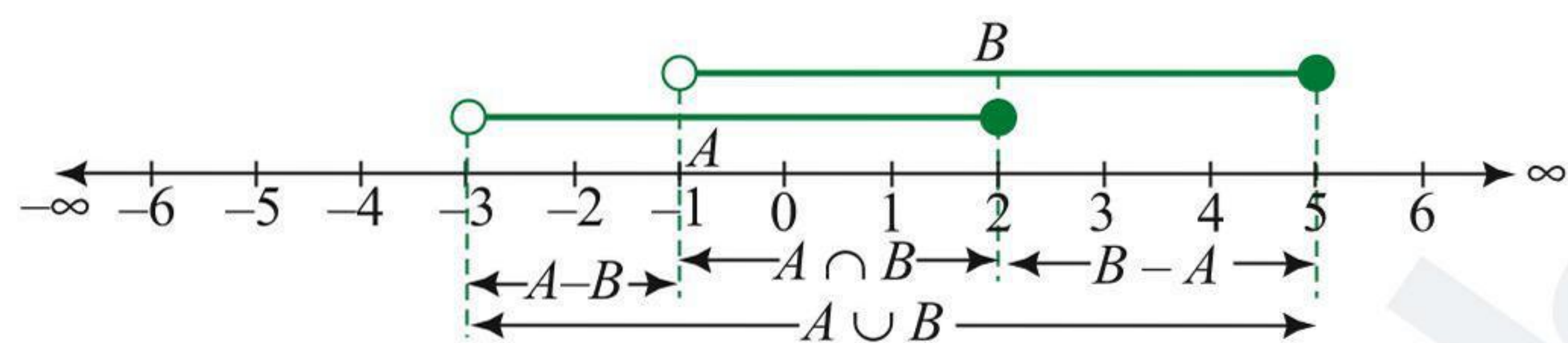
ILLUSTRATION 1.13

If sets $A = (-3, 2]$ and $B = (-1, 5]$, then find the following sets:

- (i) $A \cap B$ (ii) $A \cup B$ (iii) $A - B$ (iv) $B - A$

Sol. We have sets $A = (-3, 2]$ and $B = (-1, 5]$

These sets are plotted on real number line as shown in the following figure.



From the figure

- (i) $A \cap B = (-1, 2]$ (as $-1 \in A$ but $-1 \notin B$)
(ii) $A \cup B = (-3, 5]$
(iii) $A - B = (-3, -1]$ (as $-1 \notin B$, so $-1 \in A - B$)
(iv) $B - A = (2, 5]$ (as $2 \in A$, so $2 \notin B - A$)

CARDINAL NUMBER OF SET

The number of distinct elements or members in a finite set is called the cardinal number of a set. Through cardinality, we define the size of a set. The cardinal number of a set A is denoted as $n(A)$, where $n(A)$ represents the number of elements in set A .

Equivalent sets

Two or more sets having the same number of elements are called equivalent sets.

Thus, if sets A and B are equivalent sets then $n(A) = n(B)$.

Also, we understand that equal sets have same number of elements so are equivalent.

But equivalent sets may not be equal sets as they may have different elements.

Cardinal Number of Union of Two Sets

Consider sets $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6, 8, 10\}$

Here, sets A and B are disjoint.

$$n(A) = 4 \text{ and } n(B) = 5$$

Also, $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 10\}$

$$\therefore n(A \cup B) = 9 = n(A) + n(B)$$

Thus, when sets A and B are disjoint, $n(A \cup B) = n(A) + n(B)$.

Also, if sets A , B and C are pairwise disjoint, then

$$n(A \cup B \cup C) = n(A) + n(B) + n(C)$$

Now, consider sets $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{4, 5, 6, 7, 8\}$.

Here, $A \cap B = \{4, 5, 6\}$.

Now, $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$$\therefore n(A \cup B) = 8$$

$$\text{Also, } n(A) + n(B) = 6 + 5 = 11$$

$$\text{Thus, } n(A \cup B) \neq n(A) + n(B)$$

This is because common elements 4, 5 and 6 are counted twice in $n(A) + n(B)$, with $n(A)$ and with $n(B)$.

So, actually we should have

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Above relation is for any types of sets. When sets A and B are disjoint, $n(A \cap B) = 0$.

$$\therefore n(A \cup B) = n(A) + n(B)$$

Cardinal Number of Union of more than Two Sets

For three sets A , B and C , which are not disjoint pairwise, let us calculate $n(A \cup B \cup C)$.

Clearly, first we need to take the sum $n(A) + n(B) + n(C)$.

Here, in sum $n(A) + n(B)$, $n(A \cap B)$ is taken twice, once with $n(A)$ and once with $n(B)$, so $n(A \cap B)$ needs to be subtracted once. Similarly, $n(B \cap C)$ and $n(C \cap A)$ are also taken twice and hence, each needs to be subtracted once.

Also, $n(A \cap B \cap C)$ is taken once in each of $n(A)$, $n(B)$, $n(C)$, $n(A \cap B)$, $n(B \cap C)$, $n(C \cap A)$. So, in the sum $n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A)$, term $n(A \cap B \cap C)$ vanishes. Hence, $n(A \cap B \cap C)$ needs to be added once.

So, finally we have

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$

The concept used in above formula is called Principle of Inclusion and Exclusion. Here, the sum $n(A) + n(B) + n(C)$ gives greater value, so we exclude $n(A \cap B) + n(B \cap C) + n(C \cap A)$. But, doing this, we lose $n(A \cap B \cap C)$, so we include it.

With the help of above logic,

$$\begin{aligned} n(A \cup B \cup C \cup D) &= [\text{Sum of number of elements in sets } A, B, C \text{ and } D] \\ &\quad - [\text{Sum of number of elements of intersection of sets taken two at a time}] \\ &\quad + [\text{Sum of number of elements of intersection of sets taken three at a time}] \\ &\quad - [\text{Elements of intersection of all sets}] \\ &= n(A) + n(B) + n(C) + n(D) - n(A \cap B) \\ &\quad - n(A \cap C) - n(A \cap D) - n(B \cap C) \\ &\quad - n(B \cap D) - n(C \cap D) + n(A \cap B \cap C) \\ &\quad + n(B \cap C \cap D) + n(C \cap D \cap A) \\ &\quad + n(D \cap A \cap B) - n(A \cap B \cap C \cap D) \end{aligned}$$

Thus, $n(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n)$

$$\begin{aligned} &= \sum_{i=1}^n n(A_i) - \sum_{1 \leq i < j \leq n} n(A_i \cap A_j) \\ &\quad + \sum_{1 \leq i < j < k \leq n} n(A_i \cap A_j \cap A_k) + \dots \\ &\quad + (-1)^{n+1} \sum n(A_1 \cap A_2 \cap \dots \cap A_n) \end{aligned}$$

From the above discussion, it is easy to establish the following:

- (i) $n(A - B) = n(A) - n(A \cap B) = n(A \cap B')$
- (ii) $n(A') = n(U) - n(A)$
- (iii) $n(A' \cup B') = n(U) - n(A \cap B)$
- (iv) $n(A' \cap B') = n(U) - n(A \cup B)$

ILLUSTRATION 1.14

In a group of 500 people, 350 speak Hindi and 300 speak English. It is given that each person speaks at least one language.

- (i) How many people can speak both Hindi and English?
- (ii) How many people can speak Hindi only?
- (iii) How many people can speak English only?

Sol. Let H be the set of people who speak Hindi, and E be the set of people who speak English.

$$\therefore n(H \cup E) = 500, n(H) = 350, n(E) = 300$$

We have to find $n(H \cap E)$.

$$\text{Now, } n(H \cup E) = n(H) + n(E) - n(H \cap E)$$

$$\therefore 500 = 350 + 300 - n(H \cap E)$$

$$\Rightarrow 500 = 650 - n(H \cap E)$$

$$\Rightarrow n(H \cap E) = 650 - 500$$

$$\therefore n(H \cap E) = 150$$

Thus, 150 people can speak both Hindi and English.

$$\begin{aligned} \text{Number of people who speak Hindi only} &= n(H) - n(H \cap E) \\ &= 350 - 150 = 200 \end{aligned}$$

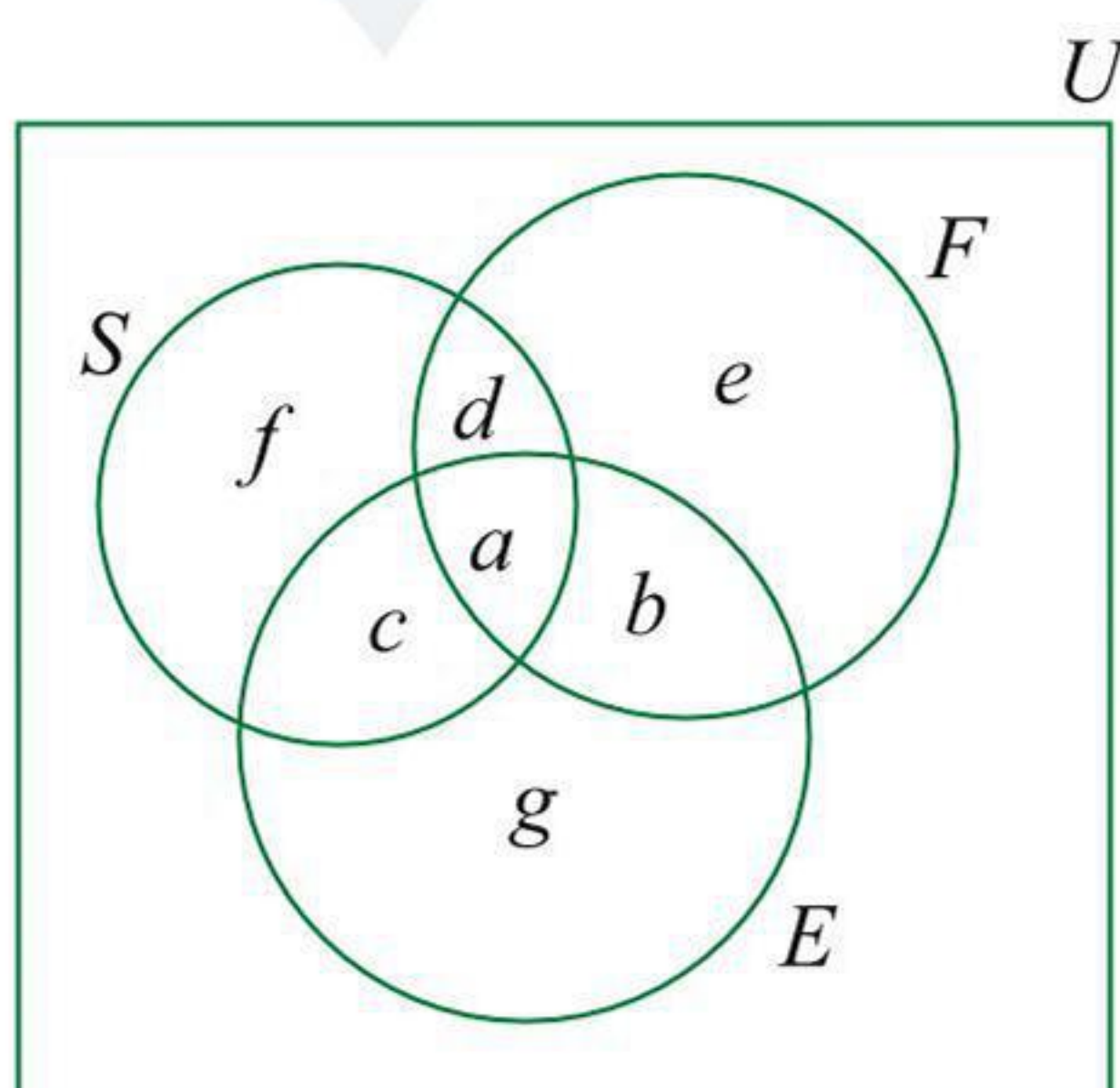
$$\begin{aligned} \text{Number of people who speak English only} &= n(E) - n(H \cap E) \\ &= 300 - 150 = 150 \end{aligned}$$

ILLUSTRATION 1.15

In a group of 50 students, the number of students learning French, English, Sanskrit were found to be as follows: French = 17, English = 13, Sanskrit = 15, French and English = 9, English and Sanskrit = 4, French and Sanskrit = 5, English, French and Sanskrit = 3. Find the number of students who are learning

- (i) French only
- (ii) English only
- (iii) Sanskrit only
- (iv) English and Sanskrit but not French
- (v) French and Sanskrit but not English
- (vi) French and English but not Sanskrit
- (vii) at least one of the three languages
- (viii) none of the three languages
- (ix) exactly one language
- (x) exactly two languages

Sol.



Let F , E and S be the sets of students who learn French, English and Sanskrit, respectively.

$$\text{Then, } n(U) = 50, n(F) = 17, n(E) = 13, n(S) = 15,$$

$$n(F \cap E) = 9, n(E \cap S) = 4, n(F \cap S) = 5,$$

$$\text{and } n(F \cap E \cap S) = 3.$$

From the figure, we have

$$a = n(E \cap F \cap S) = 3 \text{ and } a + d = n(F \cap S) = 5 \Rightarrow d = 2$$

$$a + b = n(F \cap E) = 9 \Rightarrow b = 6$$

$$a + c = n(E \cap S) = 4 \Rightarrow c = 1$$

$$a + b + d + e = n(F) = 17 \Rightarrow 3 + 6 + 2 + e = 17 \Rightarrow e = 6$$

$$a + b + c + g = n(E) = 13 \Rightarrow 3 + 6 + 1 + g = 13 \Rightarrow g = 3$$

$$a + c + d + f = n(S) = 15 \Rightarrow 3 + 1 + 2 + f = 15 \Rightarrow f = 9$$

- (i) Number of students learning French only = $e = 6$
- (ii) Number of students learning English only = $g = 3$
- (iii) Number of students learning Sanskrit only = $f = 9$
- (iv) Number of students learning English and Sanskrit but not French = $c = 1$
- (v) Number of students learning French and Sanskrit but not English = $d = 2$
- (vi) Number of students learning French and English but not Sanskrit = $b = 6$
- (vii) Number of students learning at least one of the three languages = $a + b + c + d + e + f + g = 30$
- (viii) Number of students learning none of the three languages = $50 - 30 = 20$
- (ix) Number of students learning exactly one language = $e + f + g = 6 + 3 + 9 = 18$
- (x) Number of students learning exactly two languages = $b + c + d = 6 + 2 + 1 = 9$

CONCEPT APPLICATION EXERCISE 1.1

- Let $A = \{1, 2, \{3, 4\}, 5\}$. Which of the following statements are incorrect and why?
 - (a) $\{3, 4\} \subset A$
 - (b) $\{3, 4\} \in A$
 - (c) $\{\{3, 4\}\} \subset A$
 - (d) $1 \in A$
 - (e) $1 \subset A$
 - (f) $\{1, 2, 5\} \subset A$
 - (g) $\{1, 2, 5\} \in A$
 - (h) $\{1, 2, 3\} \subset A$
 - (i) $\phi \in A$
 - (j) $\phi \subset A$
 - (k) $\{\phi\} \subset A$
- If $X = \{1, 2, 3, \dots, 10\}$, and a represents any element of X , then write the following sets, containing all the elements satisfying the given conditions.
 - (a) $a \in X$ but $a^2 \notin X$
 - (b) $a \in X$ but $a/2 \notin X$
 - (c) a is factor of 24
- Write down all the subsets of the following sets:
 - (a) $\{p\}$
 - (b) $\{p, q\}$
 - (c) $\{1, 2, 3, 4\}$
 - (d) ϕ
- If the number of elements in the power set of set A is 128, then find the number of elements in the set A .
- If $A \subset B$ and $B \in C$, then $A \in C$. Is this statement true?
- Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$. Then find

- (a) A' (b) B' (c) $(A \cup C)'$
 (d) $(A \cup B)'$ (e) $(A')'$ (f) $(B - C)'$

7. If $A - B = \phi$ and $B - A = \phi$, then what can we conclude?
 8. It is true that for any sets A and B , $P(A) \cup P(B) = P(A \cup B)$? Justify your answer.
 9. Let A , B and X be three sets such that $A \cap X = B \cap X = \phi$ and $A \cup X = B \cup X$. Then show that $A = B$.
 10. If $A = [-4, 1)$ and $B = [0, 3)$, then find the following:
 (a) $A \cap B$ (b) $A \cup B$ (c) $A - B$
 (d) $B - A$ (e) $(A \cup B)'$ (f) $(A \cap B)'$
 11. In a survey conducted on 800 students of a school, 250 students were found to like tea and 300 like coffee, 150 like both tea and coffee. Find how many students like neither tea nor coffee?
 12. Out of 100 students; 15 passed in English, 12 in Mathematics, 8 in Science, 6 in English and Mathematics, 7 in Mathematics and Science, 4 in English and Science and 4 in all the three. Find how many students passed
 (a) in English and Mathematics but not in Science
 (b) in Mathematics and Science but not in English
 (c) in Mathematics only
 (d) in more than one subjects

ANSWERS

1. (a), (e), (g), (h), (i), (k) are incorrect.
 2. (a) $\{4, 5, 6, 7, 8, 9, 10\}$ (b) $\{1, 3, 5, 7, 9\}$
 (c) $\{1, 2, 3, 4, 6, 8\}$ 4. 7 5. Not true
 7. $A = B$ 8. Not true 11. 400
 12. (a) 2 (b) 3 (c) 3 (d) 9

INEQUALITIES

We have the following elementary properties of inequalities:

- If $a \leq b$, then either $a < b$ or $a = b$.
- Let $a < b$. Now if both sides are increased or decreased by same quantity say by k then L.H.S. is still less than the R.H.S..

So, $a \pm k < b \pm k$.

- If $a < b$ and $c < d$, then $a + c < b + d$ or $a - d < b - c$
- If $a < b$ then $ka < kb$, if k is positive.
- If $a < b$ then $ka > kb$, if k is negative.

Thus, if inequality is multiplied by negative quantity on both sides, sign of inequality changes.

- $a < b$ and $b < c$, then $a < b < c$. This is called transitive property of inequality.

If $a < b$ and $b \leq c$, then $a < b \leq c$.

If $a \leq b$ and $b \leq c$, then $a \leq b \leq c$.

If $a < b$ and $b > c$ then we cannot relate a and c .

Also, if $a < b < c$, then $a < c$.

- If $0 < a < b$, then (i) $a^r < b^r$ when $r > 0$ and (ii) $a^r > b^r$ when $r < 0$.
- $a + \frac{1}{a} \geq 2$ for $a > 0$ and equality holds for $a = 1$.
- $a + \frac{1}{a} \leq -2$ for $a < 0$ and equality holds for $a = -1$.

ILLUSTRATION 1.16

Solve the following inequalities. Write the solution in the form of intervals.

- (i) $3x \geq 18$ (ii) $2x + 17 < 3$
 (iii) $7 - 4x > -17$ (iv) $\frac{x}{4} + 3 \leq -2$

Sol.

- (i) $3x \geq 18$
 $\Rightarrow x \geq 6$ (dividing both sides by 3)
 $\Rightarrow x \in [6, \infty)$
 (ii) $2x + 17 < 3$
 $\Rightarrow 2x < 3 - 17$ (subtracting 17 from both sides)
 $\Rightarrow 2x < -14$
 $\Rightarrow x < -7$ (dividing both sides by 2)
 $\Rightarrow x \in (-\infty, -7)$
 (iii) $7 - 4x > -17$
 $\Rightarrow -4x > -17 - 7$ (subtracting 7 from both sides)
 $\Rightarrow 4x < 24$
 $\Rightarrow x < 6$ (dividing both sides by -4)
 $\Rightarrow x \in (-\infty, 6)$
 (iv) $\frac{x}{4} + 3 \leq -2$
 $\Rightarrow \frac{x}{4} \leq -5$ (subtracting 3 from both sides)
 $\Rightarrow x \leq -20$ (multiplying both sides by 4)
 $\Rightarrow x \in (-\infty, -20]$

ILLUSTRATION 1.17

Solve the following inequalities:

- (i) $7x + 15 \geq 9 - 4x$ (ii) $-5 \leq \frac{2 - 3x}{4} \leq 9$
 (iii) $5x - 6 \leq 4$ and $7 - 3x \geq 2x$

Sol.

- (i) $7x + 15 \geq 9 - 4x$
 Transferring $-4x$ from RHS to LHS and 15 from LHS to RHS, we get
 $7x + 4x \geq 9 - 15$
 $\Rightarrow 11x \geq -6$
 $\Rightarrow x \geq \frac{-6}{11}$ (dividing both sides by 11)
 $\Rightarrow x \in \left[\frac{-6}{11}, \infty\right)$
 (ii) We have, $-5 \leq \frac{2 - 3x}{4} \leq 9$
 $\Rightarrow -20 \leq 2 - 3x \leq 36$
 $\Rightarrow -22 \leq -3x \leq 34$
 $\Rightarrow \frac{22}{3} \geq x \geq -\frac{34}{3}$
 $\Rightarrow x \in \left[-\frac{34}{3}, \frac{22}{3}\right]$

(iii) We have $5x - 6 \leq 4$ and $7 - 3x \geq 2x$

$$5x - 6 \leq 4$$

$$\Rightarrow 5x \leq 10 \Rightarrow x \leq 2 \quad \dots(1)$$

$$\text{Also, } 7 - 3x \geq 2x$$

$$\Rightarrow 7 \geq 2x + 3x$$

$$\Rightarrow 5x \leq 7$$

$$\Rightarrow x \leq \frac{7}{5} \quad \dots(2)$$

From (1) and (2), common values of x are $x \leq 7/5$ or $x \in (-\infty, 7/5]$

ILLUSTRATION 1.18

Abhinav obtained 65 and 80 marks in first two unit tests. Find the minimum marks he should get in the third test to have an average of at least 70 marks.

Sol. Abhinav obtained 65 and 80 marks in first two unit tests.

Let x be the marks obtained by Abhinav in the third unit test.

Since he should have an average of at least 70 marks,

$$\frac{65 + 80 + x}{3} \geq 70$$

$$\Rightarrow 145 + x \geq 210$$

$$\Rightarrow x \geq 210 - 145$$

$$\Rightarrow x \geq 65$$

Thus, Abhinav must obtain minimum of 65 marks to have an average of at least 70 marks.

ILLUSTRATION 1.19

Solve $\sqrt{x-5} - \sqrt{9-x} > 1, x \in \mathbb{Z}$

Sol. $\sqrt{x-5} - \sqrt{9-x} > 1$ is meaningful, if

$$x - 5 \geq 0 \text{ and } 9 - x \geq 0$$

$$\Rightarrow x \in [5, 9]$$

Thus, the integral values of x are 5, 6, 7, 8, 9. Of these only $x = 9$ satisfies the given inequality.

ILLUSTRATION 1.20

Solve $\sqrt{x-2} \geq -1$.

Sol. We must have $x - 2 \geq 0$ for $\sqrt{x-2}$ to get defined, thus $x \geq 2$.

Now $\sqrt{x-2} \geq -1$, as square roots are always non-negative.

Hence, $x \geq 2$.

Note:

Some students solve it by squaring on both sides for which $x - 2 \geq 1$ or $x \geq 3$, which causes loss of interval $[2, 3)$.

ILLUSTRATION 1.21

Solve $\sqrt{x-1} > \sqrt{3-x}$.

Sol. $\sqrt{x-1} > \sqrt{3-x}$ is meaningful if $x - 1 \geq 0$ and $3 - x \geq 0$

$$\text{or } 1 \leq x \leq 3 \quad (1)$$

$$\text{Also } \sqrt{x-1} > \sqrt{3-x}$$

Squaring, we have $x - 1 > 3 - x$

$$\Rightarrow x > 2 \quad (2)$$

From (1) and (2), we have $2 < x \leq 3$.

ILLUSTRATION 1.22

Solve $x + \sqrt{x} \geq \sqrt{x} - 3$.

Sol. $x + \sqrt{x} \geq \sqrt{x} - 3$ is meaningful only when $x \geq 0$ (1)

$$\text{Now } x + \sqrt{x} \geq \sqrt{x} - 3$$

$$\Rightarrow x \geq -3 \quad (2)$$

From (1) and (2), we have $x \geq 0$.

VALUES OF x^2 FROM GIVEN VALUES OF x

If $a < b$ then $a^2 < b^2$ does not follow always:

Consider the following illustrations:

$$2 < 3 \Rightarrow 4 < 9, \text{ but } -4 < 3 \Rightarrow 16 > 9$$

In fact, $a < b \Rightarrow a^2 < b^2$ follows only when distance of a from 0 on real number line is less than the distance of b from 0.

Also, if $x > 2 \Rightarrow x^2 > 4$, but for $x < 2 \Rightarrow x^2 \geq 0$

$$\text{If } 2 < x < 4 \Rightarrow 4 < x^2 < 16$$

$$\text{If } -2 < x < 4 \Rightarrow 0 \leq x^2 < 16$$

$$\text{If } -5 < x < 4 \Rightarrow 0 \leq x^2 < 25$$

Note:

- If $x^2 \leq a^2$ then $x \in [-a, a]$
- If $x^2 \geq a^2$ then $x \in (-\infty, -a] \cup [a, \infty)$

ILLUSTRATION 1.23

Find the values of x^2 for the given values of x .

- (i) $x < 2$ (ii) $x > -1$ (iii) $x \geq 2$ (iv) $x < -1$

Sol.

(i) When $x < 2$, we have $x \in (-\infty, 0) \cup [0, 2)$

for $x \in [0, 2)$, $x^2 \in [0, 4)$

for $x \in (-\infty, 0)$, $x^2 \in (0, \infty)$

$$\Rightarrow \text{for } x < 2, x^2 \in [0, 4) \cup (0, \infty)$$

$$\Rightarrow x \in [0, \infty)$$

(ii) When $x > -1$, we have $x \in (-1, 0) \cup [0, \infty)$

for $x \in (-1, 0)$, $x^2 \in (0, 1)$

for $x \in [0, \infty)$, $x^2 \in [0, \infty)$

$$\Rightarrow \text{for } x > -1, x^2 \in (0, 1) \cup [0, \infty)$$

$$\Rightarrow x \in [0, \infty)$$

(iii) Here $x \in [2, \infty)$

$$\Rightarrow x^2 \in [4, \infty)$$

(iv) Here $x \in (-\infty, -1)$

$$\Rightarrow x^2 \in (1, \infty)$$

ILLUSTRATION 1.24

Find all possible values of the following expressions:

- (i) $\sqrt{x^2 - 4}$ (ii) $\sqrt{9 - x^2}$ (iii) $\sqrt{x^2 - 2x + 10}$

Sol.

$$(i) \sqrt{x^2 - 4}$$

Least value of square root is 0 when $x^2 = 4$ or $x = \pm 2$. Also

$$x^2 - 4 \geq 0$$

Hence, $\sqrt{x^2 - 4} \in [0, \infty)$.

(ii) $\sqrt{9 - x^2}$

Least value of square root is 0, when $9 - x^2 = 0$ or $x = \pm 3$.

Also, the greatest value of $9 - x^2$ is 9, when $x = 0$.

Hence, we have $0 \leq 9 - x^2 \leq 9 \Rightarrow \sqrt{9 - x^2} \in [0, 3]$.

(iii) $\sqrt{x^2 - 2x + 10} = \sqrt{(x-1)^2 + 9}$

Here, the least value of $\sqrt{(x-1)^2 + 9}$ is 3, when $x - 1 = 0$.

Also $(x-1)^2 + 9 \geq 9 \Rightarrow \sqrt{(x-1)^2 + 9} \geq 3$

Hence, $\sqrt{x^2 - 2x + 10} \in [3, \infty)$.

ILLUSTRATION 1.25

Solve $(x^2 - 4)\sqrt{x^2 - 1} < 0$.

Sol. $(x^2 - 4)\sqrt{x^2 - 1} < 0$

We must have $x^2 - 1 \geq 0$

or $x^2 \geq 1$

or $x \leq -1$ or $x \geq 1$ (1)

Also $(x^2 - 4)\sqrt{x^2 - 1} < 0$

or $x^2 - 4 < 0$

or $x^2 < 4$

or $-2 < x < 2$ (2)

From (1) and (2), we have $x \in (-2, -1] \cup [1, 2)$

VALUES OF $1/x$ FROM GIVEN VALUES OF x

Consider the following illustrations:

(i) $2 < 3 \Rightarrow \frac{1}{2} > \frac{1}{3}$

(ii) $-6 < -4 \Rightarrow -\frac{1}{6} > -\frac{1}{4}$

(iii) $-3 < 5 \Rightarrow -\frac{1}{3} < \frac{1}{5}$

Thus, if values on both sides of inequality have same sign, then on taking its reciprocal, the sign of inequality gets reversed.

i.e., $a > b > 0 \Rightarrow \frac{1}{a} < \frac{1}{b}$ and $a < b < 0 \Rightarrow \frac{1}{a} > \frac{1}{b}$

But if values on both sides of inequality have opposite signs, then on taking reciprocal, sign of inequality remains the same.

i.e., $a < 0 < b \Rightarrow \frac{1}{a} < \frac{1}{b}$

Now, consider following illustrations:

(i) $2 < x < 3 \Rightarrow \frac{1}{2} > \frac{1}{x} > \frac{1}{3}$,

(ii) $-6 < x < -2 \Rightarrow -\frac{1}{6} > \frac{1}{x} > -\frac{1}{2}$

(iii) $-3 < x < 4$

Now, $\frac{1}{x}$ is not defined for $x = 0$.

So, we consider $-3 < x < 0$ or $0 < x < 4$

$$\Rightarrow -\frac{1}{3} > \frac{1}{x} > -\infty \text{ or } \infty > \frac{1}{x} > \frac{1}{4}$$

In $-3 < x < 0$, 0 is actually 0^- (very close to zero but less than zero).

So, $\frac{1}{0^-}$ is very huge negative value that is $-\infty$.

In $0 < x < 4$, 0 is actually 0^+ (very close to zero but more than zero).

So, $\frac{1}{0^+}$ is very huge positive value that is ∞ .

Thus, $x \in [a, b]$

$$\Rightarrow \begin{cases} \frac{1}{x} \in \left[\frac{1}{b}, \frac{1}{a} \right], & \text{if } a \text{ and } b \text{ have same sign} \\ \frac{1}{x} \in \left(-\infty, \frac{1}{a} \right] \cup \left[\frac{1}{b}, \infty \right), & \text{if } a \text{ and } b \text{ have opposite sign} \end{cases}$$

ILLUSTRATION 1.26

Find the values of $1/x$ for the given values of x .

(i) $x > 3$ (ii) $x < -2$ (iii) $x \in (-1, 3) - \{0\}$

Sol.

(i) We have $3 < x < \infty$

$$\Rightarrow \frac{1}{3} > \frac{1}{x} > \frac{1}{\rightarrow \infty} \quad (\rightarrow \infty \text{ means tends to infinity})$$

$$\Rightarrow 0 < \frac{1}{x} < \frac{1}{3}$$

(ii) We have $-\infty < x < -2$

$$\Rightarrow \frac{1}{\rightarrow -\infty} > \frac{1}{x} > \frac{1}{-2}$$

$$\Rightarrow 0 > \frac{1}{x} > -\frac{1}{2} \Rightarrow -\frac{1}{2} < \frac{1}{x} < 0$$

(iii) $x \in (-1, 3) - \{0\}$

$$\Rightarrow x \in (-1, 0) \cup (0, 3)$$

For $x \in (-1, 0)$, we have

$$\frac{1}{-1} > \frac{1}{x} > \frac{1}{\rightarrow 0^-}$$

(here $\rightarrow 0^-$ means the value of x approaches to 0 from its left hand side or negative side)

$$\Rightarrow -1 > \frac{1}{x} > -\infty$$

$$\Rightarrow -\infty < \frac{1}{x} < -1 \quad (1)$$

For $x \in (0, 3)$, we have

$$\frac{1}{\rightarrow 0^+} > \frac{1}{x} > \frac{1}{3}$$

(here $\rightarrow 0^+$ means the value of x approaches to 0 from its right hand side or positive side)

$$\Rightarrow \infty > \frac{1}{x} > \frac{1}{3}$$

$$\Rightarrow \frac{1}{3} < \frac{1}{x} < \infty \quad (2)$$

$$\text{From (1) and (2), } \frac{1}{x} \in (-\infty, -1) \cup \left(\frac{1}{3}, \infty\right)$$

Note:

$$\text{For } x \in R - \{0\}, \frac{1}{x} \in R - \{0\}$$

ILLUSTRATION 1.27

Find all possible values of the following expressions:

$$(i) \frac{1}{x^2 + 2} \quad (ii) \frac{1}{x^2 - 2x + 3} \quad (iii) \frac{1}{x^2 - x - 1}$$

Sol.

(i) We know that $x^2 \geq 0 \forall x \in R$.

$$\Rightarrow x^2 + 2 \geq 2, \forall x \in R.$$

$$\text{or } 2 \leq (x^2 + 2) < \infty$$

$$\text{or } \frac{1}{2} \geq \frac{1}{x^2 + 2} > 0$$

$$\Rightarrow 0 < \frac{1}{x^2 + 2} \leq \frac{1}{2}$$

$$(ii) \frac{1}{x^2 - 2x + 3} = \frac{1}{(x-1)^2 + 2}$$

Now we know that $(x-1)^2 \geq 0 \forall x \in R$. Thus,

$$(x-1)^2 + 2 \geq 2 \forall x \in R.$$

$$\text{or } 2 \leq (x-1)^2 + 2 < \infty$$

$$\text{or } \frac{1}{2} \geq \frac{1}{(x-1)^2 + 2} > 0$$

$$\text{or } \frac{1}{x^2 - 2x + 3} \in \left(0, \frac{1}{2}\right]$$

$$(iii) \frac{1}{x^2 - x - 1} = \frac{1}{\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}}$$

$$\left(x - \frac{1}{2}\right)^2 \geq 0, \forall x \in R$$

$$\Rightarrow \left(x - \frac{1}{2}\right)^2 - \frac{5}{4} \geq -\frac{5}{4}, \forall x \in R$$

$$\text{For } \frac{1}{\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}}, \text{ we must have}$$

$$\left(x - \frac{1}{2}\right)^2 - \frac{5}{4} \in \left[-\frac{5}{4}, 0\right) \cup (0, \infty)$$

$$\Rightarrow \frac{1}{\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}} \in \left(-\infty, -\frac{4}{5}\right] \cup (0, \infty)$$

CONCEPT APPLICATION EXERCISE 1.2

1. Find the values of x which satisfy the following inequalities simultaneously:

$$(a) -3 < 2x - 1 < 19 \quad (b) -1 \leq \frac{2x+3}{5} \leq 3$$

2. The longest side of a triangle is 3 times the shortest side and the third side is 2 cm shorter than the longest side. If the perimeter of the triangle is at least 61 cm, find the minimum length of the shortest side.

3. Find the value of x^2 for the following values of x :

$$(a) [-5, -1] \quad (b) (3, 6) \quad (c) (-2, 3] \\ (d) (-3, \infty) \quad (e) (-\infty, 4)$$

4. Find the values of $1/x$ for the following values of x :

$$(a) (2, 5) \quad (b) [-5, -1] \quad (c) (3, \infty) \\ (d) (-\infty, -2] \quad (e) [-3, 4]$$

5. Find all possible values (range) of the following quadratic expressions when $x \in R$ and when $x \in [-3, 2]$.

$$(a) 4x^2 + 28x + 41 \quad (b) 1 + 6x - x^2$$

6. Find all the possible values of expression $\frac{2+x^2}{4-x^2}$.

$$7. \text{ Solve } \frac{\sqrt{x-1}}{x-2} < 0.$$

$$8. \text{ Solve } \sqrt{x-2} \leq 3.$$

ANSWERS

$$1. (-1, 6] \quad 2. 9 \text{ cm}$$

$$3. (a) [1, 5] \quad (b) (9, 36) \quad (c) [0, 9] \quad (d) [0, \infty) \quad (e) [0, \infty)$$

$$4. (a) \left(\frac{1}{5}, \frac{1}{2}\right) \quad (b) \left(-1, -\frac{1}{5}\right] \quad (c) \left(0, \frac{1}{3}\right) \quad (d) \left[-\frac{1}{2}, 0\right)$$

$$(e) \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{1}{4}, \infty\right)$$

$$5. (a) [-8, \infty) \text{ when } x \in R; [-7, 113] \text{ when } x \in [-3, 2]$$

$$(b) (-\infty, 10] \text{ when } x \in R; [-26, 9] \text{ when } x \in [-3, 2]$$

$$6. (-\infty, -1) \cup \left[\frac{1}{2}, \infty\right) \quad 7. 1 \leq x < 2 \quad 8. 2 \leq x \leq 11$$

SIGN SCHEME METHOD

We know how to solve linear inequalities. Now, we shall discuss the algorithm to solve polynomial and rational inequalities.

$$\text{Let } F(x) = (x-a_1)^{k_1} (x-a_2)^{k_2} \dots (x-a_{n-1})^{k_{n-1}} (x-a_n)^{k_n}$$

where $k_1, k_2, \dots, k_n \in Z$ and $a_1, a_2, \dots, a_n$ are fixed real numbers satisfying the condition $a_1 < a_2 < a_3 < \dots < a_{n-1} < a_n$.

For solving $F(x) > 0$ or $F(x) < 0$, consider the following algorithm:

Step I: Mark the numbers $a_1, a_2, \dots, a_n$ on the number axis and put the plus sign in the interval on the right of the largest of these numbers, i.e., on the right of a_n . Clearly, when $x > a_n$, all the factors $(x-a_i), i = 1, 2, 3, \dots, n$ are positive. So, $F(x)$ is positive.

Step II: Put the plus sign in the interval on the left of a_n if k_n is an even number and the minus sign if k_n is an odd number.

In the next interval, we put a sign according to the following rule:

When passing through the point a_{n-1} , $F(x)$ changes sign if k_{n-1} is an odd number. Then we consider the next interval and put a sign in it using the same rule.

Step III: Consider all the intervals. The solution of the inequality $F(x) > 0$ is the union of all the intervals in which we have put the plus sign and the solution of the inequality $F(x) < 0$ is the union of all the intervals in which we have put the minus sign.

Frequently used inequalities:

- (i) $(x - a)(x - b) < 0 \Rightarrow x \in (a, b)$, where $a < b$
- (ii) $(x - a)(x - b) > 0 \Rightarrow x \in (-\infty, a) \cup (b, \infty)$, where $a < b$
- (iii) $x^2 \leq a^2 \Rightarrow x \in [-a, a]$
- (iv) $x^2 \geq a^2 \Rightarrow x \in (-\infty, -a] \cup [a, \infty)$
- (v) If $ax^2 + bx + c < 0$, ($a > 0$) $\Rightarrow x \in (\alpha, \beta)$, where α and β ($\alpha < \beta$) are roots of the equation $ax^2 + bx + c = 0$.
- (vi) If $ax^2 + bx + c > 0$, ($a > 0$) $\Rightarrow x \in (-\infty, \alpha) \cup (\beta, \infty)$, where α and β ($\alpha < \beta$) are roots of the equation $ax^2 + bx + c = 0$

ILLUSTRATION 1.28

Solve $x^2 - x - 2 > 0$.

Sol. $x^2 - x - 2 > 0$

or $(x - 2)(x + 1) > 0$

Now $x^2 - x - 2 = 0 \Rightarrow x = -1, 2$.

On the number line (x -axis) mark $x = -1$ and $x = 2$.

Now when $x > 2$, $x + 1 > 0$ and $x - 2 > 0$

$\Rightarrow (x + 1)(x - 2) > 0$

when $-1 < x < 2$, $x + 1 > 0$ but $x - 2 < 0$

$\Rightarrow (x + 1)(x - 2) < 0$

when $x < -1$, $x + 1 < 0$ and $x - 2 < 0$

$\Rightarrow (x + 1)(x - 2) > 0$

Hence, sign scheme of $x^2 - x - 2$ is



From the figure, $x^2 - x - 2 > 0$, $x \in (-\infty, -1) \cup (2, \infty)$.

ILLUSTRATION 1.29

Solve $x^2 - x - 1 < 0$.

Sol. Let us first factorize $x^2 - x - 1$.

For that let $x^2 - x - 1 = 0$

$$\Rightarrow x = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Now on the number line (x -axis), mark $x = \frac{1 \pm \sqrt{5}}{2}$



From the sign scheme of $x^2 - x - 1$, which is shown in the figure, we have

$$x^2 - x - 1 < 0$$

$$\Rightarrow x \in \left(\frac{1 - \sqrt{5}}{2}, \frac{1 + \sqrt{5}}{2} \right)$$

ILLUSTRATION 1.30

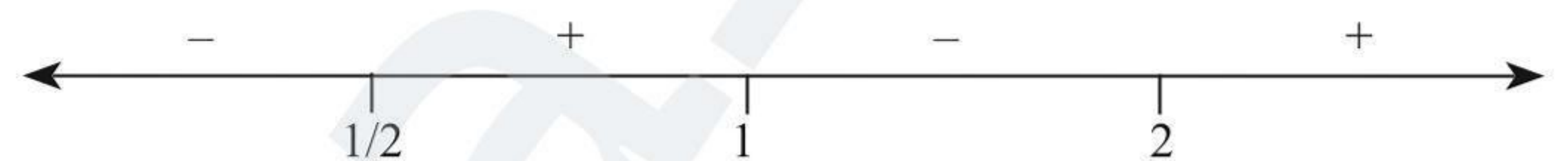
Solve $(x - 1)(x - 2)(1 - 2x) > 0$.

Sol. We have $(x - 1)(x - 2)(1 - 2x) > 0$

or $-(x - 1)(x - 2)(2x - 1) > 0$

or $(x - 1)(x - 2)(2x - 1) < 0$

On the number line, mark $x = 1/2, 1, 2$



When $x > 2$, all factors $(x - 1)$, $(2x - 1)$, and $(x - 2)$ are positive.

Hence, $(x - 1)(x - 2)(2x - 1) > 0$ for $x > 2$.

Now put positive and negative signs alternatively as shown in the figure.

Hence, solution set of $(x - 1)(x - 2)(1 - 2x) > 0$ or $(x - 1)(x - 2)(2x - 1) < 0$ is $(-\infty, 1/2) \cup (1, 2)$.

ILLUSTRATION 1.31

Solve $\frac{2}{x} < 3$.

Sol. $\frac{2}{x} < 3$

or $\frac{2}{x} - 3 < 0$ (We cannot crossmultiply with x as x can be negative or positive)

or $\frac{2 - 3x}{x} < 0$

or $\frac{3x - 2}{x} > 0$

or $\frac{(x - 2/3)}{x} > 0$

Sign scheme of $\frac{(x - 2/3)}{x}$ is as follows:



$\Rightarrow x \in (-\infty, 0) \cup (2/3, \infty)$

ILLUSTRATION 1.32

Solve $\frac{x - 2}{x + 2} > \frac{2x - 3}{4x - 1}$.

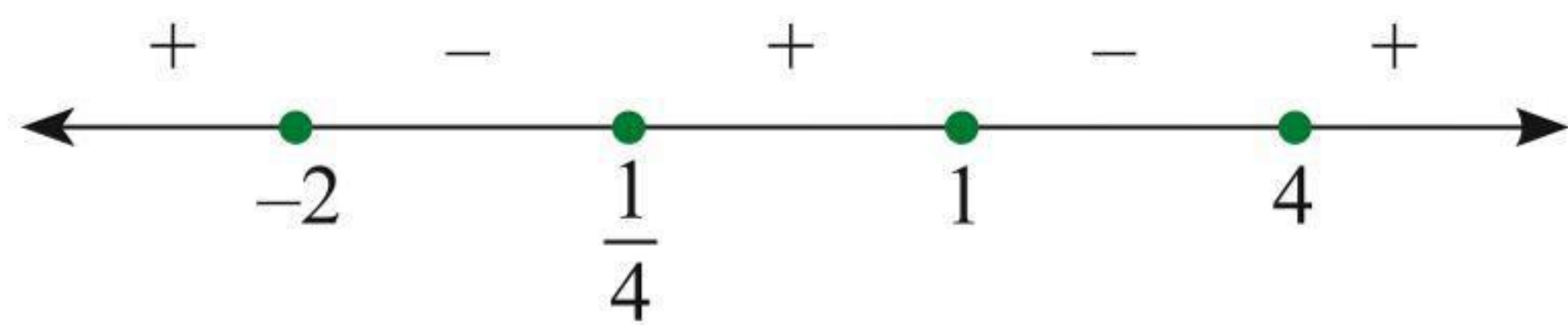
Sol. $\frac{x - 2}{x + 2} > \frac{2x - 3}{4x - 1}$

or $\frac{(x - 2)(4x - 1) - (x + 2)(2x - 3)}{(x + 2)(4x - 1)} > 0$

$$\text{or } \frac{4x^2 - 9x + 2 - 2x^2 - x + 6}{(x+2)(4x-1)} > 0$$

$$\text{or } \frac{x^2 - 5x + 4}{(x+2)(4x-1)} > 0$$

$$\text{or } \frac{(x-1)(x-4)}{(x+2)(4x-1)} > 0$$



$$\Rightarrow x \in (-\infty, -2) \cup \left(\frac{1}{4}, 1\right) \cup (4, \infty)$$

ILLUSTRATION 1.33Solve $x > \sqrt{1-x}$.**Sol.** Given inequality can be solved by squaring both sides.

But sometimes squaring gives extraneous solutions which do not satisfy the original inequality. Before squaring we must restrict x for which terms in the given inequality are well defined.

$x > \sqrt{1-x}$. Here x must be positive.

Now, $\sqrt{1-x}$ is defined only when $1-x \geq 0$ or $x \leq 1$ (1)

Squaring given inequality both sides, $x^2 > 1-x$

$$\Rightarrow x^2 + x - 1 > 0$$

$$\Rightarrow \left(x - \frac{-1-\sqrt{5}}{2}\right) \left(x - \frac{-1+\sqrt{5}}{2}\right) > 0$$



$$\Rightarrow x < \frac{-1-\sqrt{5}}{2} \text{ or } x > \frac{-1+\sqrt{5}}{2} \quad (2)$$

$$\text{From (1) and (2), } x \in \left(\frac{\sqrt{5}-1}{2}, 1\right]$$

ILLUSTRATION 1.34Solve $\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \geq 0$.

$$\text{Sol. } \frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \geq 0$$

$$\text{or } \frac{2(x+1) - (x^2-x+1) - (2x-1)}{(x+1)(x^2-x+1)} \geq 0$$

$$\text{or } \frac{-(x^2-x-2)}{(x+1)(x^2-x+1)} \geq 0$$

$$\text{or } \frac{-(x-2)(x+1)}{(x+1)(x^2-x+1)} \geq 0$$

$$\text{or } \frac{2-x}{x^2-x+1} \geq 0, \text{ where } x \neq -1$$

$$\text{or } 2-x \geq 0, x \neq -1 \quad (\text{as } x^2-x+1 > 0 \text{ for } \forall x \in \mathbb{R})$$

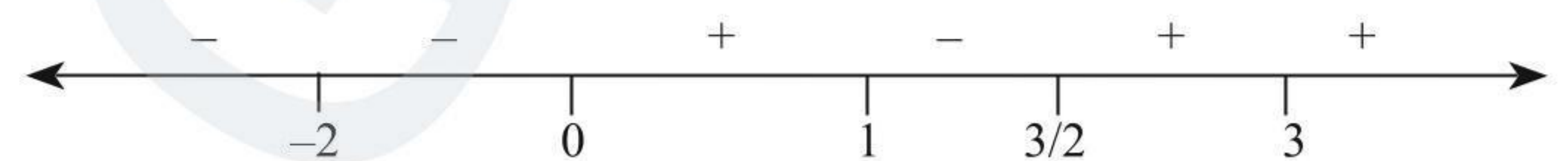
$$\Rightarrow x \leq 2, x \neq -1$$

ILLUSTRATION 1.35Solve $x(x+2)^2(x-1)^5(2x-3)(x-3)^4 \geq 0$.**Sol.** Let $E = x(x+2)^2(x-1)^5(2x-3)(x-3)^4$.

Here for x , $(x-1)$, $(2x-3)$, exponents are odd; hence, the sign of E changes while crossing $x = 0, 1, 3/2$. Also for $(x+2)$, $(x-3)$ exponents are even; hence, the sign of E does not change while crossing $x = -2$ and $x = 3$.

Further for $x > 3$, all factors are positive; hence, the sign of the expression starts with a positive sign from the right hand side.

The sign scheme of the expression is as follows:



Hence, for $E \geq 0$, we have $x \in [0, 1] \cup [3/2, \infty)$

ILLUSTRATION 1.36Solve $x(2^x-1)(3^x-9)(x-3) < 0$.**Sol.** Let $E = x(2^x-1)(3^x-9)(x-3)$

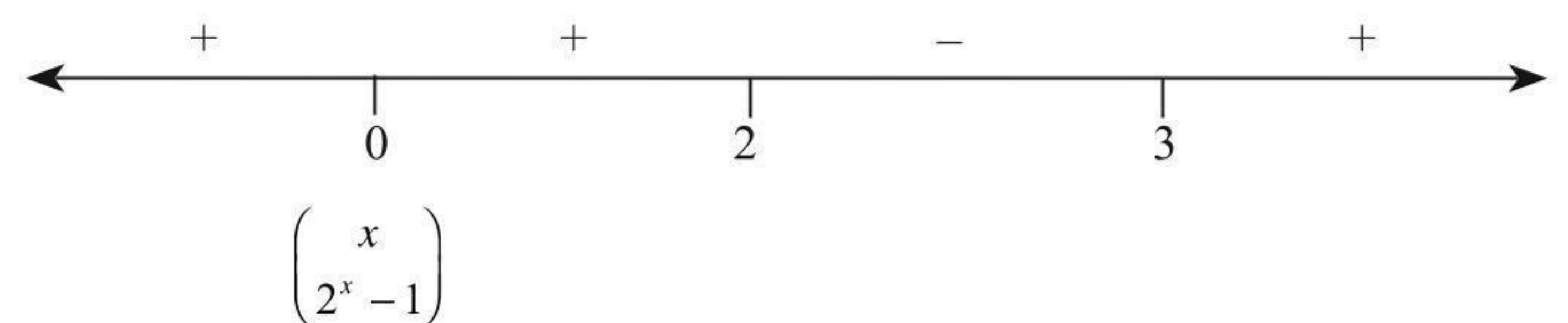
Here $2^x-1=0 \Rightarrow x=0$ and when $3^x-9=0 \Rightarrow x=2$

Now mark $x = 0, 2$, and 3 on real number line.

The sign of E starts with a positive sign from right hand side.

Also at $x = 0$, two factors vanish x and 2^x-1 ; hence, the sign of E does not change while crossing $x = 0$.

The sign scheme of E is as follows:



From the figure, we have $E < 0$ for $x \in (2, 3)$.

ILLUSTRATION 1.37Solve $(x^2-x-1)(x^2-x-7) < -5$ **Sol.** $(x^2-x-1)(x^2-x-7) + 5 < 0$

Let $x^2-x=y$

$$\Rightarrow (y-1)(y-7) + 5 < 0$$

$$\text{or } y^2 - 8y + 12 < 0$$

$$\text{or } (y-6)(y-2) < 0$$

$$\Rightarrow 2 < y < 6$$

$$\Rightarrow 2 < x^2-x < 6$$

$$\text{Now, } x^2-x > 2$$

$$\begin{aligned}
 \text{or } & x^2 - x - 2 > 0 \\
 \text{or } & (x-2)(x+1) > 0 \\
 \Rightarrow & x \in (-\infty, -1) \cup (2, \infty) \\
 & x^2 - x - 6 < 0 \\
 \Rightarrow & (x-3)(x+2) < 0 \\
 \Rightarrow & x \in (-2, 3)
 \end{aligned}
 \tag{1}$$

From (1) and (2), common values of x are $x \in (-2, -1) \cup (2, 3)$.

CONCEPT APPLICATION EXERCISE 1.3

- Solve $\frac{x(3-4x)(x+1)}{(2x-5)} < 0$.
- Solve $\frac{(2x+3)(4-3x)^3(x-4)}{(x-2)^2 x^5} \leq 0$.
- Solve $\frac{(x-3)(x+5)(x-7)}{|x-4|(x+6)} \leq 0$.
- Solve $\frac{5x+1}{(x+1)^2} < 1$.
- Solve $\frac{x+2}{x^2+1} > \frac{1}{2}$.
- Solve $4^{-x+0.5} - 7.2^{-x} < 4$.
- Solve $\frac{x^4}{(x-2)^2} > 0$.
- Solve $\frac{6x^2-5x-3}{x^2-2x+6} \leq 4$.
- Solve $\frac{(x+2)(x^2-2x+1)}{-4+3x-x^2} \geq 0$.
- Solve $\sqrt{x+2} \geq x$.
- Solve $\sqrt{2+x-x^2} > x-4$.
- Find the number of integral values of x satisfying $\sqrt{-x^2+10x-16} < x-2$.
- Find all possible values of $f(x) = \frac{1-x^2}{x^2+3}$.

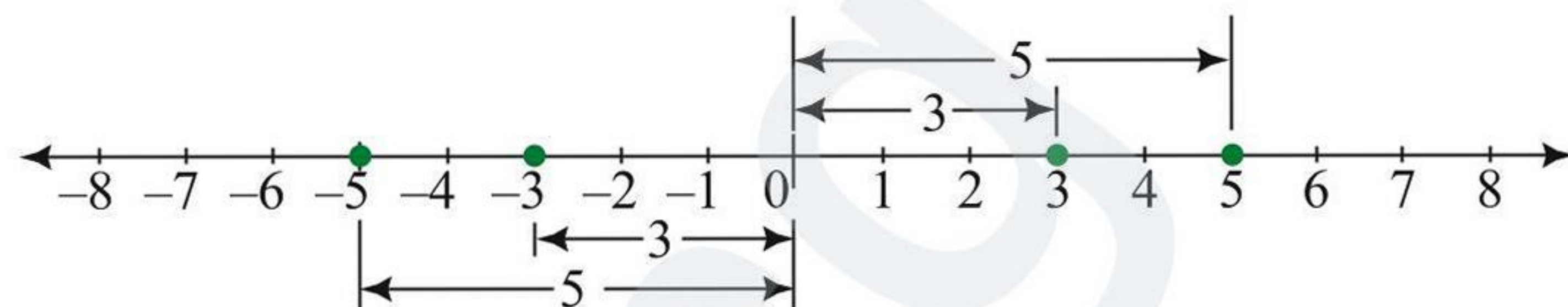
ANSWERS

- $x \in (-\infty, -1) \cup (0, 3/4) \cup (5/2, \infty)$
- $x \in (-\infty, -3/2) \cup (0, 4/3] \cup [4, \infty)$
- $x \in (-6, -5] \cup [3, 4) \cup (4, 7]$
- $x < 0$ or $x > 3, x \neq -1$
- $-1 < x < 3$
- $x \in (-2, \infty)$
- $x \in \mathbb{R} - \{0, 2\}$
- $-\frac{9}{2} \leq x \leq 3$
- $x \in (-\infty, -2] \cup \{1\}$
- $x \in [-2, 2]$
- $x \in [-1, 2]$
- 3
- $y \in (-1, 1/3]$

MODULUS OF REAL NUMBERS

We represent real numbers on real number line. The interesting feature of a number on real number line is its distance from origin. Let us define this distance and study its properties.

Consider some numbers on a real number line as shown in the following figure:



The distance of each of real numbers ' -3 ' and ' 3 ' from origin ' 0 ' is 3 units.

The distance of each of real numbers ' -5 ' and ' 5 ' from origin ' 0 ' is 5 units.

Now, which real numbers are at distance 4 units from ' 0 '?

The answer is ' 4 ' and ' -4 '.

The distance of any real number from ' 0 ' is called its absolute value or its modulus.

It is denoted by $|x|$.

Thus, $|x|$ = Distance of ' x ' from ' 0 '

Since $|x|$ denotes the distance, it is always non-negative i.e., $|x| \geq 0$.

Note:

- $x^2 = |x|^2$
- $|x| = |-x|$
- $|ab| = |a||b|$
- $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

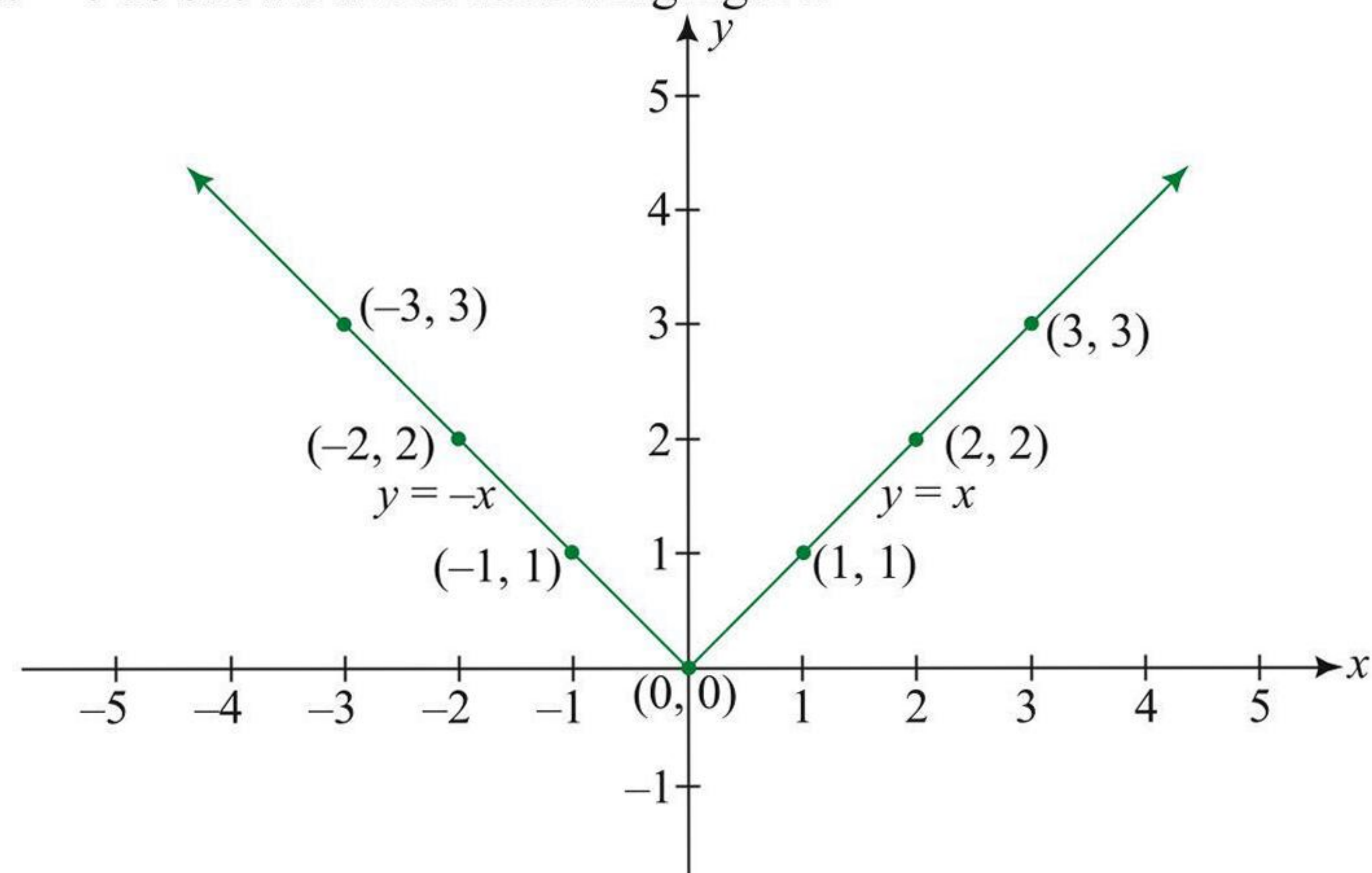
GRAPH OF MODULUS FUNCTION

Consider modulus function $y = f(x) = |x|$.

Clearly, when $x \geq 0$, $|x| = x$ and when $x < 0$, $|x| = -x$.

$$\text{So, we have } y = f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

So, the graph of $y = |x|$ is line $y = x$ for $x \geq 0$ and line $y = -x$ for $x < 0$ as shown in the following figure:



ABSOLUTE VALUE AS SQUARE ROOT FUNCTION

We know that square root of real number x exists only if x is zero or positive.

Also, square root of positive real number is always positive.

Now, observe the following calculations:

$$2 = \sqrt{2^2} = \sqrt{(-2)^2} = [(-2)^2]^{\frac{1}{2}} = -2.$$

We know that this is absolutely absurd. There is some mathematical error in one of the steps.

Actually, $(-2)^2$ is positive and square root of positive is always positive.

$$\text{So, we must have } [(-2)^2]^{\frac{1}{2}} = 2.$$

$$\text{or we can write } [(-2)^2]^{\frac{1}{2}} = |-2| = 2.$$

$$\text{Thus, } \sqrt{x^2} = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\text{e.g., } \sqrt{(1-\sqrt{2})^2} = |1-\sqrt{2}| = \sqrt{2}-1.$$

Note:

$$a^2 < b^2 \Rightarrow \sqrt{a^2} < \sqrt{b^2} \Rightarrow |a| < |b|$$

ILLUSTRATION 1.38

Solve the following:

$$(i) |x| = 5 \quad (ii) x^2 - |x| - 2 = 0$$

Sol.

(i) $|x| = 5$, i.e., those points on real number line which are at distance five units from '0' which are -5 and 5 . Thus,

$$|x| = 5 \Rightarrow x = \pm 5$$

$$(ii) x^2 - |x| - 2 = 0$$

$$\Rightarrow |x|^2 - |x| - 2 = 0 \quad (\because x^2 = |x|^2)$$

$$\Rightarrow (|x| - 2)(|x| + 1) = 0$$

$$\Rightarrow |x| = 2 \quad (\because |x| + 1 \neq 0)$$

$$\Rightarrow x = \pm 2$$

ILLUSTRATION 1.39

Find the value of x for which following expressions are defined:

$$(i) \frac{1}{\sqrt{x-|x|}} \quad (ii) \frac{1}{\sqrt{x+|x|}}$$

Sol.

$$(i) x - |x| = \begin{cases} x - x = 0, & \text{if } x \geq 0 \\ x + x = 2x, & \text{if } x < 0 \end{cases}$$

$$\Rightarrow x - |x| \leq 0, \text{ for all } x$$

$$\text{Thus, } \frac{1}{\sqrt{x-|x|}} \text{ does not take real values for any } x \in \mathbb{R}$$

and $\frac{1}{\sqrt{x-|x|}}$ is not defined for any $x \in \mathbb{R}$.

$$(ii) x + |x| = \begin{cases} x + x = 2x, & \text{if } x \geq 0 \\ x - x = 0, & \text{if } x < 0 \end{cases}$$

Thus, $\frac{1}{\sqrt{x+|x|}}$ is defined only when $x > 0$

ILLUSTRATION 1.40

Find all the possible values of following expressions:

$$(i) \frac{|x|}{x} + \frac{|y|}{y} \quad (ii) \frac{|x|}{x} + \frac{|y|}{y} + \frac{|z|}{z}$$

Sol.

$$(i) \frac{|x|}{x} + \frac{|y|}{y}$$

$$\frac{|x|}{x} = \frac{x}{x} = 1, \text{ if } x \text{ is positive}$$

$$\frac{|x|}{x} = \frac{-x}{x} = -1, \text{ if } x \text{ is negative}$$

Similarly, $\frac{|y|}{y} = \pm 1$, depending upon the value of y as positive or negative.

$$\therefore \frac{|x|}{x} + \frac{|y|}{y} = 1 + 1 \text{ or } -1 + 1 \text{ or } -1 - 1 \\ = 2, 0, -2$$

$$(ii) \frac{|x|}{x} + \frac{|y|}{y} + \frac{|z|}{z} = -1 - 1 - 1 \text{ or } -1 - 1 + 1 \text{ or } -1 + 1 + 1 \\ \text{or } 1 + 1 + 1 \\ = -3, -1, 1 \text{ or } 3$$

ILLUSTRATION 1.41

Solve $|x| = x^2 - 1$.

$$\text{Sol. } x^2 - 1 = |x|$$

$$\Rightarrow x^2 - 1 = x, \text{ when } x \geq 0$$

$$\text{or } x^2 - 1 = -x, \text{ when } x < 0$$

$$\Rightarrow x^2 - x - 1 = 0 \Rightarrow x = \frac{1+\sqrt{5}}{2} \text{ (as } x \geq 0)$$

$$x^2 + x - 1 = 0 \Rightarrow x = \frac{-1-\sqrt{5}}{2} \text{ (as } x < 0)$$

ILLUSTRATION 1.42

(i) For $2 < x < 4$, find the values of $|x|$.

(ii) For $-3 \leq x \leq -1$, find the values of $|x|$.

(iii) For $-3 \leq x < 1$, find the values of $|x|$.

(iv) For $-5 < x < 7$, find the values of $|x-2|$.

(v) For $1 \leq x \leq 5$, find the values of $|2x-7|$.

Sol.

(i) $2 < x < 4$

Here values on real number line whose distance from zero lies between 2 and 4.

Here values of x are positive $\Rightarrow |x| \in (2, 4)$

(ii) $-3 \leq x \leq -1$

Here values on real number line whose distance from zero lies between 1 and 3 or equal to 1 or 3.

$\Rightarrow 1 \leq |x| \leq 3$

(iii) $-3 \leq x < 1$

For $-3 \leq x < 0$, $|x| \in (0, 3]$

For $0 \leq x < 1$, $|x| \in [0, 1)$

So for $-3 \leq x < 1$, $|x| \in [0, 1) \cup (0, 3]$ or $|x| \in [0, 3]$

(iv) $-5 < x < 7$

or $-7 < x - 2 < 5$

$\Rightarrow 0 \leq |x - 2| < 7$

(v) $1 \leq x \leq 5$

or $2 \leq 2x \leq 10$

$\Rightarrow -5 \leq 2x - 7 \leq 3$

$\Rightarrow |2x - 7| \in [0, 5]$

MODULUS OF ANY EXPRESSION

Consider two points $A(x_1, 0)$ and $B(x_2, 0)$ on the x -axis.

$$\begin{aligned} \text{Now, distance } AB &= \sqrt{(x_1 - x_2)^2 + (0 - 0)^2} \\ &= \sqrt{(x_1 - x_2)^2} = |x_1 - x_2| \end{aligned}$$

Thus, expression $|x_1 - x_2|$ represents the distance between two real numbers x_1 and x_2 on the real number line, just like $|x| = |x - 0|$ represents the distance between two real numbers x and 0.

Now, we shall extend the definition of absolute value.

$$|x - a| = \begin{cases} x - a, & x - a \geq 0 \\ a - x, & x - a < 0 \end{cases}$$

$$\text{or } |x - a| = \begin{cases} x - a, & x \geq a \\ a - x, & x < a \end{cases}$$

The least value of $|x - a|$ is zero when $x = a$.

Note that $|x - a| = |a - x|$.

In general, for any expression $E(x)$,

$$E(x) = \begin{cases} E(x), & E(x) \geq 0 \\ -E(x), & E(x) < 0 \end{cases}$$

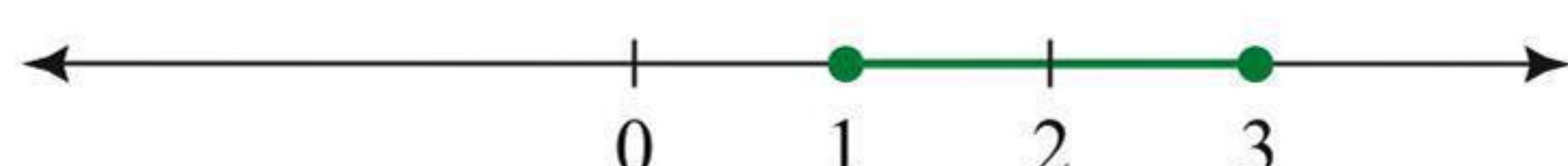
ILLUSTRATION 1.43

Solve the following:

(i) $|x - 2| = 1$ (ii) $2|x + 1|^2 - |x + 1| = 3$

Sol.

(i) $|x - 2| = 1$, i.e., those points on real number line which are at distance 1 units from 2.



As shown in the figure $x = 1$ and $x = 3$ are at distance 1 unit from 2.

Hence, $x = 1$ or $x = 3$.

Also, $|x - 2| = 1$

$\Rightarrow x - 2 = \pm 1$

$\Rightarrow x = 1 \text{ or } x = 3$

(ii) $2|x + 1|^2 - |x + 1| = 3$

$\Rightarrow 2|x + 1|^2 - |x + 1| - 3 = 0$

$\Rightarrow 2|x + 1|^2 - 3|x + 1| + 2|x + 1| - 3 = 0$

$\Rightarrow (2|x + 1| - 3)(|x + 1| + 1) = 0$

$\Rightarrow 2|x + 1| - 3 = 0$

$\Rightarrow |x + 1| = 3/2$

$\Rightarrow x + 1 = \pm 3/2$

$\Rightarrow x = 1/2 \text{ or } x = -5/2$

ILLUSTRATION 1.44

Find the values of a for which the equation $||x - 2| + a| = 4$ can have four distinct real solutions.

Sol. $|x - 2| + a = \pm 4$

$\Rightarrow |x - 2| = \pm 4 - a$

For four real roots, $4 - a > 0$ and $-4 - a > 0$

$\Rightarrow a \in (-\infty, -4)$

ILLUSTRATION 1.45

Solve the following:

(i) $|x - 2| = (x - 2)$

(ii) $|x + 3| = -x - 3$

(iii) $|x^2 - x| = x^2 - x$

(iv) $|x^2 - x - 2| = 2 + x - x^2$

Sol.

(i) $|x - 2| = (x - 2)$, if $x - 2 \geq 0$ or $x \geq 2$

(ii) $|x + 3| = -x - 3$, if $x + 3 \leq 0$ or $x \leq -3$

(iii) $|x^2 - x| = x^2 - x$, if $x^2 - x \geq 0$

or $x(x - 1) \geq 0$

$\Rightarrow x \in (-\infty, 0] \cup [1, \infty)$

(iv) $|x^2 - x - 2| = 2 + x - x^2$

or $x^2 - x - 2 \leq 0$

or $(x - 2)(x + 1) \leq 0$

or $-1 \leq x \leq 2$

ILLUSTRATION 1.46

Solve $1 - x = \sqrt{x^2 - 2x + 1}$.

Sol. $1 - x = \sqrt{x^2 - 2x + 1}$

or $1 - x = \sqrt{(x - 1)^2}$

or $1 - x = |x - 1|$

$\Rightarrow 1 - x \geq 0$

or $x \leq 1$

ILLUSTRATION 1.47Solve $|3x - 2| = x$.**Sol.** $|3x - 2| = x$ **Case (i)**When $3x - 2 \geq 0$ or $x \geq 2/3$ For which we have $3x - 2 = x$ or $x = 1$.**Case (ii)**When $3x - 2 < 0$ or $x < 2/3$ For which we have $2 - 3x = x$ or $x = 1/2$.Hence, solution set is $\{1/2, 1\}$.**ILLUSTRATION 1.48**Solve $\sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{x-1}} = 1$ **Sol.** $\sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{x-1}} = 1$ or $\sqrt{x-1-4\sqrt{x-1}+4} + \sqrt{x-1-6\sqrt{x-1}+9} = 1$ or $\sqrt{|\sqrt{x-1}-2|^2} + \sqrt{|\sqrt{x-1}-3|^2} = 1$ or $|\sqrt{x-1}-2| + |\sqrt{x-1}-3| = 1$ or $|\sqrt{x-1}-2| + |\sqrt{x-1}-3| = (\sqrt{x-1}-2) - (\sqrt{x-1}-3)$ $\Rightarrow \sqrt{x-1}-2 \geq 0$ and $\sqrt{x-1}-3 \leq 0$ $\Rightarrow 2 \leq \sqrt{x-1} \leq 3$ $\Rightarrow 4 \leq x-1 \leq 9$ $\Rightarrow 5 \leq x \leq 10$ **ILLUSTRATION 1.49**Prove that $\sqrt{x^2+2x+1} - \sqrt{x^2-2x+1} = \begin{cases} -2, & x < -1 \\ 2x, & -1 \leq x \leq 1 \\ 2, & x > 1 \end{cases}$ **Sol.** $\sqrt{x^2+2x+1} - \sqrt{x^2-2x+1}$

$$= \sqrt{(x+1)^2} - \sqrt{(x-1)^2}$$

$$= |x+1| - |x-1|$$

$$= \begin{cases} -x-1-(1-x), & x < -1 \\ x+1-(1-x), & -1 \leq x \leq 1 \\ x+1-(x-1), & x > 1 \end{cases} = \begin{cases} -2, & x < -1 \\ 2x, & -1 \leq x \leq 1 \\ 2, & x > 1 \end{cases}$$

ILLUSTRATION 1.50For $x \in \mathbb{R}$, find all possible values of**(i)** $|x-3|-2$ **(ii)** $4-|2x+3|$ **Sol.****(i)** We know that $|x-3| \geq 0 \forall x \in \mathbb{R}$

$$\Rightarrow |x-3|-2 \geq -2$$

$$\Rightarrow |x-3|-2 \in [-2, \infty)$$

(ii) We know that $|2x+3| \geq 0 \forall x \in \mathbb{R}$

$$\Rightarrow -|2x+3| \leq 0$$

$$\Rightarrow 4-|2x+3| \leq 4$$

$$\text{or } 4-|2x+3| \in (-\infty, 4]$$

ILLUSTRATION 1.51

Find all possible values of

(i) $\sqrt{|x|-2}$ **(ii)** $\sqrt{3-|x-1|}$ **(iii)** $\sqrt{4-\sqrt{x^2}}$ **Sol.****(i)** $\sqrt{|x|-2}$

We know that square roots are defined for non-negative values only.

It implies that we must have $|x|-2 \geq 0$. Thus,

$$\sqrt{|x|-2} \geq 0$$

(ii) $\sqrt{3-|x-1|}$ is defined when $3-|x-1| \geq 0$ But the maximum value of $3-|x-1|$ is 3, when $|x-1|$ is 0.Hence, for $\sqrt{3-|x-1|}$ to get defined, $0 \leq 3-|x-1| \leq 3$. Thus,

$$\sqrt{3-|x-1|} \in [0, \sqrt{3}]$$

Alternatively, $|x-1| \geq 0$

$$\Rightarrow -|x-1| \leq 0$$

$$\Rightarrow 3-|x-1| \leq 3$$

But for $\sqrt{3-|x-1|}$ to get defined, we must have

$$0 \leq 3-|x-1| \leq 3$$

$$\Rightarrow 0 \leq \sqrt{3-|x-1|} \leq \sqrt{3}$$

(iii) $\sqrt{4-\sqrt{x^2}} = \sqrt{4-|x|}$

$$|x| \geq 0$$

$$\Rightarrow -|x| \leq 0$$

$$\Rightarrow 4-|x| \leq 4$$

But for $\sqrt{4-|x|}$ to get defined $0 \leq 4-|x| \leq 4$

$$\therefore 0 \leq \sqrt{4-|x|} \leq 2$$

ILLUSTRATION 1.52Solve $|x-3| + |x-2| = 1$.**Sol.** $|x-3| + |x-2| = 1$

$$\Rightarrow |x-3| + |x-2| = (3-x) + (x-2)$$

$$\Rightarrow x-3 \leq 0 \text{ and } x-2 \geq 0$$

$$\Rightarrow x \leq 3 \text{ and } x \geq 2$$

$$\Rightarrow 2 \leq x \leq 3$$

CONCEPT APPLICATION EXERCISE 1.4

- Which of the following is always true?
 - If $a < b$, then $a^2 < b^2$
 - If $a < b$, then $\frac{1}{a} > \frac{1}{b}$
 - If $a < b$, then $|a| < |b|$
- Which of the following equations has maximum number of real roots?
 - $x^2 - |x| - 2 = 0$
 - $x^2 - 2|x| + 3 = 0$
 - $x^2 - 3|x| + 2 = 0$
 - $x^2 + 3|x| + 2 = 0$
- Find the number of solutions of the system of equation $x + 2y = 6$ and $|x - 3| = y$.
- Find the values of x for which the following function is defined:

$$f(x) = \sqrt{\frac{1}{|x-2| - (x-2)}}$$
- Find all values of $f(x)$ for which $f(x) = x + \sqrt{x^2}$.
- Solve $\left| \frac{x+2}{x-1} \right| = 2$.
- Solve $|x| = 2x - 1$.
- Solve $|2^x - 1| + |2^x + 1| = 2$.
- Solve $|x^2 + 4x + 3| = x + 1$.
- Solve $|4 - |x - 1|| = 3$
- Solve $|x - 1| - |2x - 5| = 2x$.

ANSWERS

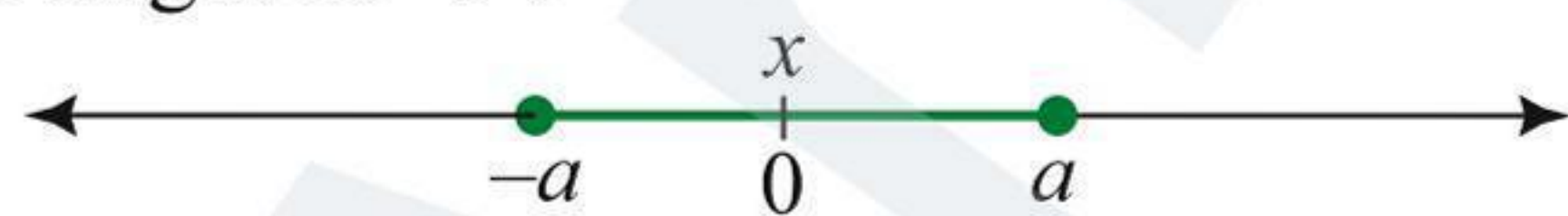
- None
- (c)
- 2
- $(-\infty, 2)$
- $[0, \infty)$
- $x = 0, 4$
- $x = 1/3$
- $x \leq 0$
- $x = -1$
- $x = -6, 0, 2, 8$
- $x = -4$

INEQUALITIES INVOLVING MODULUS

We have following inequalities involving modulus:

- (i) $|x| \leq a$ (where $a > 0$)

The values of x satisfying above inequality are those values of x on real number line which are at distance a or less than a from origin or '0'.



$$\Rightarrow -a \leq x \leq a$$

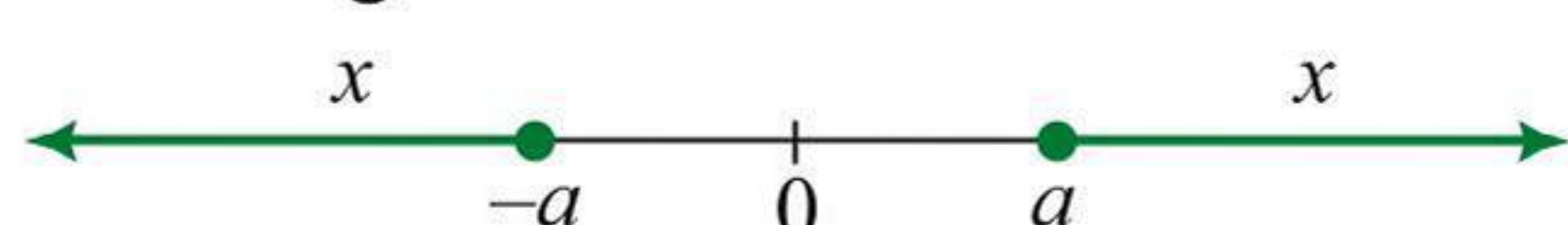
e.g., $|x| \leq 2 \Rightarrow -2 \leq x \leq 2$

$|x| < 3 \Rightarrow -3 < x < 3$ etc.

In general, $|f(x)| \leq a$ (where $a > 0$) $\Rightarrow -a \leq f(x) \leq a$

- (ii) $|x| \geq a$ (where $a > 0$)

The values of x satisfying above inequality are those values of x on real number line which are at distance a or more than a from origin or 0.



$$\Rightarrow x \leq -a \text{ or } x \geq a$$

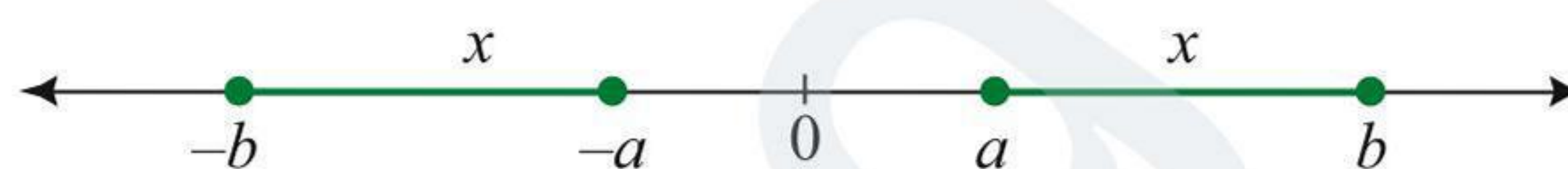
e.g., $|x| \geq 3 \Rightarrow x \leq -3 \text{ or } x \geq 3$.

$|x| > 2 \Rightarrow x < -2 \text{ or } x > 2$

In general, $|f(x)| \geq a \Rightarrow f(x) \leq -a \text{ or } f(x) \geq a$

- (iii) $a \leq |x| \leq b$ (where $a, b > 0$)

The values of x satisfying above inequality are those values of x on real number line which are at distance equal to a or b or between a and b .



$$\Rightarrow [-b, -a] \cup [a, b]$$

e.g., $2 \leq |x| \leq 4$

$$\Rightarrow x \in [-4, -2] \cup [2, 4]$$

ILLUSTRATION 1.53

Solve $x^2 - 4|x| + 3 < 0$.

Sol. $x^2 - 4|x| + 3 < 0$

$$\Rightarrow (|x| - 1)(|x| - 3) < 0$$

$$\Rightarrow 1 < |x| < 3$$

$$\Rightarrow -3 < x < -1 \text{ or } 1 < x < 3$$

$$\Rightarrow x \in (-3, -1) \cup (1, 3)$$

ILLUSTRATION 1.54

Solve $0 < |x| < 2$.

Sol. We know that $|x| \geq 0, \forall x \in R$

But given $|x| > 0 \Rightarrow x \neq 0$

Now $0 < |x| < 2$

$$\Rightarrow x \in (-2, 2), x \neq 0$$

$$\Rightarrow x \in (-2, 2) - \{0\}$$

ILLUSTRATION 1.55

Solve $|3x - 2| < 4$.

Sol. $|3x - 2| < 4$

$$\Rightarrow -4 < 3x - 2 < 4$$

$$\Rightarrow -2 < 3x < 6$$

$$\Rightarrow -2/3 < x < 2$$

ILLUSTRATION 1.56

Solve $1 \leq |x - 2| \leq 3$.

Sol. $1 \leq |x - 2| \leq 3$

$$\Rightarrow -3 \leq x - 2 \leq -1 \text{ or } 1 \leq x - 2 \leq 3$$

$$\Rightarrow -1 \leq x \leq 1 \text{ or } 3 \leq x \leq 5$$

$$\Rightarrow x \in [-1, 1] \cup [3, 5]$$

ILLUSTRATION 1.57

Solve $0 < |x - 3| \leq 5$.

Sol. $0 < |x - 3| \leq 5$

$$\Rightarrow -5 \leq x - 3 < 0 \text{ or } 0 < x - 3 \leq 5$$

$$\Rightarrow -2 \leq x < 3 \text{ or } 3 < x \leq 8$$

$$\Rightarrow x \in [-2, 3) \cup (3, 8]$$

ILLUSTRATION 1.58Solve $|x-1| - 2 < 5$.

$$\begin{aligned}
 \text{Sol. } & |x-1| - 2 < 5 \\
 \Rightarrow & -5 < |x-1| - 2 < 5 \\
 \Rightarrow & -3 < |x-1| < 7 \\
 \Rightarrow & 0 \leq |x-1| < 7 \\
 \Rightarrow & -7 < x-1 < 7 \\
 \Rightarrow & -6 < x < 8
 \end{aligned}$$

ILLUSTRATION 1.59Solve $|x-3| \geq 2$.

$$\begin{aligned}
 \text{Sol. } & |x-3| \geq 2 \\
 \Rightarrow & x-3 \leq -2 \text{ or } x-3 \geq 2 \\
 \Rightarrow & x \leq 1 \text{ or } x \geq 5
 \end{aligned}$$

ILLUSTRATION 1.60Solve $||x| - 3| > 1$.

$$\begin{aligned}
 \text{Sol. } & ||x| - 3| > 1 \\
 \Rightarrow & |x| - 3 < -1 \text{ or } |x| - 3 > 1 \\
 \Rightarrow & |x| < 2 \text{ or } |x| > 4 \\
 \Rightarrow & -2 < x < 2 \text{ or } x < -4 \text{ or } x > 4
 \end{aligned}$$

ILLUSTRATION 1.61Solve $\left| \frac{x-3}{x+1} \right| \leq 1$.

$$\begin{aligned}
 \text{Sol. } & \left| \frac{x-3}{x+1} \right| \leq 1 \\
 \text{or } & -1 \leq \frac{x-3}{x+1} \leq 1 \\
 \Rightarrow & \frac{x-3}{x+1} - 1 \leq 0 \text{ and } 0 \leq \frac{x-3}{x+1} + 1 \\
 \Rightarrow & \frac{-4}{x+1} \leq 0 \text{ and } 0 \leq \frac{2x-2}{x+1} \\
 \Rightarrow & x > -1 \text{ and } \{x < -1 \text{ or } x \geq 1\} \\
 \Rightarrow & x \geq 1
 \end{aligned}$$

ILLUSTRATION 1.62Solve $\left| 1 + \frac{3}{x} \right| > 2$.

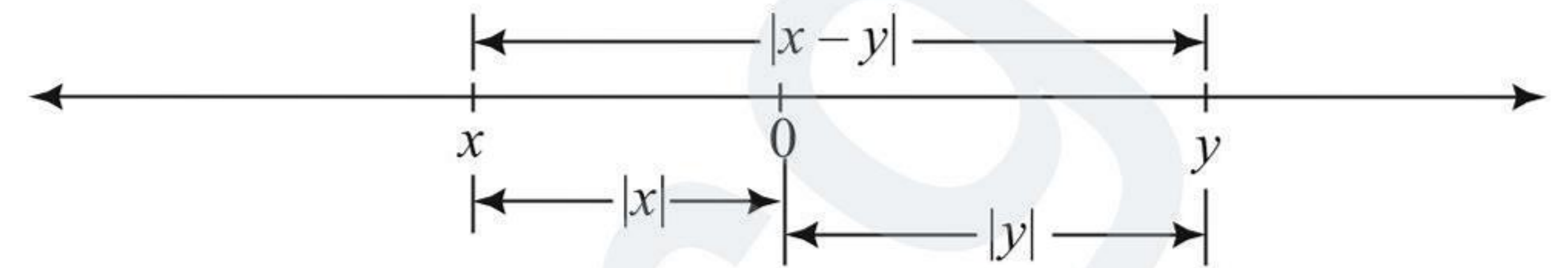
$$\begin{aligned}
 \text{Sol. } & \left| 1 + \frac{3}{x} \right| > 2 \\
 \Rightarrow & 1 + \frac{3}{x} > 2 \text{ or } 1 + \frac{3}{x} < -2 \\
 \Rightarrow & \frac{3-x}{x} > 0 \text{ or } \frac{3(x+1)}{x} < 0 \\
 \Rightarrow & \frac{x-3}{x} < 0 \text{ or } \frac{x+1}{x} < 0 \\
 \Rightarrow & x \in (0, 3) \text{ or } x \in (-1, 0) \\
 \therefore & x \in (-1, 0) \cup (0, 3)
 \end{aligned}$$

MODULUS OF ADDITION AND SUBTRACTION

We know that modulus is distributive over multiplication and division. But, it is not distributive over addition and subtraction always.

Also, we know that $|x - y|$ represents the distance between two real numbers x and y on real number line.

Let x be negative and y be positive.



Clearly, from the figure,

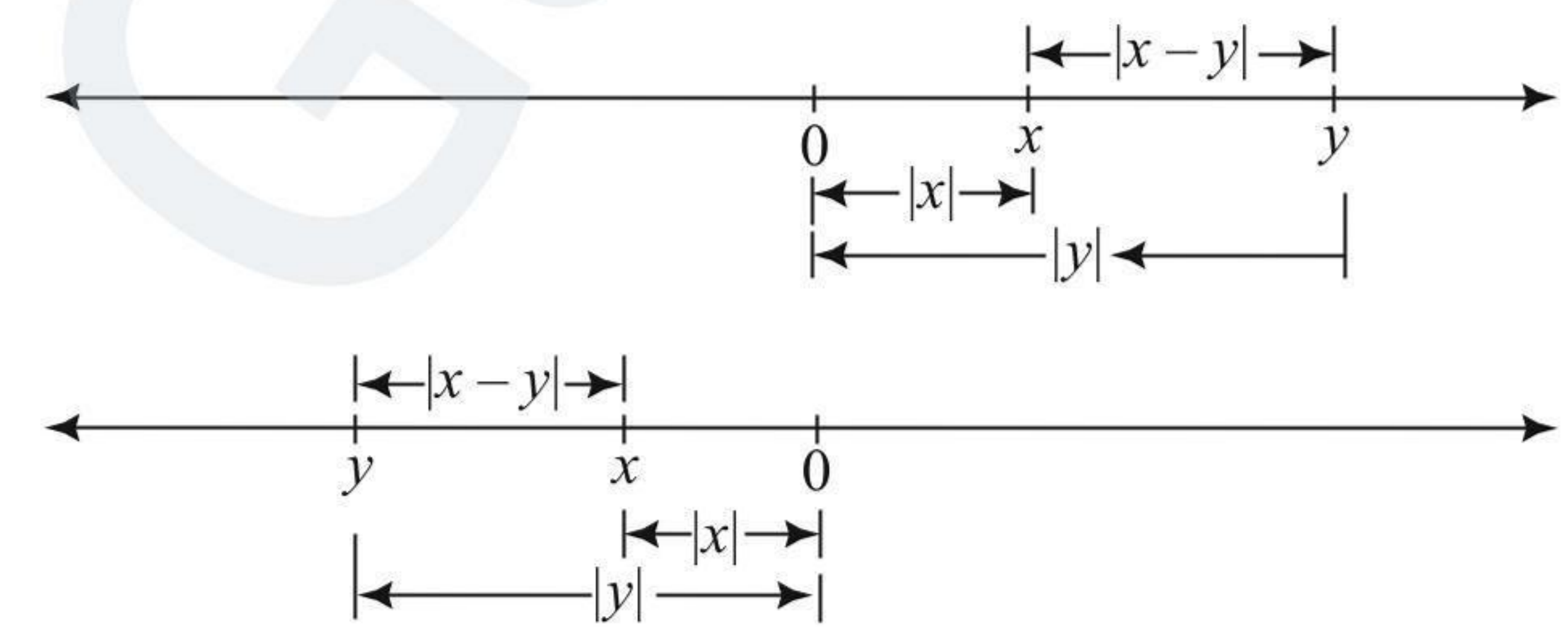
$$|x - y| = |x| + |y|$$

Now, replacing y by $-y$ in above equation, we get

$$|x - (-y)| = |x| + |-y|$$

$$\text{or } |x + y| = |x| + |y|$$

Now, let x and y have the same sign.



Clearly, from the figure,

$$|x - y| < |x| + |y|$$

Replacing y by $-y$, we get

$$|x - (-y)| < |x| + |-y|$$

$$\text{or } |x + y| < |x| + |y|$$

Thus, we get following results:

- $|x + y| < |x| + |y|$ if x and y have opposite signs.
- $|x - y| < |x| + |y|$ if x and y have the same sign.
- $|x + y| = |x| + |y|$ if x and y have the same sign or at least one of x and y is zero.
- $|x - y| = |x| + |y|$ if x and y have opposite signs or at least one of x and y is zero.

Hence, in general, $|x \pm y| \leq |x| + |y|$.

ILLUSTRATION 1.63Solve $|x| + |x-2| = 2$.

$$\begin{aligned}
 \text{Sol. } & \text{We have } |x| + |x-2| = 2 \\
 \Rightarrow & |x| + |x-2| = |x - (x-2)| \\
 \Rightarrow & x(x-2) \leq 0 \\
 \Rightarrow & 0 \leq x \leq 2
 \end{aligned}$$

ILLUSTRATION 1.64Solve $|2x-3| + |x-1| = |x-2|$.

$$\begin{aligned}
 \text{Sol. } & |2x-3| + |x-1| = |(2x-3) - (x-1)| \\
 \Rightarrow & (2x-3)(x-1) \leq 0 \\
 \Rightarrow & 1 \leq x \leq 3/2
 \end{aligned}$$

ILLUSTRATION 1.65Solve $|x^2 + x - 4| = |x^2 - 4| + |x|$.

Sol. $|(x^2 - 4) + x| = |x^2 - 4| + |x|$
 $\Rightarrow x(x^2 - 4) \geq 0$
 $\Rightarrow x(x - 2)(x + 2) \geq 0$
 $\Rightarrow x \in [-2, 0] \cup [2, \infty)$

ILLUSTRATION 1.66If $|\sin x + \cos x| = |\sin x| + |\cos x|$ ($\sin x, \cos x \neq 0$), then in which quadrant does x lie?

Sol. Here we have $|\sin x + \cos x| = |\sin x| + |\cos x|$.
 It implies that $\sin x$ and $\cos x$ must have the same sign.
 Therefore, x lies in the first or third quadrant.

ILLUSTRATION 1.67Is $|\tan x + \cot x| < |\tan x| + |\cot x|$ true for any x ? If it is true, then find the values of x .

Sol. Since $\tan x$ and $\cot x$ have always the same sign,
 $|\tan x + \cot x| < |\tan x| + |\cot x|$ does not hold true for any value of x .

ILLUSTRATION 1.68Solve $\left| \frac{x+1}{x} \right| + |x+1| = \frac{(x+1)^2}{|x|}$.

Sol. $\left| \frac{x+1}{x} \right| + |x+1| = \frac{(x+1)^2}{|x|}$
 or $\left| \frac{x+1}{x} \right| + |x+1| = \left| \frac{x+1}{x} + (x+1) \right|$
 or $\frac{(x+1)}{x} \times (x+1) \geq 0$
 or $\frac{(x+1)^2}{x} \geq 0$
 $\Rightarrow x \in (0, \infty) \cup \{-1\}$

ILLUSTRATION 1.69Solve $|x^2 - 2x| + |x - 4| > |x^2 - 3x + 4|$.

Sol. We have $|x^2 - 2x| + |4 - x| > |x^2 - 2x + 4 - x|$
 $(x^2 - 2x)(4 - x) < 0$
 $\Rightarrow x(x - 2)(x - 4) > 0$
 $\Rightarrow x \in (0, 2) \cup (4, \infty)$

ILLUSTRATION 1.70Solve $|2^x - 1| + |4 - 2^x| < 3$

Sol. $|2^x - 1| + |4 - 2^x| < 3$
 $2^x - 1 = 0$, if $x = 0$
 $4 - 2^x = 0$, if $x = 2$

Case (i): $x < 0$

$\Rightarrow 1 - 2^x + 4 - 2^x < 3$
 $\Rightarrow 2 \times 2^x > 2 \Rightarrow 2^x > 1$, not possible when $x < 0$

Case (ii):

$0 \leq x < 2$
 $\Rightarrow 2^x - 1 + 4 - 2^x < 3$ not possible

Case (iii):

$x \geq 2$
 $\Rightarrow 2 \times 2^x < 8$
 $\Rightarrow 2^x < 4$
 $\Rightarrow x < 2$ not possible

Alternative Method:We have $|2^x - 1| + |4 - 2^x| < 3$ But $|(2^x - 1) + (4 - 2^x)| \leq |2^x - 1| + |4 - 2^x| < 3$ $\therefore |(2^x - 1) + (4 - 2^x)| < 3$ or $3 < 3$, which is not possible.

Hence, there is no real solution.

CONCEPT APPLICATION EXERCISE 1.5

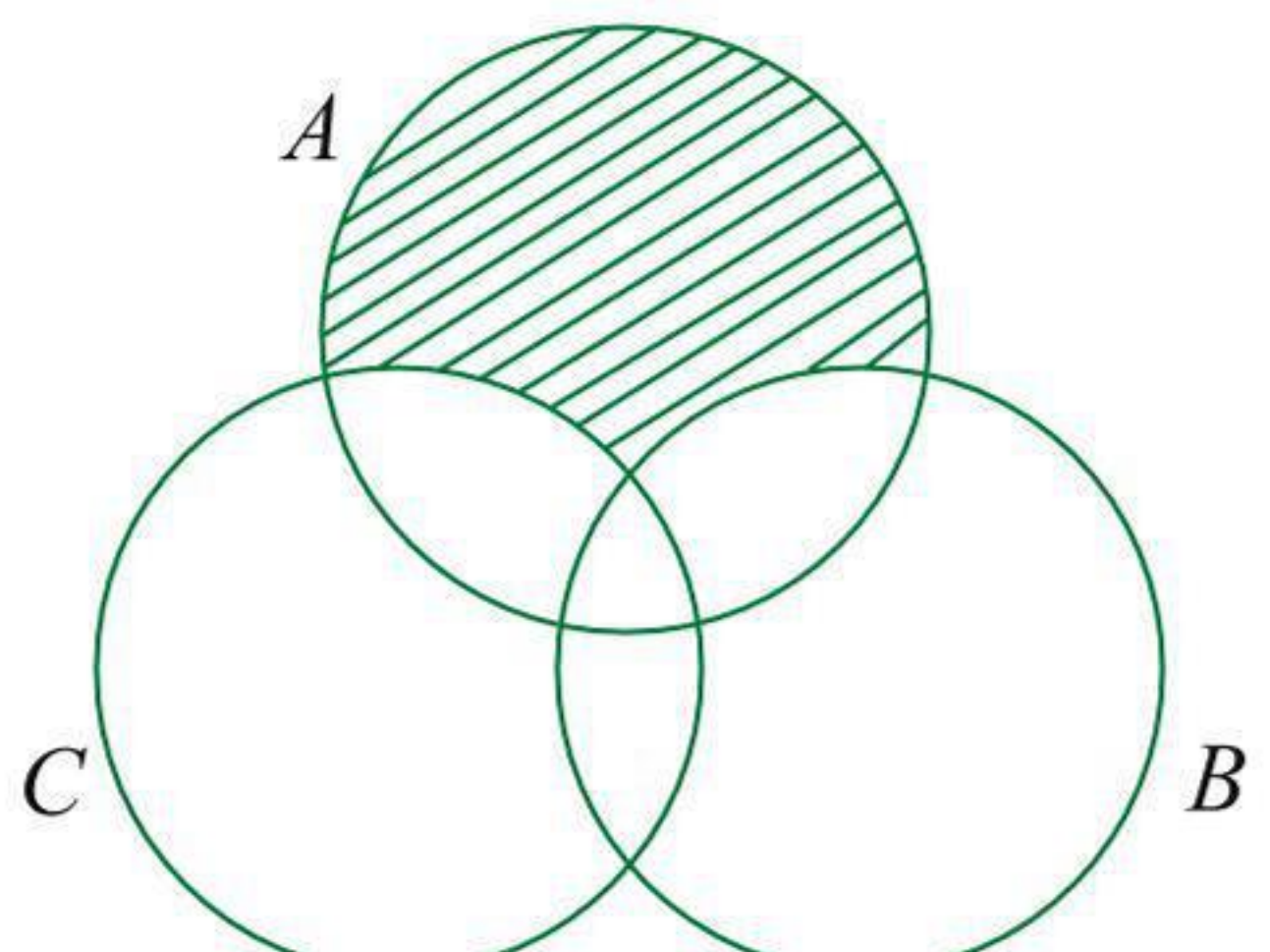
- If $|x^2 - 7| \leq 9$, then find the values of x .
- Solve $||x - 2| - 3| < 5$.
- Which of the following is/are true?
 - If $|x + y| = |x| + |y|$, then points (x, y) lie in 1st or 3rd quadrant or any of the x -axis or y -axis.
 - If $|x + y| < |x| + |y|$, then points (x, y) lie in 2nd or 4th quadrant.
 - If $|x - y| = |x| + |y|$, then points (x, y) lie in 2nd or 4th quadrant.
- Find the values of x for which $\sqrt{5 - |2x - 3|}$ is defined.
- Solve $|x^2 - x - 2| + |x + 6| = |x^2 - 2x - 8|$.
- Solve $|x^2 + x - 6| < 6$
- Solve $|x| + \left| \frac{4 - x^2}{x} \right| = \left| \frac{4}{x} \right|$
- Solve $\frac{1}{|x| - 3} < \frac{1}{2}$
- Solve $|x - \pi| + |x^2 - \pi^2| \leq 0$
- Solve $\left| 1 - \frac{|x|}{1 + |x|} \right| \geq \frac{1}{2}$
- Solve $\frac{|x - 1|}{x + 2} > 1$
- Solve $|x^2 - 1| + |x^2 - 4| > 3$.

ANSWERS

- $x \in [-4, 4]$
- $-6 < x < 10$
- All
- $x \in [-1, 4]$
- $x \in (-\infty, -6] \cup [-1, 2]$
- $x \in (-4, -1) \cup (0, 3)$
- $x \in [-2, 2] - \{0\}$
- $x \in (-\infty, 5) \cup (-3, 3) \cup (5, \infty)$
- $x = \pi$
- $-1 \leq x \leq 1$
- $x \in \left(-2, -\frac{1}{2}\right)$
- $-1 < x < 1$ or $x < -2$ or $x > 2$

Exercises

Single Correct Answer Type

- Let F_1 be the set of parallelograms, F_2 the set of rectangles, F_3 be the set of rhombuses, F_4 be the set of squares and F_5 be the set of trapeziums in a plane. Then F_1 may be equal to
 - $F_2 \cap F_3$
 - $F_3 \cap F_4$
 - $F_2 \cup F_5$
 - $F_2 \cup F_3 \cup F_4 \cup F_1$
- If $n(A) = 3$, $n(B) = 6$ and $A \subseteq B$. Then the number of elements in $A \cup B$ is equal to
 - 3
 - 9
 - 6
 - None of these
- If the sets A and B are defined as
 $A = \{(x, y) \mid y = 1/x, x \neq 0, x \in R\}$,
 $B = \{(x, y) \mid y = -x, x \in R\}$. Then
 - $A \cap B = A$
 - $A \cap B = B$
 - $A \cap B = \phi$
 - $A \cup B = A$
- Two finite sets have m and n ($m > n$) elements. The number of subsets of the first set is 112 more than that of the second set. The value of mn is
 - 18
 - 28
 - 32
 - 36
- Let A and B be two non-empty subsets of a set X such that A is not a subset of B . Then
 - A is a subset of complement of B
 - B is a subset of A
 - A and B are disjoint sets
 - A and complement of B are non-disjoint sets
- If $aN = \{ax : x \in N\}$, then the set $4N \cap 6N$ is
 - $8N$
 - $10N$
 - $12N$
 - None of these
- The set $(A \cap B')' \cup (B \cap C)$ is equal to
 - $A' \cup B \cup C$
 - $A' \cup B$
 - $A' \cup C'$
 - $A' \cap B$
- For sets A and B , $(A \cup B)' \cup (A' \cap B)$ equals
 - A'
 - B'
 - A
 - None of these
- Let U be the universal set and $A \cup B \cup C = U$. Then $[(A - B) \cup (B - C) \cup (C - A)]'$ equals
 - $A \cup B \cup C$
 - $A \cap B \cap C$
 - $A \cup (B \cap C)$
 - $A \cap (B \cup C)$
- The shaded region in the given figure is
 
- $A \cap (B \cup C)$
 - $A \cup (B \cap C)$
 - $A \cap (B - C)$
 - $A - (B \cup C)$
- Which is the simplified representation of $(A' \cap B' \cap C) \cup (B \cap C) \cup (A \cap C)$ where A, B and C are subsets of set X ?
 - A
 - B
 - C
 - $X \cap (A \cup B \cup C)$
- In statistical survey of 1003 families of Kolkata, it was found that 63 families has neither a radio nor a TV, 794 families has a radio and 187 has TV. The number of families in that group having both a radio and a TV is
 - 36
 - 41
 - 32
 - None of these
- A survey shows that 63% of the people watch a news channel whereas, 76% watch an entertainment channel at a particular time. If $x\%$ of the people watch both types of channels, then
 - $x = 35$
 - $x \geq 63$
 - $39 \leq x \leq 63$
 - $x = 39$
- In a town of 10,000 families, it was found that 40% families buy newspaper A, 20% buy newspaper B and 10% buy newspaper C. Also, 5% families buy newspapers A and B, 3% buy newspapers B and C and 4% buy newspapers A and C. If 2% families buy all the three newspapers, then number of families which buy newspaper A only is
 - 3100
 - 3300
 - 2900
 - 1400
- Complete solution set of inequality $\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1$ is
 - $(-\infty, 0]$
 - $(-\infty, 0] \cup (2, 3)$
 - $[2, 3]$
 - $(-\infty, 2) \cup (3, \infty)$
- The number of integral values of x if $5x - 1 < (x + 1)^2 < 7x - 3$, is
 - 0
 - 1
 - 2
 - 3
- If set $A = \left\{x \mid \frac{x^2(5-x)(1-2x)}{(5x+1)(x+2)} < 0\right\}$ and
 set $B = \left\{x \mid \frac{3x+1}{6x^3+x^2-x} > 0\right\}$, then $A \cap B$ does not contain
 - (1, 4)
 - (5, 11)
 - $\left(-\frac{3}{2}, \frac{-1}{2}\right)$
 - none of these
- Number of integers satisfying the inequality, $x^4 - 29x^2 + 100 \leq 0$ is
 - 2
 - 4
 - 6
 - 8
- If $n > 0$ and exactly 15 integers satisfy $(x+6)(x-4)(x-5)(2x-n) \leq 0$, then sum of digits of the least possible value of n is
 - 10
 - 12
 - 14
 - 16

20. The solution of the inequality $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$ is
 (1) $[1, 3] \cup (5, \infty)$ (2) $(1, 3) \cup (5, \infty)$
 (3) $(-\infty, 1) \cup (5, \infty)$ (4) none of these
21. The complete solution set of inequality $\frac{(x-5)^{1005}(x+8)^{1008}(x-1)}{x^{1006}(x-2)^3(x-3)^5(x-6)(x+9)^{1010}} \leq 0$ is
 (1) $(-\infty, -9) \cup (-8, 0) \cup (0, 1) \cup (2, 3) \cup [5, 6)$
 (2) $(-\infty, -9) \cup (-9, 0) \cup (0, 1) \cup (2, 3) \cup (5, 6)$
 (3) $(-\infty, -9) \cup (-9, 0) \cup (0, 1] \cup (2, 3) \cup [5, 6)$
 (4) $(-\infty, 0) \cup (0, 1] \cup (2, 3) \cup [5, 6]$
22. Sum of solutions of the equation $|x|^3 - 4|x|^2 + 3|x| = 0$ is
 (1) 4 (2) 3 (3) 0 (4) 1
23. Number of integral roots of $|x-1| |x^2-2| = 2$ is
 (1) 0 (2) 1 (3) 2 (4) 3
24. The solution set of the inequality $\frac{|x+2|-x}{x} < 2$ is
 (1) $(0, 1)$ (2) $[0, 2]$
 (3) $(-\infty, 0) \cup (1, \infty)$ (4) none of these
25. Number of solutions of the equation $|2 - |x|| = x + 4$ is
 (1) 0 (2) 1
 (3) 2 (4) Infinite
26. Number of integral values of x satisfying the inequality $\frac{x^2+6x-7}{|x+2||x+3|} < 0$ is
 (1) 5 (2) 6 (3) 7 (4) 8
27. If $-4 \leq x < 2$, then $||x+2|-3|$ lies in the interval
 (1) $(1, 3]$ (2) $[1, 3]$
 (3) $[0, 3]$ (4) $[0, \infty)$
28. Complete set of values of x satisfying inequality $||x-1|-5| < 2x-5$ is
 (1) $(5/2, \infty)$ (2) $(11/3, \infty)$
 (3) $(-1, \infty)$ (4) $(-\infty, 1/3)$
29. If $|x^2 - 2x + 2| - |2x^2 - 5x + 2| = |x^2 - 3x|$, then the set of values of x is
 (1) $(-\infty, 0] \cup [3, \infty)$
 (2) $\left[0, \frac{1}{2}\right] \cup [2, 3]$
 (3) $(-\infty, 0] \cup \left[\frac{1}{2}, 2\right] \cup [3, \infty)$
 (4) $[0, 2] \cup [3, \infty)$
30. The complete solution set of the equation $|x^2 - 5x + 6| + |x^2 + 12x + 27| = |17x + 21|$ is
 (1) $x \in [-9, 3]$ (2) $x \in [-3, 2) \cup (2, 3]$
 (3) $x \in [-9, -3] \cup [2, 3]$ (4) $x \in (-2, 3)$

Answers Key

EXERCISES

Single Correct Answer Type

1. (4) 2. (3) 3. (3) 4. (2) 5. (4)
 6. (3) 7. (2) 8. (1) 9. (2) 10. (4)

11. (3) 12. (2) 13. (3) 14. (2) 15. (2)
 16. (2) 17. (2) 18. (4) 19. (4) 20. (1)
 21. (3) 22. (3) 23. (4) 24. (3) 25. (2)
 26. (1) 27. (3) 28. (2) 29. (2) 30. (3)

Solutions

Chapter 1

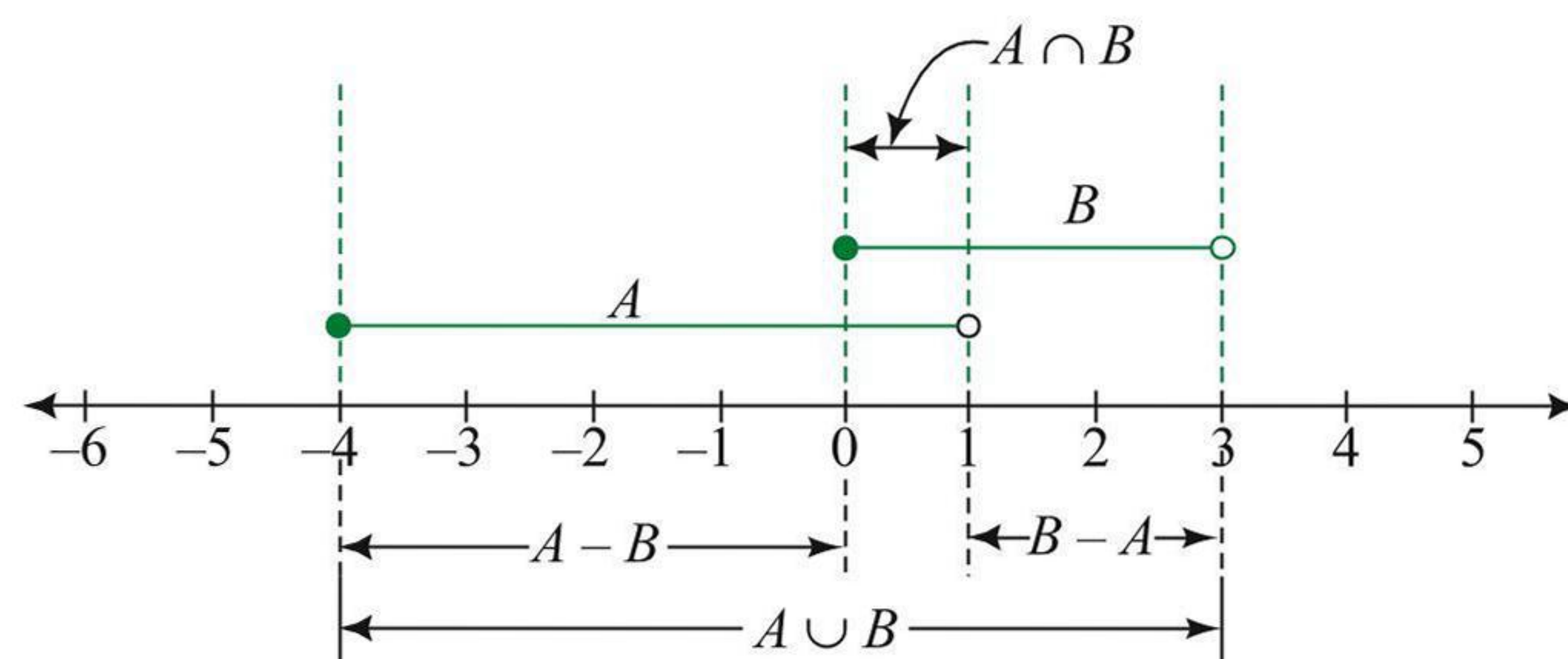
Concept Application Exercises

Exercise 1.1

- We have set, $A = \{1, 2, \{3, 4\}, 5\}$
 - The statement $\{3, 4\} \subset A$ is incorrect because $3 \in \{3, 4\}$, however, $3 \notin A$.
 - We know that the element of the set can be a set. So, the statement $\{3, 4\} \in A$ is correct because $\{3, 4\}$ is an element of A .
 - The statement $\{\{3, 4\}\} \subset A$ is correct because $\{3, 4\} \in \{\{3, 4\}\}$ and $\{3, 4\} \in A$.
 - The statement $1 \in A$ is correct because 1 is an element of A .
 - The statement $1 \subset A$ is incorrect because an element of a set can never be a subset of the set.
 - The statement $\{1, 2, 5\} \subset A$ is correct because each element of $\{1, 2, 5\}$ is element of the set A .
 - The statement $\{1, 2, 5\} \in A$ is incorrect because set $\{1, 2, 5\}$ is not an element of the set A .
 - The statement $\{1, 2, 3\} \subset A$ is incorrect because $3 \in \{1, 2, 3\}$, however $3 \notin A$.
 - The statement $\phi \in A$ is incorrect because ϕ is not an element of the set A .
 - The statement $\phi \subset A$ is correct because ϕ is a subset of every set.
 - The statement $\{\phi\} \subset A$ is incorrect because $\phi \in \{\phi\}$, however $\phi \notin A$.
- Given, $X = \{1, 2, 3, \dots, 10\}$
 - Since $1^2, 2^2, 3^2 \in X$
 $\therefore \{a \mid a \in X \text{ and } a^2 \notin X\} = \{4, 5, 6, 7, 8, 9, 10\}$
 - $a \in X$ but $a/2 \notin X$
 $\Rightarrow a = 1, 3, 5, 7, 9$
 $\therefore \{a \mid a \in X \text{ but } a/2 \notin X\} = \{1, 3, 5, 7, 9\}$
 - a is factor of 24
 $\Rightarrow a = 1, 2, 3, 4, 6, 8$
 $\therefore \{a \mid a \text{ is factor of } 24\} = \{1, 2, 3, 4, 6, 8\}$
- The subsets of $\{p\}$ are ϕ and $\{p\}$.
 - The subsets of $\{p, q\}$ are ϕ , $\{p\}$, $\{q\}$, and $\{p, q\}$.
 - The subsets of $\{1, 2, 3, 4\}$ are ϕ , $\{1\}$, $\{2\}$, $\{3\}$, $\{4\}$, $\{1, 2\}$, $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$, $\{1, 2, 3\}$, $\{2, 3, 4\}$, $\{3, 4, 1\}$, $\{4, 1, 2\}$, $\{1, 2, 3, 4\}$
 - The only subset of ϕ is ϕ .
- Let set A have n elements.
 Then number of elements in its power set $P(A)$ is 2^n .
 According to the question,
 $2^n = 128 = 2^7$
 $\therefore n = 7$
- Let $A = \{2\}$, $B = \{0, 2\}$, and $C = \{1, \{0, 2\}, 3\}$
 Here, $A \subset B$, $B \in C$
 However, $A \notin C$
 So, the given statement is false.

- $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
 $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$, $C = \{3, 4, 5, 6\}$
 - $A' = \{5, 6, 7, 8, 9\}$
 - $B' = \{1, 3, 5, 7, 9\}$
 - $A \cup C = \{1, 2, 3, 4, 5, 6\}$
 $\therefore (A \cup C)' = \{7, 8, 9\}$
 - $A \cup B = \{1, 2, 3, 4, 6, 8\}$
 $\therefore (A \cup B)' = \{5, 7, 9\}$
 - $(A')' = A = \{1, 2, 3, 4\}$
 - $B - C = \{2, 8\}$
 $\therefore (B - C)' = \{1, 3, 4, 5, 6, 7, 9\}$
- We have $A - B = \phi$, which means that all the elements of set A are common with elements of set B .
 Also, $B - A = \phi$, which means that all the elements of set B are common with elements of set A .
 So, $A = B$.
- Let $A = \{0, 1\}$ and $B = \{1, 2\}$
 $\therefore A \cup B = \{0, 1, 2\}$
 $P(A) = \{\phi, \{0\}, \{1\}, \{0, 1\}\}$
 $P(B) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$
 $P(A \cup B) = \{\phi, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{1, 2\}, \{0, 2\}, \{0, 1, 2\}\}$
 $P(A) \cup P(B) = \{\phi, \{0\}, \{1\}, \{0, 1\}, \{2\}, \{1, 2\}\}$
 $\therefore P(A) \cup P(B) \neq P(A \cup B)$
- We have $A \cap B = B \cap X = \phi$ and $A \cup X = B \cup X$
 It can be seen that
 $A = A \cap (A \cup X) = A \cap (B \cup X)$ [Distributive law]
 $= (A \cap B) \cup (A \cap X)$ [Distributive law]
 $= (A \cap B) \cup \phi$ [Distributive law]
 $= A \cap B$... (1)
 Now, $B = B \cap (B \cup X)$ [Distributive law]
 $= B \cap (A \cup X)$ [Distributive law]
 $= (B \cap A) \cup (B \cap X)$ [Distributive law]
 $= (B \cap A) \cup \phi$ [Distributive law]
 $= B \cap A$ [Distributive law]
 $= A \cap B$... (2)
 Hence, from (1) and (2), we obtain $A = B$.

10.



- $A \cap B = [0, 1)$
- $A \cup B = [-4, 3)$
- $A - B = [-4, 0)$

- (d) $B - A = [1, 3)$
 (e) $(A \cup B)' = (-\infty, -4) \cup [3, \infty)$
 (f) $(A \cap B)' = (-\infty, 0) \cup [1, \infty)$

11. Let

U be the set of all students who took part in the survey,
 T be the set of students who like tea, and
 C be the set of students who like coffee.

According to the question,

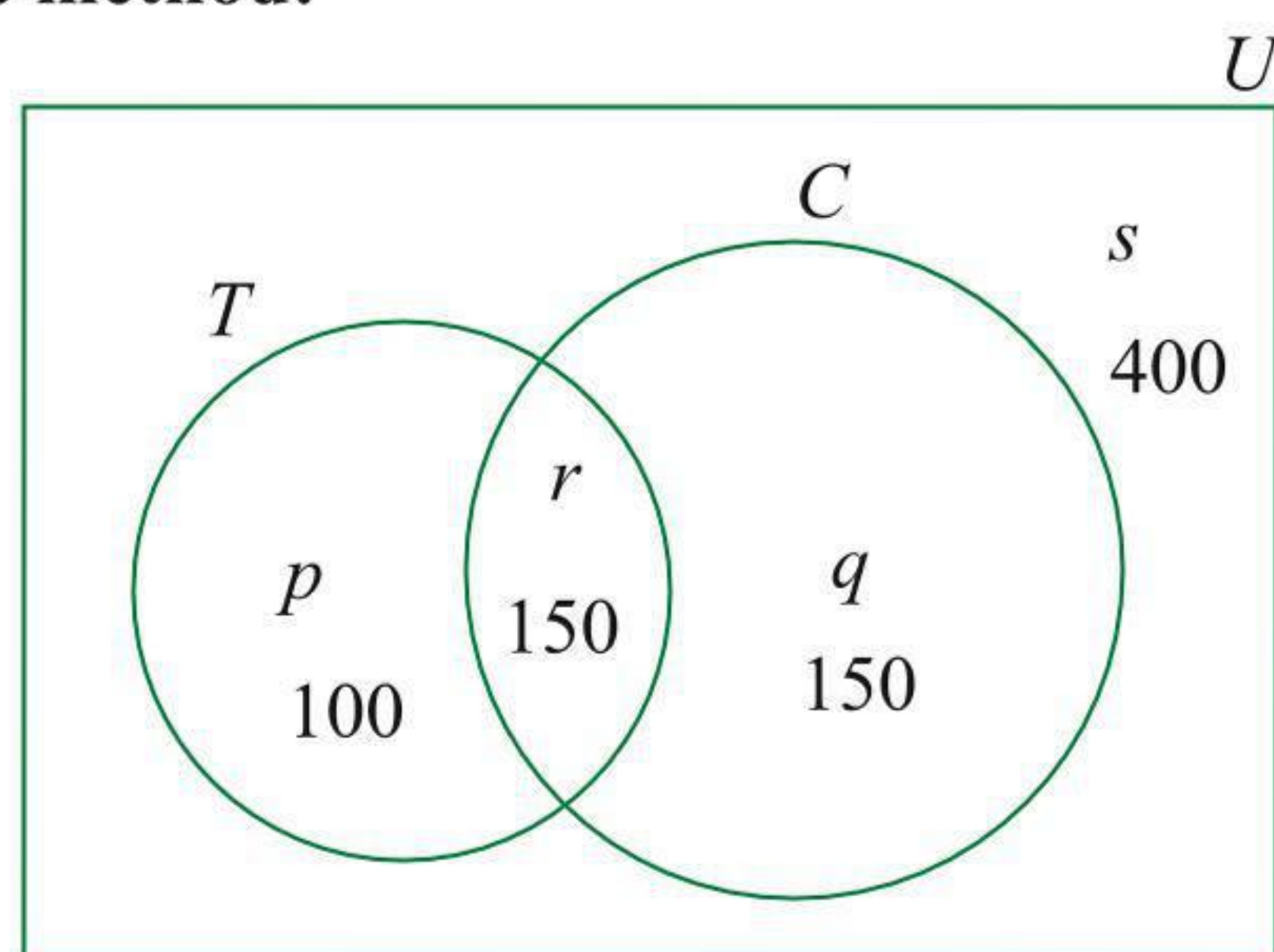
$$n(U) = 800, n(T) = 250, n(C) = 300, n(T \cap C) = 150$$

To find the number of students taking neither tea nor coffee i.e., we have to find $n(T' \cap C')$

$$\begin{aligned} n(T' \cap C') &= n((T \cup C)') \\ &= n(U) - n(T \cup C) \\ &= n(U) - [n(T) + n(C) - n(T \cap C)] \\ &= 800 - [250 + 300 - 150] = 400 \end{aligned}$$

Hence, 400 students like neither tea nor coffee.

Alternative method:



We can use Venn diagram as shown in the figure.

In the above figure regions are

- $r \rightarrow$ students liking tea and coffee both $= n(T \cap C)$
 $p \rightarrow$ students liking tea only $= n(T) - n(T \cap C)$
 $q \rightarrow$ students liking coffee only $= n(C) - n(T \cap C)$
 $s \rightarrow$ students liking neither tea nor coffee $= n(T' \cap C')$

Here, $r = 150$ (given)

Now, $p + r =$ students liking tea $= 250$ (given)

$$\Rightarrow p = 250 - 150 = 100$$

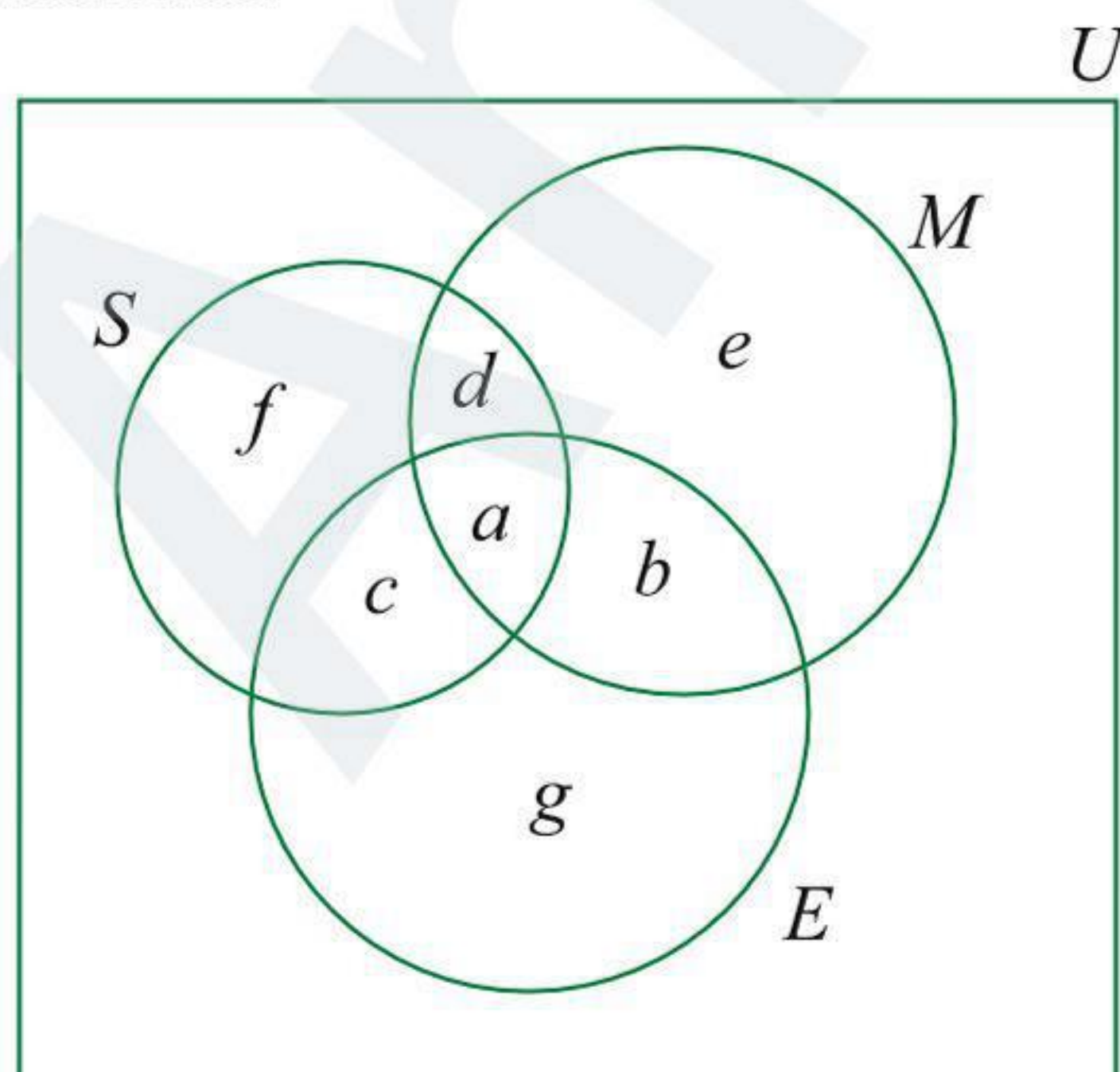
$q + r =$ students liking coffee $= 300$ (given)

$$\Rightarrow q = 300 - 150 = 150$$

$$\begin{aligned} p + q + r &= \text{students liking at least one of tea and coffee} \\ &= 100 + 150 + 150 = 400 \end{aligned}$$

$$s = \text{students liking neither tea nor coffee} = 800 - 400 = 400$$

12. Let M be the set of students who passed in Mathematics, E be the set of students who passed in English and S be the set of students who passed in Science.



Given

$$\begin{aligned} n(U) &= 100, \\ n(E) &= 15, n(M) = 12, n(S) = 8, \\ n(E \cap M) &= 6, n(M \cap S) = 7, \\ n(E \cap S) &= 4, \text{ and } n(E \cap M \cap S) = 4, \end{aligned}$$

From the Venn diagram, we have

$$\begin{aligned} a &= n(E \cap M \cap S) = 4 \text{ and } a + d = n(M \cap S) = 7 \Rightarrow d = 3 \\ a + b &= n(M \cap E) = 6 \Rightarrow b = 2 \\ a + c &= n(S \cap E) = 4 \Rightarrow c = 0 \\ a + b + d + e &= n(M) = 12 \\ \Rightarrow 4 + 2 + 3 + e &= 12 \Rightarrow e = 3 \\ a + b + c + g &= 15 \\ \Rightarrow 4 + 2 + 0 + g &= n(E) = 15 \Rightarrow g = 9 \\ a + c + d + f &= n(S) = 8 \\ \Rightarrow 4 + 0 + 3 + f &= 8 \Rightarrow f = 1 \end{aligned}$$

Number of students passed in

- (a) English and Mathematics but not in Science $= b = 2$
 (b) Mathematics and Science but not in English $= d = 3$
 (c) Mathematics only $= e = 3$
 (d) more than one subject $= a + b + c + d = 4 + 2 + 0 + 3 = 9$

Exercise 1.2

$$1. -3 < 2x - 1 < 19$$

$$\text{or } -2 < 2x < 20$$

$$\text{or } -1 < x < 10 \quad (1)$$

$$-1 \leq \frac{2x+3}{5} \leq 3$$

$$\text{or } -5 \leq 2x + 3 \leq 15$$

$$\text{or } -8 \leq 2x \leq 12$$

$$\text{or } -4 \leq x \leq 6 \quad (2)$$

Now from (1) and (2), common values are $(-1, 6]$

2. Let the length of the shortest side of the triangle be x cm.

Then, length of the longest side $= 3x$ cm

Length of the third side $= (3x - 2)$ cm

Since the perimeter of the triangle is at least 61 cm.

$$x + 3x + (3x - 2) \geq 61$$

$$\Rightarrow 7x - 2 \geq 61$$

$$\Rightarrow 7x \geq 61 + 2$$

$$\Rightarrow 7x \geq 63$$

$$\Rightarrow x \geq 9$$

Thus, the minimum length of the shortest side is 9 cm.

3. (a) $-5 \leq x \leq -1$

$$\Rightarrow x^2 \in [1, 5]$$

- (b) $3 < x < 6$

$$\Rightarrow x^2 \in (9, 36)$$

- (c) $-2 < x \leq 3$

$$\Rightarrow -2 < x < 0 \text{ or } 0 \leq x \leq 3$$

$$\text{For } -2 < x < 0, x^2 \in (0, 4)$$

$$\text{and for } 0 \leq x \leq 3, x^2 \in [0, 9] \quad (1)$$

$$\text{From (1) and (2), } x^2 \in [0, 9]$$

Alternatively, $x \in (-2, 3]$, now least value of x^2 is 0 which occurs when $x = 0$

Greatest value of x^2 is 9 for $x = 3$

$$\Rightarrow x^2 \in [0, 9]$$

- (d) $(-3, \infty)$

Here least value of x^2 is 0 for $x = 0$, and when x goes up to infinity, x^2 also goes up to infinity

$$\Rightarrow x^2 \in [0, \infty)$$

- (e) $(-\infty, 4)$

Here least value of x^2 is 0 for $x = 0$ and $x^2 \rightarrow \infty$, when $x \rightarrow -\infty$

$$\text{Hence, } x^2 \in [0, \infty)$$

4. (a) $2 < x < 5$

$$\Rightarrow \frac{1}{2} > \frac{1}{x} > \frac{1}{5}$$

$$\Rightarrow \frac{1}{x} \in \left(\frac{1}{5}, \frac{1}{2}\right)$$

(b) $-5 \leq x < -1$

$$\Rightarrow -\frac{1}{5} \geq \frac{1}{x} > -1$$

$$\Rightarrow \frac{1}{x} \in \left(-1, -\frac{1}{5}\right]$$

(c) $x > 3$ or $3 < x < \infty$

$$\Rightarrow \frac{1}{3} > \frac{1}{x} > 0$$

$$\Rightarrow \frac{1}{x} \in \left(0, \frac{1}{3}\right)$$

(d) $x \leq -2$ or $-\infty < x \leq -2$

$$\Rightarrow 0 > \frac{1}{x} \geq -\frac{1}{2}$$

$$\Rightarrow \frac{1}{x} \in \left[-\frac{1}{2}, 0\right)$$

(e) $x \in [-3, 4]$ or $-3 \leq x \leq 4$

Now for $\frac{1}{x}$ to get defined, we must have $-3 \leq x < 0$ or $0 < x \leq 4$

$$\Rightarrow -\frac{1}{3} \geq \frac{1}{x} > -\infty \text{ or } \frac{1}{4} \leq \frac{1}{x} < \infty$$

$$\Rightarrow \frac{1}{x} \in \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{1}{4}, \infty\right)$$

5. (a) $4x^2 + 28x + 41 = (2x + 7)^2 - 8$

Now, $(2x + 7)^2 \geq 0 \forall x \in R$.

$$\Rightarrow (2x + 7)^2 - 8 \geq -8 \forall x \in R$$

So, the values of the expression are $[-8, \infty)$.

For $x \in [-3, 2]$

$$-3 \leq x \leq 2$$

$$\Rightarrow -6 \leq 2x \leq 4$$

$$\Rightarrow 1 \leq 2x + 7 \leq 11$$

$$\Rightarrow 1 \leq (2x + 7)^2 \leq 121$$

$$\Rightarrow -7 \leq (2x + 7)^2 - 8 \leq 113$$

Thus, for $x \in [-3, 2]$, the values of the expression are $[-7, 113]$.

(b) $1 + 6x - x^2 = 10 - (x - 3)^2$

Now, $(x - 3)^2 \geq 0 \forall x \in R$.

$$\Rightarrow -(x - 3)^2 \leq 0 \forall x \in R$$

$$\Rightarrow 10 - (x - 3)^2 \leq 10 \forall x \in R$$

So, the values of the expression are $(-\infty, 10]$.

For $x \in [-3, 2]$

$$-3 \leq x \leq 2$$

$$\Rightarrow -6 \leq x - 3 \leq -1$$

$$\Rightarrow 1 \leq (x - 3)^2 \leq 36$$

$$\Rightarrow -36 \leq -(x - 3)^2 \leq -1$$

$$\Rightarrow -26 \leq 10 - (x - 3)^2 \leq 9$$

Thus, for $x \in [-3, 2]$ values of the expression are $[-26, 9]$.

6. $\frac{2+x^2}{4-x^2} = \frac{6-(4-x^2)}{4-x^2} = \frac{6}{4-x^2} - 1$

Now, $x^2 \geq 0$

$$\therefore -x^2 \leq 0$$

$$\Rightarrow 4 - x^2 \leq 4$$

$$\Rightarrow \frac{4-x^2}{6} \leq \frac{2}{3}$$

$$\Rightarrow -\infty < \frac{4-x^2}{6} < 0 \text{ or } 0 < \frac{4-x^2}{6} \leq \frac{2}{3}$$

$$\Rightarrow 0 > \frac{6}{4-x^2} > -\infty \text{ or } \infty > \frac{6}{4-x^2} \geq \frac{3}{2}$$

$$\Rightarrow \frac{6}{4-x^2} \in (-\infty, 0) \cup \left[\frac{3}{2}, \infty\right)$$

$$\Rightarrow \left(\frac{6}{4-x^2} - 1\right) \in (-\infty, -1) \cup \left[\frac{1}{2}, \infty\right)$$

7. $\frac{\sqrt{x-1}}{x-2} < 0$ is meaningful for $x-1 \geq 0$ or $x \geq 1$ (1)

Also $\frac{\sqrt{x-1}}{x-2} < 0$

$$\Rightarrow x-2 < 0 \text{ or } x < 2$$
 (2)

$$\therefore 1 \leq x < 2$$

8. $\sqrt{x-2} \leq 3$ is meaningful for $x-2 \geq 0$ or $x \geq 2$ (1)

Also $\sqrt{x-2} \leq 3$

$$\Rightarrow x-2 \leq 9$$

$$\Rightarrow x \leq 11$$
 (2)

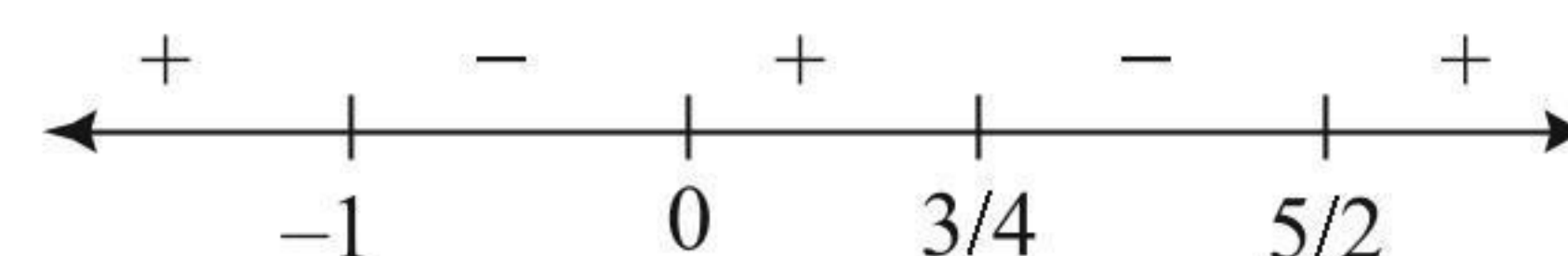
From (1) and (2), we have $2 \leq x \leq 11$

Exercise 1.3

1. $\frac{x(3-4x)(x+1)}{(2x-5)} < 0$

or $-\frac{x(4x-3)(x+1)}{(2x-5)} < 0$

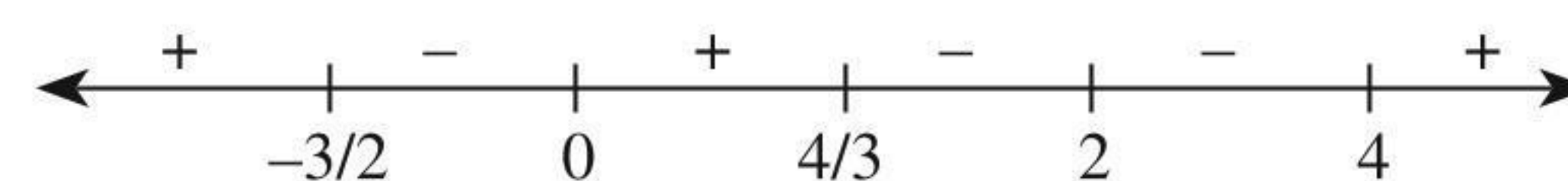
or $\frac{x(4x-3)(x+1)}{(2x-5)} > 0$



$$\Rightarrow x \in (-\infty, -1) \cup (0, 3/4) \cup (5/2, \infty)$$

2. $\frac{(2x+3)(4-3x)^3(x-4)}{(x-2)^2 x^5} \leq 0$

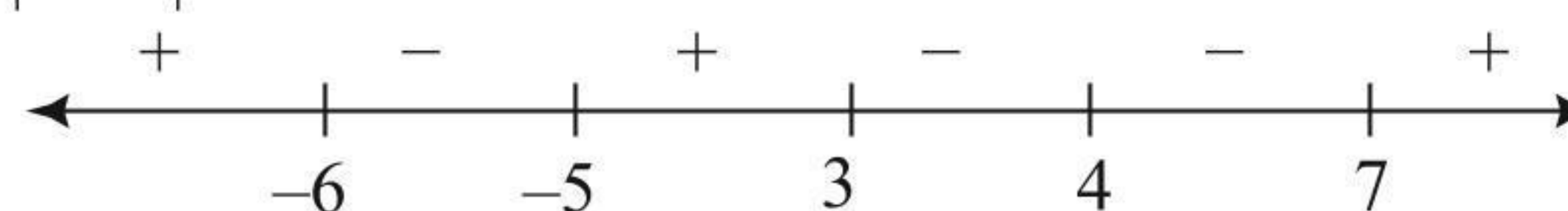
or $\frac{(2x+3)(3x-4)^3(x-4)}{(x-2)^2 x^5} \geq 0$



From the sign scheme, we have

$$x \in (-\infty, -3/2) \cup (0, 4/3] \cup [4, \infty)$$

3. Sign of expression does not change while crossing $x = 4$ as we have $|x - 4|$.



From the sign scheme, we have $x \in (-6, -5] \cup [3, 4) \cup (4, 7]$

$$4. \frac{5x+1}{(x+1)^2} - 1 < 0$$

or $\frac{5x+1-(x+1)^2}{(x+1)^2} < 0$

or $\frac{-x^2+3x}{(x+1)^2} < 0$

or $\frac{x(x-3)}{(x+1)^2} > 0$

$$\Rightarrow x < 0 \text{ or } x > 3, x \neq -1$$

$$5. \frac{x+2}{x^2+1} > \frac{1}{2}$$

or $\frac{x+2}{x^2+1} - \frac{1}{2} > 0$

or $\frac{-x^2-1+2x+4}{2(x^2+1)} > 0$

or $\frac{3+2x-x^2}{2(x^2+1)} > 0$

or $x^2-2x-3 < 0$

or $(x-3)(x+1) < 0$

$$\Rightarrow -1 < x < 3$$

$$6. 4^{-x+0.5} - 7 \cdot 2^{-x} < 4 \quad (1)$$

$$\text{Let } 2^{-x} = t$$

$$\therefore t > 0 \quad (2)$$

$$\text{Now (1) reduces to } 2t^2 - 7t < 4$$

$$\text{or } 2t^2 - 7t - 4 < 0$$

$$\text{or } (2t+1)(t-4) < 0$$

$$\text{or } -\frac{1}{2} < t < 4 \quad (3)$$

From (2) and (3), we have

$$0 < t < 4$$

$$\Rightarrow 0 < 2^{-x} < 4$$

$$\Rightarrow -\infty < -x < 2$$

$$\Rightarrow -2 < x < \infty$$

$$\Rightarrow x \in (-2, \infty)$$

$$7. \frac{x^4}{(x-2)^2} > 0 \text{ is always true except for } x = 0, 2.$$

$$\text{Hence, } x \in R - \{0, 2\}$$

$$8. \frac{6x^2-5x-3}{x^2-2x+6} \leq 4$$

$$\text{or } \frac{6x^2-5x-3}{x^2-2x+6} - 4 \leq 0$$

$$\text{or } \frac{2x^2+3x-27}{(x-1)^2+5} \leq 0$$

$$\text{or } 2x^2+3x-27 \leq 0$$

$$\text{or } (2x+9)(x-3) \leq 0$$

$$\text{or } -\frac{9}{2} \leq x \leq 3$$

$$9. \frac{(x+2)(x^2-2x+1)}{-x^2+3x-4} \geq 0$$

$$\text{or } \frac{(x+2)(x-1)^2}{x^2-3x+4} \leq 0$$

$$\text{Now } x^2-3x+4 > 0 \forall x \in R.$$

$$\therefore (x+2)(x-1)^2 \leq 0$$

$$\Rightarrow x \in (-\infty, -2] \cup \{1\}$$

$$10. \sqrt{x+2} \geq x$$

$$\Rightarrow x+2 \geq x^2 \text{ for } x \geq 0$$

$$\Rightarrow x^2-x-2 \leq 0$$

$$\Rightarrow (x-2)(x+1) \leq 0$$

$$\Rightarrow x \in [0, 2]$$

$$x < 0 \text{ always valid but } x+2 \geq 0$$

$$\Rightarrow x \geq -2$$

$$x \in [-2, 2]$$

$$11. \text{ Since } \sqrt{2+x-x^2} \text{ is defined only if}$$

$$2+x-x^2 \geq 0$$

$$\text{or } (x-2)(x+1) \leq 0$$

$$\text{or } -1 \leq x \leq 2.$$

For these values of x , R.H.S. < 0 ; hence, $x \in [-1, 2]$

$$12. \sqrt{-x^2+10x-16} < x-2$$

$$\text{We must have } -x^2+10x-16 \geq 0$$

$$\text{or } x^2-10x+16 \leq 0$$

$$\Rightarrow 2 \leq x \leq 8 \quad (1)$$

$$\text{Also } -x^2+10x-16 < x^2-4x+4$$

$$\text{or } 2x^2-14x+20 > 0$$

$$\text{or } x^2-7x+10 > 0$$

$$\Rightarrow x > 5 \text{ or } x < 2 \quad (2)$$

From (1) and (2), $5 < x \leq 8 \Rightarrow x = 6, 7, 8$

$$13. f(x) = y = \frac{1-x^2}{x^2+3}$$

$$\text{or } yx^2+3y=1-x^2$$

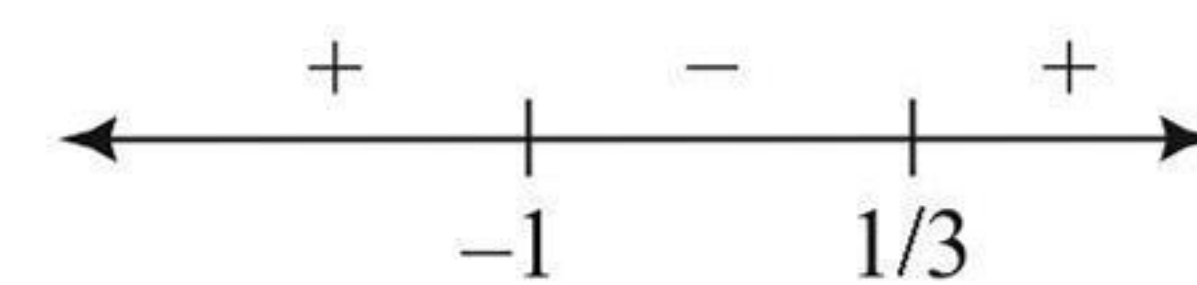
$$\text{or } x^2 = \frac{1-3y}{y+1}$$

$$\text{Now } x^2 \geq 0$$

$$\Rightarrow \frac{1-3y}{y+1} \geq 0$$

$$\Rightarrow \frac{3y-1}{y+1} \leq 0$$

$$\text{Hence, } y \in (-1, 1/3]$$



Exercise 1.4

1. (a) is not always true: for $a = -4$ and $b = 3$, given statement is not true

(b) is not always true: for $a = -3$ and $b = 2$, given statement is not true

(c) is not always true: for $a = -5$ and $b = 3$, given statement is not true

$$2. x^2 - |x| - 2 = 0$$

$$\Rightarrow |x|^2 - |x| - 2 = 0$$

$$\Rightarrow (|x| - 2)(|x| + 1) = 0$$

$$\Rightarrow |x| = 2 \Rightarrow x = \pm 2.$$

$$x^2 - 2|x| + 3 = 0$$

Thus, $|x|^2 - 2|x| + 3 = 0$ has no roots as discriminant is negative.

$$x^2 - 3|x| + 2 = 0$$

$$\Rightarrow |x|^2 - 3|x| + 2 = 0$$

$$\Rightarrow (|x| - 2)(|x| - 1) = 0$$

$$\Rightarrow |x| = 1, 2 \Rightarrow x = \pm 1, \pm 2.$$

$$x^2 + 3|x| + 2 = 0$$

$$\Rightarrow |x|^2 + 3|x| + 2 = 0$$

$$\Rightarrow (|x| + 2)(|x| + 1) = 0$$

$$\Rightarrow |x| = -1, -2 \text{ (not possible)}$$

Thus, the equation (c) has maximum real roots.

3. **Case I:** Let $x \geq 3$

$$\therefore \begin{cases} x + 2y = 6 \\ x - 3 = y \end{cases}$$

$$\Rightarrow 3x = 12$$

$$\Rightarrow x = 4, y = 1$$

Case II: Let $x < 3$

$$\begin{cases} x + 2y = 6 \\ 3 - x = y \end{cases}$$

$$\Rightarrow -x = 0 \text{ or } x = 0, y = 3$$

Hence, the solutions are (0, 3) and (4, 1)

$$4. |x - 2| = \begin{cases} x - 2, & x \geq 2 \\ 2 - x, & x < 2 \end{cases}$$

$$\Rightarrow |x - 2| - (x - 2) = \begin{cases} 0, & x \geq 2 \\ 4 - 2x, & x < 2 \end{cases}$$

Therefore, given expression is defined for $(-\infty, 2)$

$$5. f(x) = x + \sqrt{x^2} = x + |x|$$

$$= \begin{cases} x + x, & x \geq 0 \\ x - x, & x < 0 \end{cases}$$

$$= \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Hence, $f(x)$ takes values $[0, \infty)$

$$6. \left| \frac{x+2}{x-1} \right| = 2$$

$$\text{or } \frac{x+2}{x-1} = \pm 2$$

$$\Rightarrow x + 2 = 2x - 2 \text{ or } x + 2 = 2 - 2x$$

$$\Rightarrow x = 4 \text{ or } x = 0$$

Here, both $x = 0$ and 4 satisfy the original equation.

$$7. 2x - 1 = x \text{ if } x \geq 0$$

For which $x = 1$

Also $2x - 1 = -x$ if $x < 0$

For which $x = 1/3$ which is not the solution as $x < 0$

Hence, the only solution is $x = 1/3$

$$8. 2^x + 1 > 0, x \in R$$

Therefore, given equation is $|2^x - 1| + 2^x + 1 = 2$

$$\text{or } |2^x - 1| = 1 - 2^x$$

$$\text{or } 2^x - 1 \leq 0$$

$$\text{or } 2^x \leq 1$$

$$\text{or } x \leq 0$$

$$9. |x^2 + 4x + 3| = x + 1$$

$$\text{or } |(x+1)(x+3)| = x + 1$$

$$|(x+1)(x+3)| = (x+1)(x+3), \text{ when } (x+1)(x+3) \geq 0$$

$$\text{or } x \leq -3 \text{ or } x \geq -1$$

Hence, given equation reduces to $(x+1)(x+3) = x+1$

$$\Rightarrow x = -1 \text{ (} x = -2 \text{ is rejected as } x \leq -3 \text{ or } x \geq -1)$$

$$|(x+1)(x+3)| = -(x+1)(x+3), \text{ when } (x+1)(x+3) < 0$$

$$\text{or } -3 < x < -1$$

Hence, given equation reduces to $-(x+1)(x+3) = x+1$

$$\Rightarrow x = -4 \text{ which is rejected as } -3 < x < -1$$

$$10. |4 - |x - 1|| = 3$$

$$\Rightarrow |x - 1| - 4 = \pm 3$$

$$\Rightarrow |x - 1| = 7, |x - 1| = 1$$

$$\Rightarrow x = -6, 0, 2, 8$$

$$11. \text{ Let } f(x) = |x - 1| - |2x - 5|$$

A. x	B. $f(x)$	C. x for which $f(x) = 2x$	D. $A \cap C$
$x < 1$	$1 - x - (5 - 2x)$	$x - 4 = 2x$ or $x = -4$	$x = -4$
$1 \leq x \leq 5/2$	$x - 1 - (5 - 2x)$	$3x - 6 = 2x$ or $x = 6$	No such x exists
$x > 5/2$	$x - 1 - (2x - 5)$	$4 - x = 2x$ or $x = 4/3$	No such x exists

Hence, solutions set is $\{-4\}$

Exercise 1.5

$$1. |x^2 - 7| \leq 9$$

$$\text{or } -9 \leq x^2 - 7 \leq 9$$

$$\text{or } -2 \leq x^2 \leq 16$$

$$\text{or } x^2 \leq 16$$

$$\Rightarrow x \in [-4, 4]$$

$$2. ||x - 2| - 3| < 5$$

$$\text{or } -5 < |x - 2| - 3 < 5$$

$$\text{or } -2 < |x - 2| < 8$$

$$\text{or } |x - 2| < 8$$

$$\text{or } -8 < x - 2 < 8$$

$$\text{or } -6 < x < 10$$

$$3. |x + y| = |x| + |y|$$

$$\Rightarrow x \text{ and } y \text{ have same sign or at least one of } x \text{ and } y \text{ is zero.}$$

$$\Rightarrow (x, y) \text{ lies in 1st or 3rd quadrant or any of the } x\text{-axis or } y\text{-axis}$$

$$|x + y| < |x| + |y|$$

$$\Rightarrow x \text{ and } y \text{ have opposite sign}$$

$$\Rightarrow (x, y) \text{ lies in 2nd or 4th quadrant}$$

$$|x - y| = |x| + |y|$$

$$\Rightarrow x \text{ and } -y \text{ have same sign}$$

$$\Rightarrow x \text{ and } y \text{ have opposite sign}$$

$$\Rightarrow (x, y) \text{ lies in 2nd or 4th quadrant at least one of } x \text{ and } y \text{ is zero.}$$

$$4. \text{ We must have } 5 - |2x - 3| \geq 0$$

$$\text{or } |2x - 3| \leq 5$$

$$\text{or } -5 \leq 2x - 3 \leq 5$$

$$\Rightarrow x \in [-1, 4]$$

$$5. |x^2 - x - 2| + |x + 6| = |x^2 - 2x - 8|$$

$$\text{or } |x^2 - x - 2| + |x + 6| = |(x^2 - x - 2) - (x + 6)|$$

$$\text{or } |x^2 - x - 2| + |-x - 6| = |(x^2 - x - 2) + (-x - 6)|$$

$$\text{or } (x^2 - x - 2)(-x - 6) \geq 0$$

$$\text{or } (x^2 - x - 2)(x + 6) \leq 0$$

$$\text{or } (x - 2)(x + 1)(x + 6) \leq 0$$

$$\text{or } x \in (-\infty, -6] \cup [-1, 2]$$

$$\begin{aligned}
 6. \quad & |x^2 + x - 6| < 6 \\
 \Rightarrow & -6 < x^2 + x - 6 < 6 \\
 \text{Now } & -6 < x^2 + x - 6 \\
 \text{or } & x^2 + x > 0 \\
 \text{or } & x(x+1) > 0 \\
 \Rightarrow & x \in (-\infty, -1) \cup (0, \infty) \quad (1) \\
 & x^2 + x - 6 < 6
 \end{aligned}$$

$$\begin{aligned}
 \text{or } & x^2 + x - 12 < 0 \\
 \text{or } & (x+4)(x-3) < 0 \\
 \Rightarrow & x \in (-4, 3) \quad (2)
 \end{aligned}$$

From (1) and (2), common values are $x \in (-4, -1) \cup (0, 3)$

$$\begin{aligned}
 7. \quad & |x| + \left| \frac{4-x^2}{x} \right| = \left| \frac{4}{x} \right| \\
 \Rightarrow & |x| + \left| \frac{4}{x} - x \right| = \left| x + \left(\frac{4}{x} - x \right) \right| \\
 \therefore & x \left(\frac{4}{x} - x \right) \geq 0, x \neq 0 \\
 \Rightarrow & x^2 - 4 \leq 0 \\
 \Rightarrow & x \in [-2, 2] - \{0\}
 \end{aligned}$$

$$\begin{aligned}
 8. \quad & \frac{1}{|x|-3} - \frac{1}{2} < 0 \\
 \text{or } & \frac{2-|x|+3}{|x|-3} < 0 \\
 \text{or } & \frac{|x|-5}{|x|-3} > 0 \\
 \Rightarrow & |x| > 5 \text{ or } |x| < 3 \\
 \Rightarrow & x > 5 \text{ or } x < -5 \text{ or } -3 < x < 3 \\
 \Rightarrow & x \in (-\infty, -5) \cup (-3, 3) \cup (5, \infty)
 \end{aligned}$$

$$\begin{aligned}
 9. \quad & |x - \pi| + |x^2 - \pi^2| \leq 0 \\
 \text{or } & |x - \pi|(1 + |x + \pi|) \leq 0 \\
 \text{or } & |x - \pi| = 0 \\
 \text{or } & x = \pi
 \end{aligned}$$

$$\begin{aligned}
 10. \quad & \left| 1 - \frac{|x|}{1+|x|} \right| \geq \frac{1}{2} \\
 \text{or } & \left| \frac{1}{1+|x|} \right| \geq \frac{1}{2} \\
 \text{or } & 1 + |x| \leq 2 \\
 \text{or } & |x| \leq 2 \\
 & -1 \leq x \leq 1
 \end{aligned}$$

$$\begin{aligned}
 11. \quad & \frac{|x-1|}{x+2} > 1 \\
 \Rightarrow & \frac{|x-1| - (x+2)}{x+2} > 0
 \end{aligned}$$

Case I: when $x \geq 1$

$$\frac{(x-1) - (x+2)}{(x+2)} > 0$$

Hence, $x < -2$, which is not possible as $x \geq 1$.

Case II: when $x < 1$

$$\Rightarrow \frac{(1-x) - (x+2)}{(x+2)} > 0$$

$$\text{or } \frac{-1-2x}{(x+2)} > 0$$

$$\text{or } \frac{2x+1}{(x+2)} < 0$$

$$\Rightarrow x \in \left(-2, -\frac{1}{2} \right) \quad (\text{as } x < 1)$$

$$\begin{aligned}
 12. \quad & \text{We have } |x^2 - 1| + |x^2 - 4| > 3 \\
 \Rightarrow & |x^2 - 1| + |4 - x^2| > |x^2 - 1 + 4 - x^2| \\
 \Rightarrow & (x^2 - 1)(4 - x^2) < 0 \\
 \Rightarrow & (x^2 - 1)(x^2 - 4) > 0 \\
 \Rightarrow & x^2 < 1 \text{ or } x^2 > 4 \\
 \Rightarrow & -1 < x < 1 \text{ or } x < -2 \text{ or } x > 2
 \end{aligned}$$

Exercises

Single Correct Answer Type

1. (4) Every rectangle, rhombus, square in a plane is a parallelogram but every trapezium is not a parallelogram.

$$\therefore F_1 = F_2 \cup F_3 \cup F_4 \cup F_1$$

2. (3) Since $A \subseteq B$, $A \cup B = B$

$$\text{So, } n(A \cup B) = n(B) = 6$$

3. (3) For $A \cap B$, we have

$$-x = \frac{1}{x}$$

$$\Rightarrow -x^2 = 1$$

$$\Rightarrow x^2 = -1, \text{ which is not possible}$$

$$\therefore A \cap B = \phi$$

4. (2) According to the question,

$$2^m - 2^n = 112$$

$$\Rightarrow 2^n(2^{m-n} - 1) = 2^4 \times 7$$

Comparing exponents on both sides, we get

$$2^n = 2^4 \text{ and } 2^{m-n} - 1 = 7$$

$$\text{Now, } 2^n = 2^4$$

$$\Rightarrow n = 4$$

$$\text{And } 2^{m-n} = 8$$

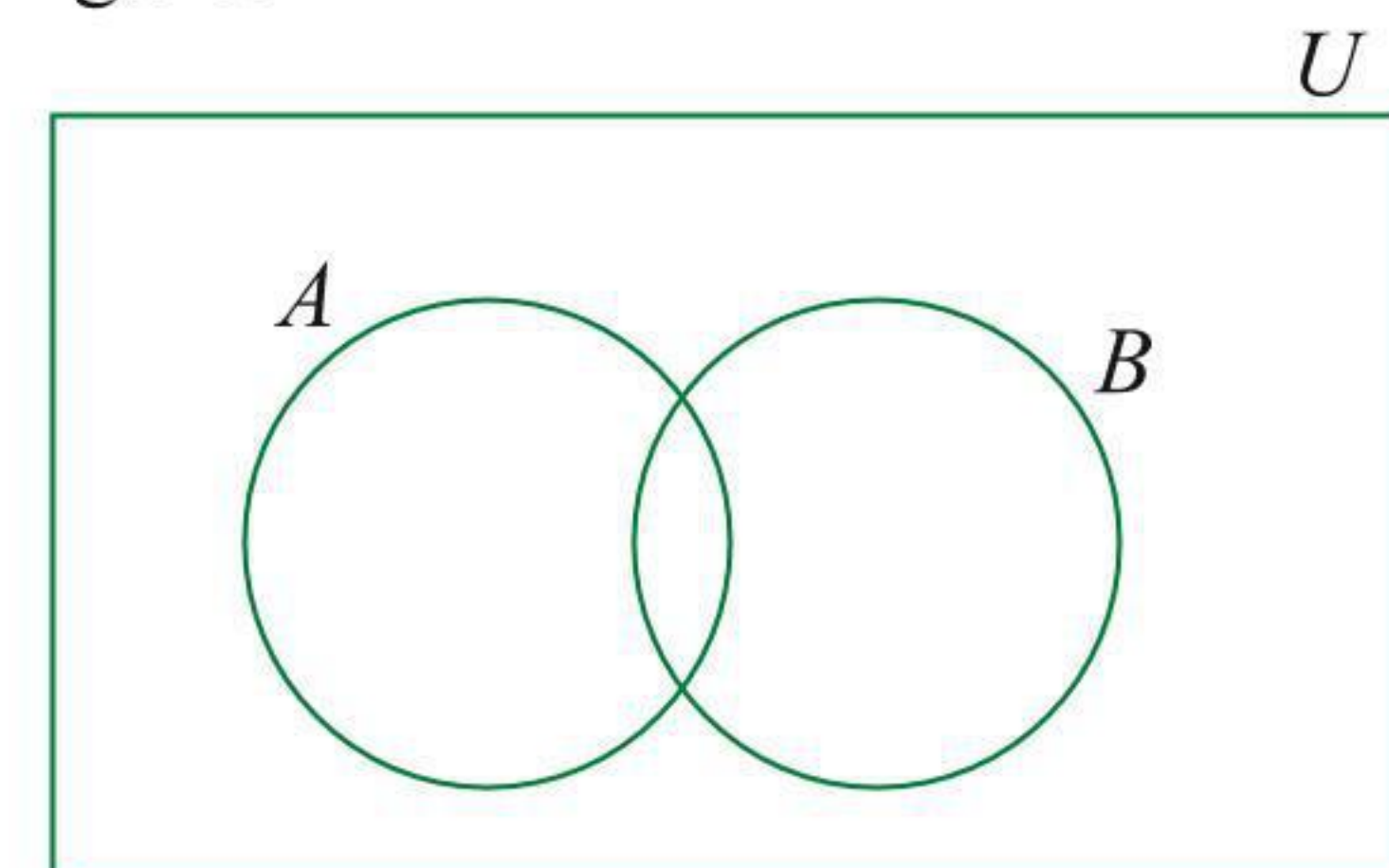
$$\Rightarrow 2^{m-n} = 2^3$$

$$\Rightarrow m - n = 3$$

$$\Rightarrow m - 4 = 3$$

$$\therefore m = 7$$

5. (4) Given that $A \not\subset B$. Then it is not necessary that B is subset of A as shown in figure.



Also, it is not necessary that A and B are disjoint.

Obviously, A and complement of B are not disjoint.

6. (3) $4N = \{4, 8, 12, 16, 20, 24, \dots\}$ and $6N = \{6, 12, 18, 24, \dots\}$

$$\text{Hence, } 4N \cap 6N = \{12, 24, 36, \dots\} = 12N$$

7. (2) $(A \cap B')' \cup (B \cap C) = ((A' \cup (B')') \cup (B \cap C))$

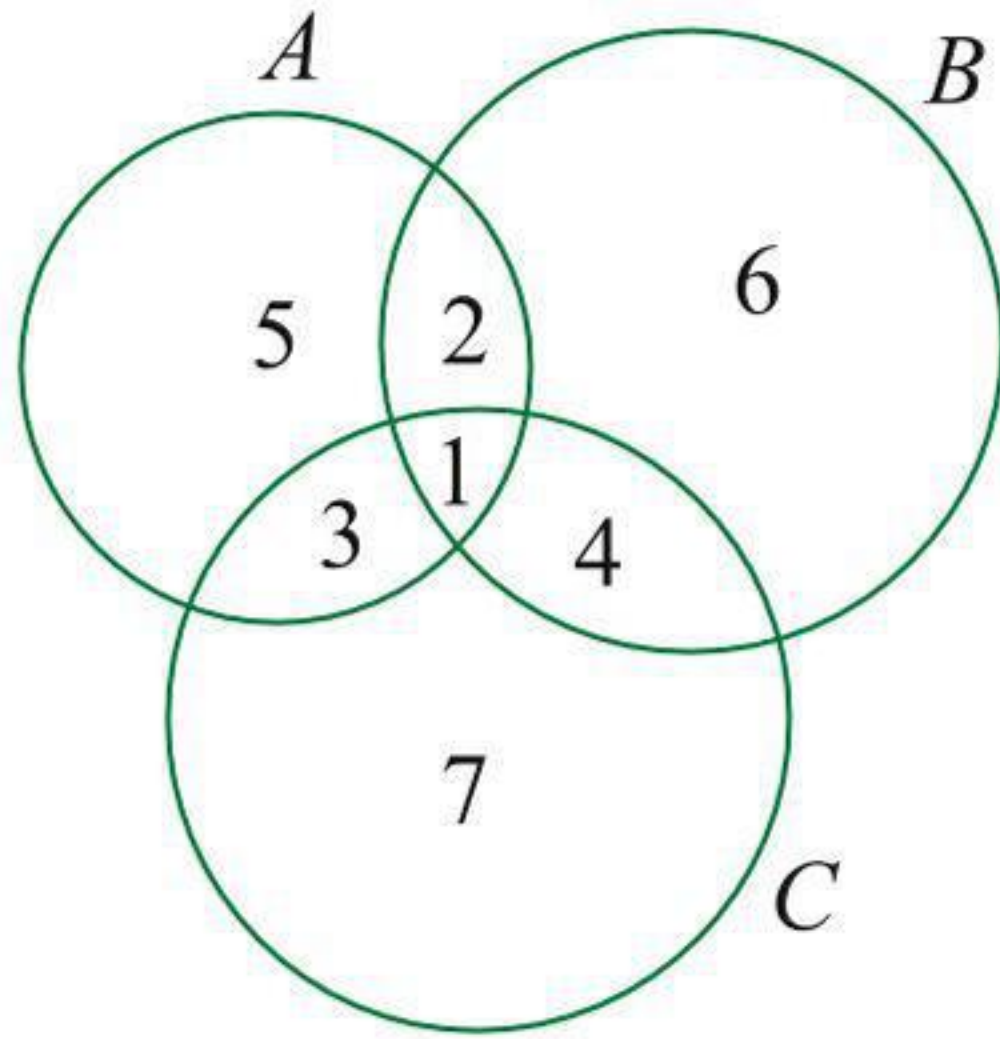
$$= (A' \cup B) \cup (C \cap B)$$

$$= A' \cup (B \cup (C \cap B))$$

$$= A' \cup B$$

$$\begin{aligned}
8. (1) & (A \cup B)' \cup (A' \cap B) \\
&= (A' \cap B') \cup (A' \cap B) \\
&= (A' \cup A') \cap (A' \cup B) \cap (B' \cup A') \cap (B' \cup B) \\
&= A' \cap \{A' \cup (B \cap B')\} \cap U \\
&= A' \cap \{A' \cup \phi\} \cap U \\
&= A' \cap A' \cap U \\
&= A' \cap U = A'
\end{aligned}$$

9. (2)

 $A - B \rightarrow$ Regions 3 and 5. $B - C \rightarrow$ Regions 2 and 6. $C - A \rightarrow$ Regions 4 and 7. $\therefore (A - B) \cup (B - C) \cup (C - A) \rightarrow$ Regions 2, 3, 4, 5, 6, 7 $\therefore [(A - B) \cup (B - C) \cup (C - A)]' \rightarrow$ Region 1, which is $A \cap B \cap C$ 10. (4) Shaded region in the figure is clearly $A - (B \cup C)$.

$$\begin{aligned}
11. (3) & [(A' \cap B' \cap C)] \cup [(B \cap C) \cup (A \cap C)] \\
&= [(A \cup B)' \cap C] \cup [(B \cup A) \cap C] \\
&= [(A \cup B)' \cup (B \cup A)] \cap C \\
&= X \cap C \\
&= C
\end{aligned}$$

12. (2) Let R be the set of families having a radio and T be the set of families having a TV.

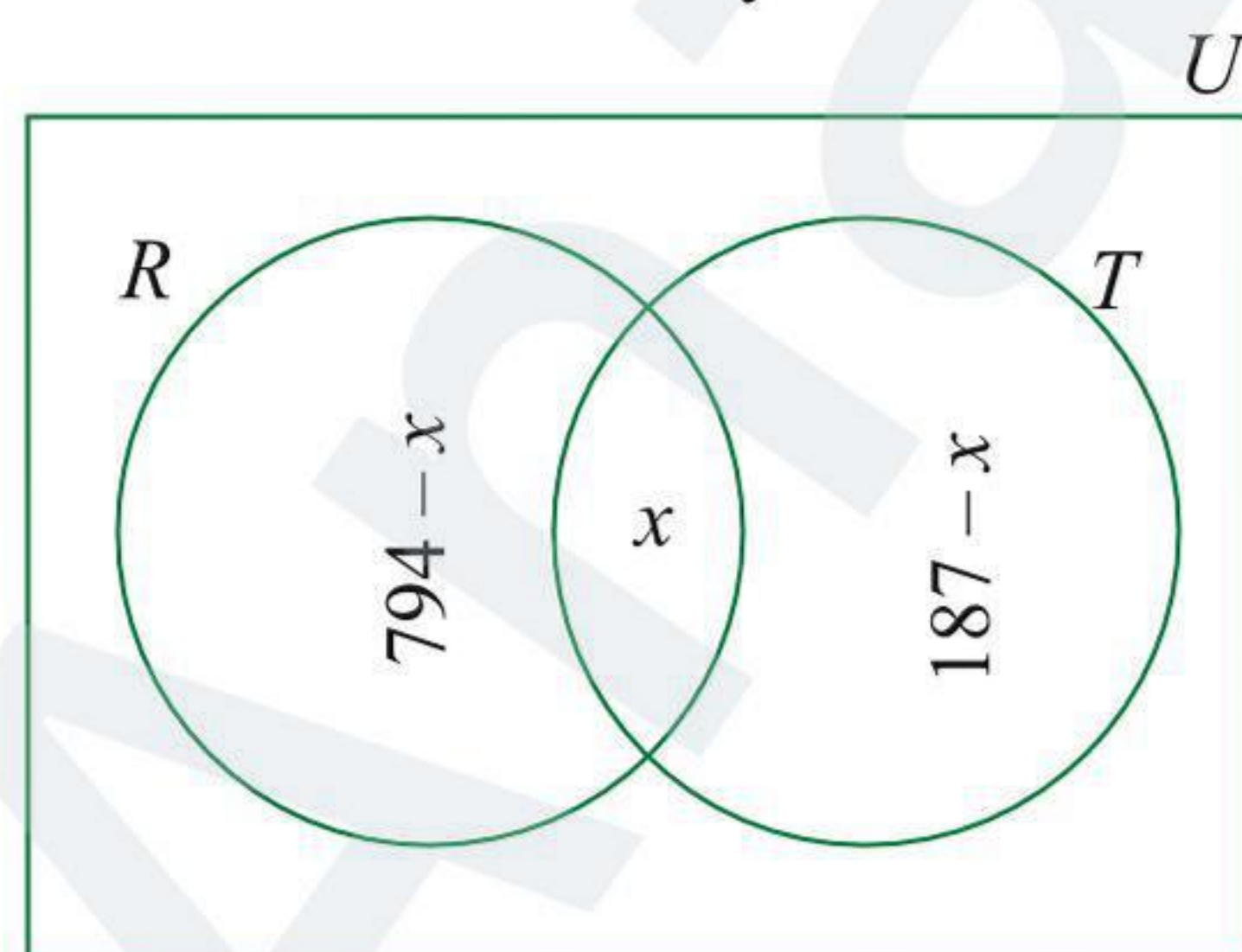
Then

$$n(R \cup T)$$

= Number of families having at least one of the radio and TV

$$= 1003 - 63 = 940$$

$$n(R) = 794 \text{ and } n(T) = 187$$

Let x be the number of families having both a radio and a TV.Then number of families who have only Radio = $794 - x$ Number of families who have only TV = $187 - x$ 

From Venn diagram,

$$794 - x + x + 187 - x = 940$$

$$\Rightarrow 981 - x = 940$$

$$\therefore x = 981 - 940 = 41$$

Hence, the required number of families having both a radio and a TV is 41.

13. (3) Let A be the set of those people who watch news channel and B be the set of those people who watch entertainment channel.

Then

$$n(A) = 63, n(B) = 76, \text{ and } n(A \cap B) = x$$

$$\therefore n(A \cup B) \leq 100$$

$$\Rightarrow n(A) + n(B) - n(A \cap B) \leq 100$$

$$\Rightarrow 63 + 76 - x \leq 100$$

$$\Rightarrow x \geq 39$$

$$\text{Also, } n(A \cap B) \leq \min\{n(A), n(B)\}$$

$$\Rightarrow x \leq 63$$

$$\therefore 39 \leq x \leq 63$$

14. (2) $n(A) = 40\%$ of 10,000 = 4,000

$$n(B) = 20\%$$
 of 10,000 = 2,000

$$n(C) = 10\%$$
 of 10,000 = 1,000

$$n(A \cap B) = 5\%$$
 of 10,000 = 500

$$n(B \cap C) = 3\%$$
 of 10,000 = 300

$$n(C \cap A) = 4\%$$
 of 10,000 = 400

$$n(A \cap B \cap C) = 2\%$$
 of 10,000 = 200

Number of families which buy newspaper A only

$$= n(A \cap B^c \cap C^c)$$

$$= n[A \cap (B \cup C)^c]$$

$$= n(A) - n[A \cap (B \cup C)]$$

$$= n(A) - n[(A \cap B) \cup (A \cap C)]$$

$$= n(A) - [n(A \cap B) + n(A \cap C) - n(A \cap B \cap C)]$$

$$= 4000 - [500 + 400 - 200]$$

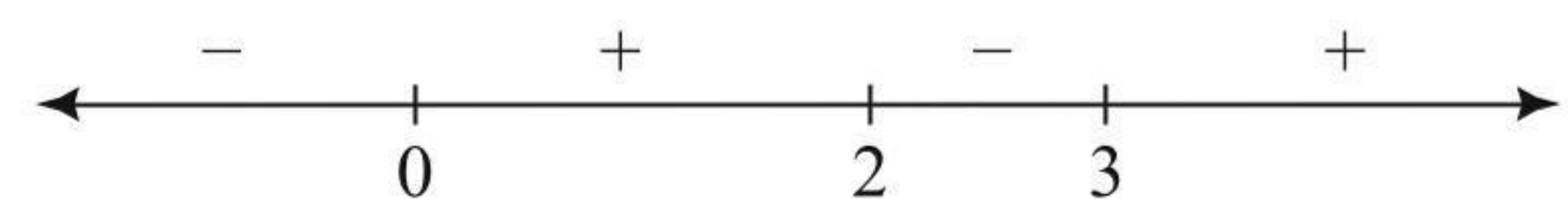
$$= 4000 - 700$$

$$= 3300$$

15. (2) $\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1$

$$\Rightarrow \frac{x^2 + 5x + 6}{x^2 - 5x + 6} - 1 \leq 0$$

$$\Rightarrow \frac{10x}{(x-2)(x-3)} \leq 0$$



$$x \in (-\infty, 0] \cup (2, 3)$$

16. (2) We have $(x+1)^2 > 5x - 1$

$$\Rightarrow x^2 - 3x + 2 > 0$$

$$\Rightarrow (x-1)(x-2) > 0$$

$$\Rightarrow x < 1 \text{ or } x > 2$$

... (i)

And $(x+1)^2 < 7x - 3$

$$\Rightarrow x^2 - 5x + 4 < 0$$

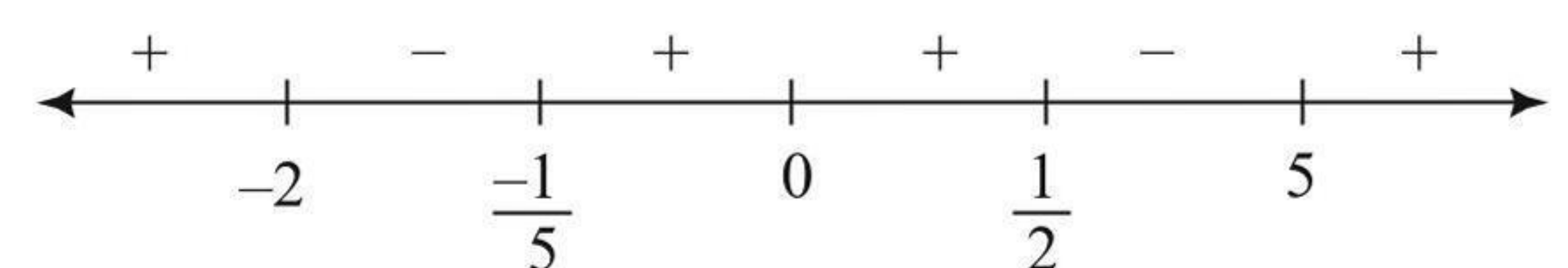
$$\Rightarrow (x-1)(x-4) < 0$$

$$\Rightarrow 1 < x < 4$$

... (ii)

From (i) and (ii), common values are, $x \in (2, 4)$ Therefore, there is only one integral value of x which is 3.

17. (2) We have, $\frac{x^2(x-5)(2x-1)}{(5x+1)(x+2)} < 0$



From the sign scheme, solution is

$$\left(-2, -\frac{1}{5}\right) \cup \left(\frac{1}{2}, 5\right)$$

... (i)

Now $\frac{3x+1}{x(6x^2+x-1)} > 0$

$$\Rightarrow \frac{3x+1}{x(3x-1)(2x+1)} > 0$$

From the sign scheme solution is

$$\left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{3}, 0\right) \cup \left(\frac{1}{3}, \infty\right) \quad \dots(ii)$$

Clearly, from (i) and (ii), option(s) (1) and (3) are correct.

18. (4) We have $x^4 - 29x^2 + 100 \leq 0$

$$\Rightarrow (x^2 - 4)(x^2 - 25) \leq 0$$

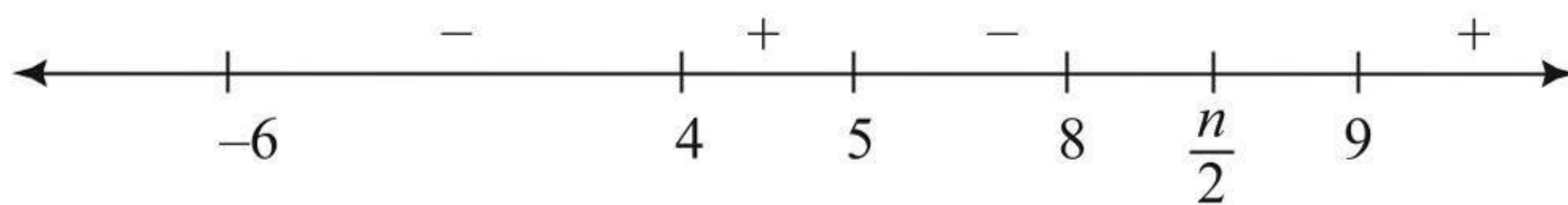
$$\Rightarrow 4 \leq x^2 \leq 25$$

$$\Rightarrow x \in [-5, -2] \cup [2, 5]$$

Integers satisfying above are $\pm 5, \pm 4, \pm 3, \pm 2$.

So, there are eight integers.

19. (4) $(x+6)(x-4)(x-5)(2x-n) \leq 0$



$$\text{Hence, } x \in [-6, 4] \cup \left[5, \frac{n}{2}\right]$$

Now, there are 11 integers in $[-6, 4]$.

So, there must be 4 integers in $\left[5, \frac{n}{2}\right]$

$$\text{So, } \frac{n}{2} \in [8, 9)$$

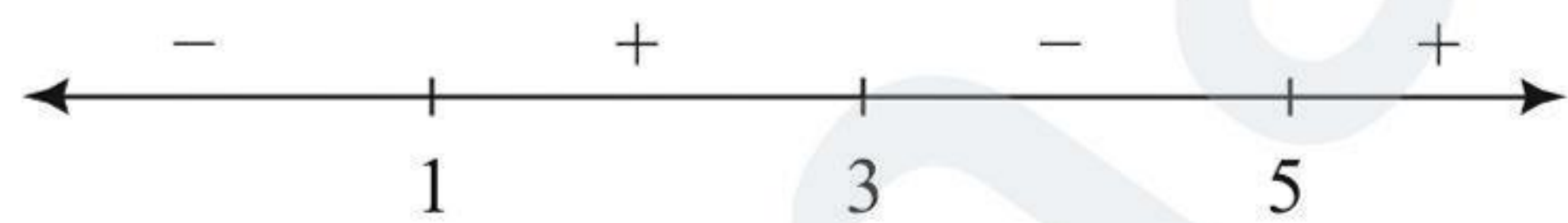
$$\Rightarrow n \in [16, 18)$$

20. (1) We have $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$

$$\Rightarrow \frac{2x+14+3x^2-14x-5}{2(x-5)} \geq 0$$

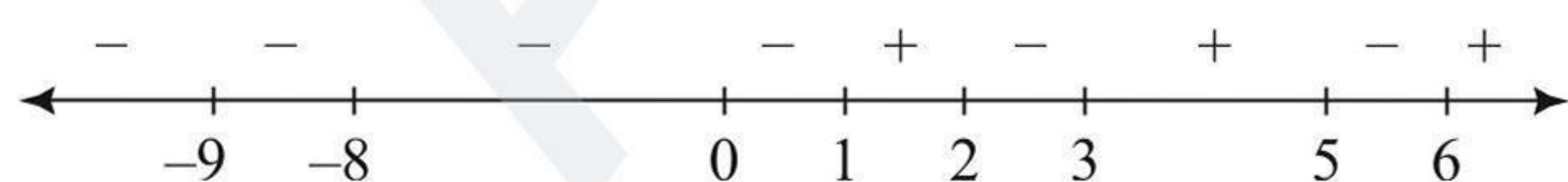
$$\Rightarrow \frac{3x^2-12x+9}{2(x-5)} \geq 0$$

$$\Rightarrow \frac{(x-1)(x-3)}{(x-5)} \geq 0$$



From the sign scheme, $x \in [1, 3] \cup (5, \infty)$.

21. (3) We have $\frac{(x-5)^{1005}(x+8)^{1008}(x-1)}{x^{1006}(x-2)^3(x-3)^5(x-6)(x+9)^{1010}} \leq 0$



From the sign scheme,

$$x \in (-\infty, 9) \cup (-9, 0) \cup (0, 1] \cup (2, 3) \cup [5, 6)$$

22. (3) We have $|x|^3 - 4|x|^2 + 3|x| = 0$

$$\Rightarrow |x|(|x|^2 - 4|x| + 3) = 0$$

$$\Rightarrow |x|(|x| - 1)(|x| - 3) = 0$$

$$\Rightarrow |x| = 0, 1, 3$$

$$\Rightarrow x = 0, 1, 3, -1, -3$$

23. (4) We have $|x-1||x^2-2| = 2$

Since x is an integer, we have

$$|x-1| = 1 \text{ and } |x^2-2| = 2$$

$$\text{or } |x-1| = 2 \text{ and } |x^2-2| = 1$$

$$\therefore x = 0, 2 \text{ and } x = 0, \pm 2$$

$$\text{or } x = -1, 3 \text{ and } x = \pm 1$$

Therefore, solutions are $-1, 0, 2$.

24. (3) We have $\frac{|x+2|-x}{x} < 2$

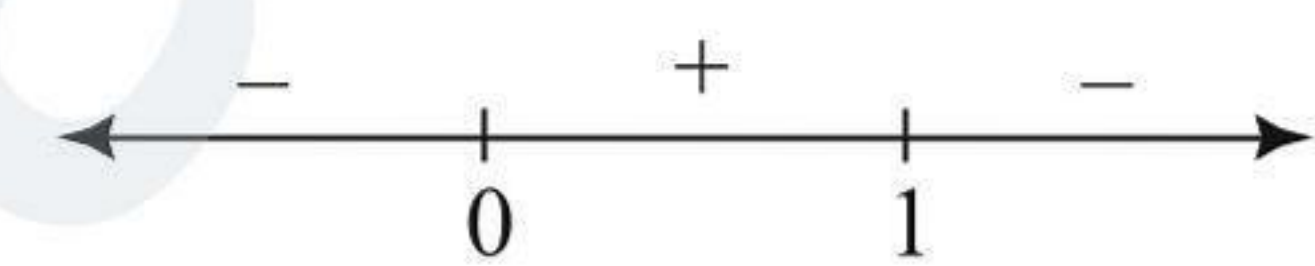
$$\Rightarrow \frac{|x+2|-3x}{x} < 0$$

$$\text{Let } |x+2|-3x = 0$$

$$\Rightarrow x+2 = \pm 3x$$

$$\Rightarrow x = 1$$

Sign scheme of the expression $\frac{|x+2|-3x}{x}$ is as shown in the following figure:



From the sign scheme, $x \in (-\infty, 0) \cup (1, \infty)$.

25. (2) $|2-|x|| = x+4 \quad \dots(i)$

$$\Rightarrow 2-|x| = x+4 \quad \dots(ii)$$

$$\text{or } |x|-2 = x+4 \quad \dots(iii)$$

From (i), $2-x = x+4$ or $x = -1$.

But this value does not satisfy (i).

From (iii), $-x-2 = x+4$ or $x = -3$, which satisfies (i).

So, equation has only one solution.

26. (1) We have $\frac{x^2+6x-7}{|x+2||x+3|} < 0$.

Clearly, $x \neq -2, -3$.

$$\text{Now, } x^2+6x-7 < 0$$

$$\Rightarrow (x+7)(x-1) < 0$$

$$\Rightarrow x \in (-7, 1) - \{-3, -2\}$$

So, integers are $\{-6, -5, -4, -1, 0\}$

27. (3) We have $-4 \leq x < 2$

$$\Rightarrow -2 \leq x+2 < 4$$

$$\Rightarrow 0 \leq |x+2| < 4$$

$$\Rightarrow -3 \leq |x+2|-3 < 1$$

$$\Rightarrow 0 \leq ||x+2|-3| \leq 3$$

28. (2) We have, $||x-1|-5| < 2x-5$

$$\therefore 2x-5 > 0$$

$$\Rightarrow x > \frac{5}{2} \quad (1)$$

Now from (1),

$$-2x+5 < |x-1|-5 < 2x-5$$

$$\Rightarrow -2x+10 < |x-1| < 2x$$

Case I: $x \geq 1$

$$\therefore -2x+10 < x-1 < 2x$$

$$\Rightarrow 3x > 11 \text{ and } x > -1$$

$$\Rightarrow x > 11/3 \quad (2)$$

Case II: $x < 1$

$$\therefore -2x+10 < 1-x < 2x$$

$$\Rightarrow x > 9 \text{ and } x > 1/3$$

$$\Rightarrow x > 9, \text{ not possible as } x < 1$$

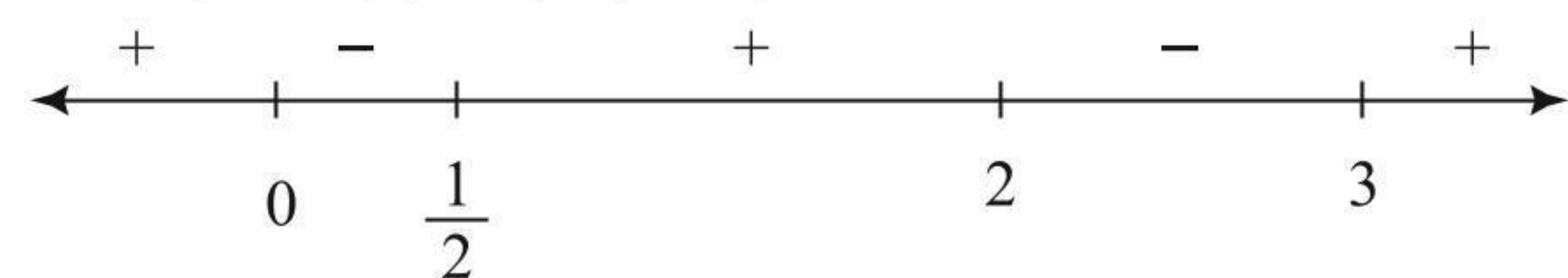
Therefore, $x \in (11/3, \infty)$

29. (2) We have,

$$\begin{aligned} |2x^2 - 5x + 2| + |x^2 - 3x| &= |x^2 - 2x + 2| \\ &= |(2x^2 - 5x + 2) - (x^2 - 3x)| \end{aligned}$$

$$\therefore (2x^2 - 5x + 2)(x^2 - 3x) \leq 0$$

$$\Rightarrow (2x - 1)(x - 2)x(x - 3) \leq 0$$



From the sign scheme, we have $x \in \left[0, \frac{1}{2}\right] \cup [2, 3]$

30. (3) We have $|x^2 - 5x + 6| + |x^2 + 12x + 27| = |17x + 21|$

$$\begin{aligned} \text{or } |x^2 - 5x + 6| + |x^2 + 12x + 27| &= |(x^2 + 12x + 27) - (x^2 - 5x + 6)| \\ &= |(x^2 + 12x + 27) - (x^2 - 5x + 6)| \end{aligned}$$

$$\Rightarrow (x^2 - 5x + 6)(x^2 + 12x + 27) \leq 0$$

$$\Rightarrow (x - 2)(x - 3)(x + 3)(x + 9) \leq 0$$



From the sign scheme, $x \in [-9, -3] \cup [2, 3]$